

The Environmental Cost of Place-Based Industrial Policy: Evidence from Industrial Zones in Vietnam

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Abstract

Special economic zones are widely adopted across countries to boost growth, yet their environmental cost is far less documented. This paper examines the local air pollution footprint of a zone program in Vietnam, which has opened more than 300 industrial zones since the early 1990s. Exploiting their staggered establishment, I first show that a zone's operation increases satellite-detected industrial combustion within a 5 km radius more than twofold. Using wind-driven variation in downwind exposure, I find that it also raises PM_{2.5} by approximately $0.70 \mu\text{g}/\text{m}^3$, a 25-day loss of life expectancy per exposed person. A back-of-the-envelope calculation suggests this health cost can be as high as the program's local income gains for the average resident. The pollution effect does not decline over a zone's life but differs across cohorts. A scale-composition-technique decomposition attributes the variation mainly to the industrial mix each cohort admitted rather than to cleaner production or zone scale, thereby suggesting that where enforcement on operating firms is weak, the environmental cost of a zone is set by the industries admitted at approval rather than by standards on incumbents.

Keywords: place-based policy, industrial zones, air pollution, Vietnam

JEL Codes: Q53, O14, R11

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1 INTRODUCTION

Special economic zones are widely adopted across countries to attract investment and boost growth, with more than 5,000 operating worldwide (UNCTAD, 2019). Evaluations of a small set of programs typically find higher firm entry, productivity, employment, wages, and household welfare (Busso et al., 2013; J. Wang, 2013; Kline & Moretti, 2014; Y. Lu et al., 2019; F. Lu et al., 2023; Abagna et al., 2025; Tafese et al., 2025; Pham, 2026). Much less is known about their environmental cost, even though zones concentrate the industrial activity that drives local air pollution. This paper examines how much pollution a zone program imposes on nearby communities and what determines the size of that cost.

I answer these questions with the industrial zone program of Vietnam, which has opened over 300 zones since the early 1990s. The program is a useful case for both questions. For the first, the staggered opening of zones across provinces provides the variation that identifies their causal effect on local air pollution. For the second, zones were approved under different regimes, and the cost can be compared across them: environmental regulation arrived in stages and government promotion directed investment toward export-oriented manufacturing, while enforcement on the firms already operating remained weak. This gap between *de jure* regulation and *de facto* enforcement is common to many developing countries in which zone policies operate.

To answer the first question, I proceed in two steps that link the source of the pollution to the population exposed to it. In the first step, I establish that industrial zones generate a localized increase in industrial combustion, measured by satellite-detected thermal anomalies from NASA’s FIRMS record (NASA FIRMS, 2021). A distance-based analysis comparing cells at varying distances from zone centroids to cells 45 to 50 km away shows that the effect is concentrated within 5 km: coefficients attenuate by an order of magnitude between the 0 to 5 km and 5 to 10 km bins and are indistinguishable from zero beyond 10 km, consistent with combustion as a spatially concentrated, point-source phenomenon. Using this result to define treatment, I then exploit the staggered establishment of zones to trace the effect over time. The effect accumulates progressively over the zone lifecycle as tenants fill in rather than appearing discretely when a zone opens, and the event-study pattern is qualitatively similar across two-way fixed effects and the heterogeneity-robust estimators of de Chaisemartin and d’Haultfoeuille (2024) and Callaway and Sant’Anna (2021). Because satellite-detected thermal anomalies are rare events with substantial mass at zero, I estimate a Poisson specification to obtain a proper percentage-change interpretation (J. Chen & Roth, 2024): zone operation more than doubles the frequency of detections. The result is not driven by any single province in leave-one-out estimates and is robust to potential violations of parallel trends (Rambachan & Roth, 2023).

In the second step, I ask whether zone establishment raises ambient air pollution for the surrounding population, treating the 5 km ring from the first step as the source area from which emissions travel downwind. Unlike thermal anomalies, which are confined to the source pixel, fine particulate matter disperses with prevailing winds and accumulates at a distance from where it is emitted. Following a literature that exploits wind direction to identify the effects of spatially

fixed pollution sources (e.g., Deryugina et al., 2019; Anderson, 2020), I use variation in downwind exposure to separate zone-induced pollution from background concentrations that do not depend on a cell's position in the plume. Because a cell's position relative to the wind changes with daily weather while its distance from the zone does not, the same cell's exposure varies from month to month. I aggregate these daily fluctuations into a continuous monthly downwind share, and a cell then serves as its own control in the months when it lies mostly upwind of an operational zone. A one-standard-deviation increase in downwind exposure raises PM_{2.5} by 2.7%. The population-weighted zone-attributable PM_{2.5} increment is 0.70 $\mu\text{g}/\text{m}^3$, equivalent to 14% of the WHO annual guideline. Pre-treatment placebo tests confirm that the wind-based exposure measure does not predict PM_{2.5} before any nearby zone becomes operational, supporting temporal attribution to zone establishment. A permutation test provides complementary spatial evidence: in each of 500 iterations, I relocate every zone to a randomly drawn site in its own province, at least 25 km from the province's actual zones, reshuffle operational years, and re-estimate the baseline specification. The placebo coefficients center around zero, and the actual estimate lies entirely outside their distribution.

Translating the pollution estimates into health costs, the population-weighted PM_{2.5} increment implies a reduction of approximately 25 days of life expectancy per exposed individual, aggregating to roughly 3.2 million person-years across the 46 million residents of the transport ring. The burden falls unevenly. Residents of cells at the 95th percentile of downwind exposure face a PM_{2.5} increment of 1.39 $\mu\text{g}/\text{m}^3$, with an implied life expectancy reduction of 50 days. At the 99th percentile, the increment reaches 3.34 $\mu\text{g}/\text{m}^3$, or approximately 120 days. These are populations living close to and directly downwind of multiple operational zones simultaneously. A back-of-the-envelope calculation compares this health cost with the program's local income gains, which Pham (2026) estimates at roughly \$30 to \$50 per person per year. To make the comparison, I value the lost life expectancy using the U.S. value of a life-year scaled down for Vietnam's lower income. Under the scaling recommended for low- and middle-income countries (Robinson et al., 2019), the cost is about \$8 per exposed person per year or one fifth of the gains. Under proportional scaling (Viscusi & Masterman, 2017), it reaches about \$44 and can be as high as the income gains for the average resident.

To answer the second question, I examine how the pollution effect varies along three dimensions: over a zone's life, across establishment cohorts, and across the places where zones are sited. Estimating the dynamics separately by establishment cohort shows that the per-exposure effect of a given cohort persists essentially unchanged beyond a decade of operation. Across cohorts, the effect is largest for zones established between 2003 and 2009, during Vietnam's rapid industrialization and before the 2009 technical standards. It is roughly half as large for zones established before 2003 and indistinguishable from zero for zones established after 2009. Those zones faced the standards at the design stage, entered during the reorientation of foreign investment toward electronics and light assembly (Hollweg et al., 2017; Athukorala & Nguyen, 2022), and were still filling with tenants at the end of the study period. Across places, the effect is persistently larger

where the surrounding industrial base is more pollution-intensive. These differences could reflect the scale of zone activity (a scale effect), the industries a zone admits (composition), or how firms within an industry produce (technique) (Grossman & Krueger, 1995; Shapiro & Walker, 2018). To separate them, I combine the pollution estimates with a complementary firm-level dataset, the energy records of the Vietnam Enterprise Survey. Neither technique nor scale explains it: the energy records show no within-industry shift toward cleaner fuels or lower energy intensity across cohorts, and the effect shows no gradient in zone scale. The variation follows the industrial mix of the firms each cohort attracted. On this evidence, the environmental record of the program was set largely when its zones were approved and changed little under the standards later imposed on the firms operating in them.

This paper contributes to two literatures. The first concerns the environmental consequences of place-based development policies. The evidence concentrates on their economic effects: firm entry, productivity, employment, wages, and household welfare (Busso et al., 2013; J. Wang, 2013; Kline & Moretti, 2014; Neumark & Simpson, 2015; Y. Lu et al., 2019; F. Lu et al., 2023; Abagna et al., 2025; Tafese et al., 2025; Pham, 2026). The environmental costs have only recently drawn attention: polluting-industry growth (Fan et al., 2023; Ma et al., 2024), air pollution (Duan et al., 2024), deforestation (Hess et al., 2021), and avoided or reversed damage under programs with environmental safeguards (T. Garg & Shenoy, 2021; Lin et al., 2026). These studies compare outcomes in areas with and without a program. For air pollution, such a comparison establishes neither where the pollution comes from nor who bears it. I propose a two-step approach that answers both questions: the first step detects the combustion a program generates at its sites, and the second attributes the PM_{2.5} arriving downwind to those sites through wind-driven variation in exposure. Both steps require only satellite and reanalysis data and the location and timing of program sites. The approach therefore applies where ground monitoring is sparse, and it draws on a broader practice of measuring from space what official records miss (Henderson et al., 2012; Burke et al., 2021). I use it to examine the air pollution externality of Vietnam’s national zone program and the health burden it implies. The estimates measure the program’s local footprint (the pollution its zones impose on the people around them) rather than its net national effect. For the average resident of the nearby communities, the health cost can be as high as the program’s documented income gains.

The second contribution concerns what determines the pollution effect. A long literature on trade and growth distinguishes three channels (Grossman & Krueger, 1995): pollution can change with the scale of activity, the composition of industry, or the technique of production. In advanced economies, cleaner production within industries (the technique effect) accounts for most of the decline in manufacturing emissions under binding regulation (Levinson, 2009; Cherniwchan et al., 2017; Shapiro & Walker, 2018). That condition often fails in developing countries, where the gap between the regulatory framework and its enforcement is large (Duflo et al., 2013; Greenstone & Hanna, 2014). I apply the same decomposition to the variation in the zone effect in this weak-enforcement setting, and the evidence points to the opposite ordering. The zone effect varies

across the cohorts established under different regulatory regimes and across the places where zones are sited, but not over a zone’s life. Firm-level energy data show no within-industry movement toward cleaner fuels or lower energy intensity, across cohorts or after standards tightened. The variation instead tracks the industrial composition each cohort admitted, shaped jointly by the instruments acting at approval and the changing structure of Vietnamese manufacturing. Where firm-level enforcement is weak, the environmental footprint of a zone appears to depend more on which industries it admits than on how those industries produce. The same compositional logic reconciles the pollution increases estimated here with the damage avoided elsewhere: an Indian program excluded damaging industries outright (T. Garg & Shenoy, 2021), and a Chinese zone raised entry barriers for polluting firms under its environmental targets (Lin et al., 2026). In each setting, the outcome follows which industries the policy admits.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background, including characteristics of industrial zones and related environmental regulations in Vietnam. Section 3 describes the data and provides descriptive evidence. Section 4 presents the empirical analysis of industrial combustion at the source. Section 5 presents the wind-based analysis of ambient air pollution. Section 6 translates the estimates into a health burden and examines what accounts for the variation in the pollution effect. Section 7 concludes.

2 BACKGROUND

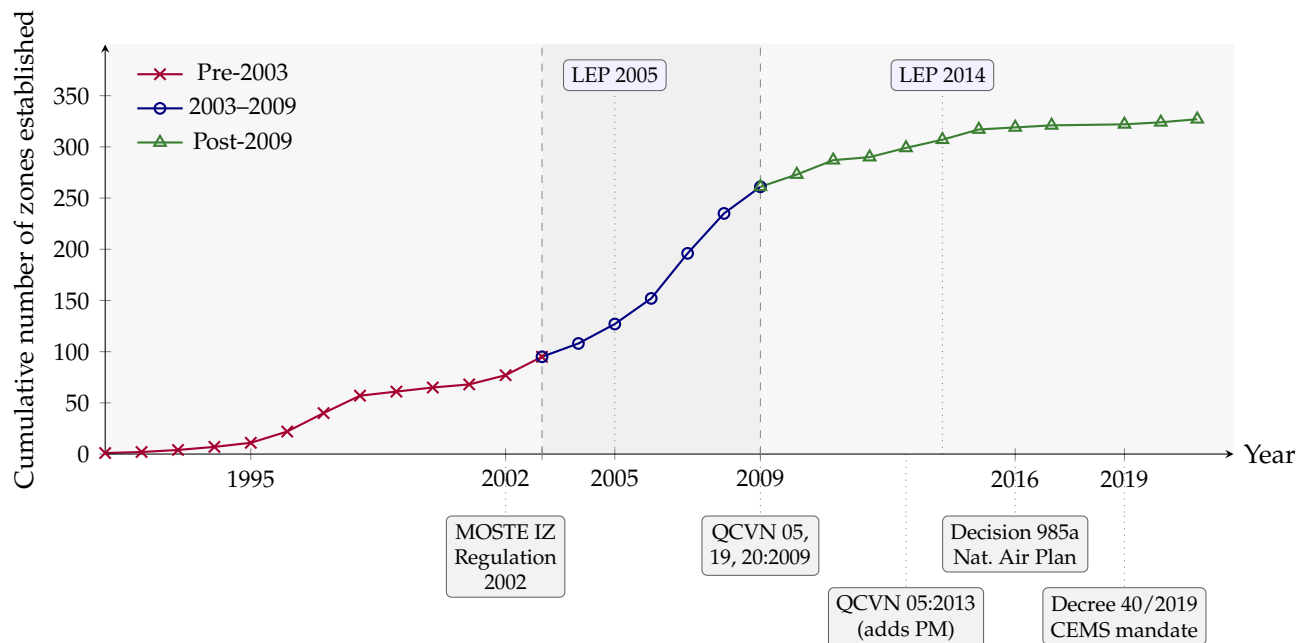
2.1 Industrial Zones and Environmental Regulations

Industrial zones have been a central instrument of Vietnam’s industrial policy since the mid-1990s. The first zones were established following Decree 192/CP of 1994, and by the end of 2020 over 300 zones were in operation across provinces (Figure 1). Zones are bounded areas set aside for manufacturing and related services, approved by the Prime Minister on proposal from the Ministry of Planning and Investment and administered by management boards operating under provincial People’s Committees.¹ Tenants receive publicly built infrastructure together with tax, duty, and land-rent incentives.

The environmental regime governing these zones has evolved through three broad periods, marked by the dashed lines in Figure 1. The first generation of zones, established before 2003, operates under minimal environmental oversight. The Ministry of Science, Technology, and Environment (MOSTE) issues the first regulation specific to industrial zone environmental management in 2002, but monitoring infrastructure and enforcement capacity are limited. The 2003 to 2009 period sees the first comprehensive framework legislation with the Law on Environmental Protection (LEP) of 2005, which formalizes the environmental impact assessment (EIA) process (Decree 80/2006/ND-CP) and is followed by the first national technical regulations on ambient air quality (QCVN 05:2009/BTNMT) and industrial emissions of inorganic dust and organic substances (QCVN 19:2009/BTNMT, QCVN 20:2009/BTNMT). Sector-specific emission caps are established

¹Decree 35/2022/ND-CP on the Management of Industrial Zones and Economic Zones.

Figure 1: Evolution of Industrial Zones and Air Pollution Regulatory Framework in Vietnam



Notes: The figure presents the evolution of Vietnam's air pollution regulatory framework. Laws (Law on Environmental Protection, LEP) are shown above the timeline; technical regulations (QCVN and TCVN standards) and action plans are shown below. The shaded region from 2016 onward marks the period following Prime Minister Decision 985a/QD-TTg, which established a coordinated air quality management regime. MOSTE refers to the Ministry of Science, Technology, and Environment; QCVN denotes National Technical Regulations on air quality and industrial emissions; CEMS denotes continuous emission monitoring systems.

for the highest-emitting industries, including thermal power (QCVN 22:2009/BTNMT), cement (QCVN 23:2009/BTNMT), and steel (QCVN 51:2013/BTNMT). The post-2009 period brings progressively tighter requirements. QCVN 05:2013 adds $PM_{2.5}$ to the ambient air quality standard for the first time, the revised LEP of 2014 strengthens EIA and inspection provisions, and Prime Minister Decision 985a/QD-TTg of June 2016 establishes a coordinated air quality management regime combining emission source control, ambient monitoring, and vehicle and construction dust standards. Decree 40/2019/ND-CP mandates continuous emissions monitoring systems (CEMS) at major industrial sources.

Throughout these three periods, regulation has mainly relied on command-and-control instruments, with market-based instruments playing a minor role. Zone-level environmental infrastructure has been focused on water rather than air: industrial zones are required to install centralized wastewater treatment plants before tenant firms can operate, but no equivalent zone-level air treatment requirement exists.² Price instruments during the study period are limited to the Law on Environmental Protection Tax of 2010, which took effect in 2012 and applies upstream to fossil fuels, coal, and selected chemicals rather than directly to emissions at the stack. Vietnam has no

²This asymmetry is partly technical: wastewater can be aggregated and treated at a common facility, while air emissions must be controlled at each stack.

operational emissions trading scheme covering industrial air pollutants.³

Most of these instruments apply before a firm operates. Projects in polluting sectors or above size thresholds must prepare an EIA, and regulators must accept it before the project receives an investment certificate. The project design must also meet the sector emission standards, and the heaviest industries need approval under their sector master plans. Once in operation, a firm faces the same standards enforced through periodic inspections and continuous emissions monitoring from 2019.

Despite increases in provincial spending on environmental protection per capita and in detected environmental violations (General Statistics Office, 2023), enforcement remains structurally weak through the end of the study period. The physical monitoring infrastructure is sparse and poorly connected: by end-2020, Vietnam operated just 108 ambient air quality monitoring stations nationwide (76 reference-grade) and 522 automatic emissions monitoring systems at polluter facilities (Ministry of Natural Resources and Environment, 2021), yet only 55% of emissions stations and 85% of ambient stations were transmitting data to the Ministry, with eight of 63 provinces not transmitting any emissions data at all.⁴ Even where monitoring systems were installed and transmitting, the data were unreliable. In March 2026, Vietnam's Investigation Police Agency charged 74 individuals across 59 enterprises with manipulating data from the mandated continuous monitoring systems, with tampering identified at approximately 55% of inspected stations nationwide, including at major thermal power plants and metal producers.⁵ Sparse infrastructure, incomplete data transmission, compromised monitoring integrity, and routine inspections that rely on self-reporting by firms to under-resourced provincial departments together imply that the effective constraint on firm-level air emissions was considerably weaker than the statutory framework alone would suggest. Regulation therefore likely acted mainly on which industries entered a zone rather than on the firms already operating.

2.2 Industrial Composition of Zones

Firms operating inside zones have a distinctive profile. Table 1, Panel A, reports firm and employment composition in the Vietnam Enterprise Survey (VES) 2016 for three groups: firms operating inside an industrial zone, firms outside zones but in districts whose territory intersects a 5 km buffer around any zone, and firms outside zones in districts beyond the 5 km buffer. Inside zones, 39% of firms are foreign-invested, 29% employ 100 or more workers, and 62% operate in manufacturing. Manufacturing employment accounts for 95% of total employment inside zones, against 23 to 36% outside. Zones therefore function as concentrated sites of foreign-invested, large-firm, manufacturing activity within a broader Vietnamese economy dominated by small domestic en-

³A direct fee on industrial exhaust gas emissions was issued only under Decree 153/2024/ND-CP in November 2024, effective January 2025. A domestic carbon market pilot under Decree 06/2022/ND-CP is scheduled for 2025 to 2028.

⁴The 108 ambient stations include 32 lower-cost sensor units. The emissions station count sits alongside 655 automatic wastewater monitoring stations.

⁵<https://vietnamnews.vn/society/1778345/viet-nam-prosecutes-74-over-widespread-manipulation-of-environmental-monitoring-data.html>

terprises and services.

Table 1: Firm and Employment Composition, by Proximity to Industrial Zones, Year 2016

Panel A: All Firms, by Proximity				
	Inside Zones		Outside Zones	
	(1)	(2)	Within 5 km	Beyond 5 km
	(1)	(2)	(3)	(4)
Share of foreign-invested firms	0.393	0.018	0.012	
Share of firms with 100+ employees	0.292	0.029	0.029	
Share of firms in manufacturing	0.619	0.144	0.105	
Share of employment in manufacturing	0.946	0.359	0.233	

Panel B: Manufacturing Firms, by Pollution-Intensity Classification				
	Inside Zones		Outside Zones	
	Light	Pollution-intensive	Light	Pollution-intensive
	(1)	(2)	(3)	(4)
Share of manufacturing firms	0.810	0.190	0.847	0.153
Share of manufacturing employment	0.905	0.095	0.862	0.138
Share of foreign-invested firms	0.615	0.424	0.035	0.030
Share of firms with 100+ employees	0.446	0.385	0.077	0.094
Mean employment	411	184	53	47
Foreign-invested firms	564	270	516	189
Domestic firms	166	120	36	42

Notes: Panel A reports group means for all firms in the Vietnam Enterprise Survey 2016, by proximity to industrial zones. Inside Zones comprises firms operating inside an industrial zone, as reported by the firm in the VES. Outside Zones (Within 5 km) comprises firms located in districts whose territory intersects a 5 km buffer around an industrial zone centroid but which are not themselves inside a zone. Outside Zones (Beyond 5 km) comprises firms in districts that do not intersect the 5 km buffer. Panel B reports means for manufacturing firms, splitting firms by location (inside a zone or outside) and by the pollution-intensity classification (Appendix B): pollution-intensive sectors are the top quartile of the energy-based intensity ranking constructed from the VES energy module, comprising non-metallic minerals, paper, coke and refined petroleum, basic metals, and textiles; light manufacturing comprises all other manufacturing sectors. The shares of manufacturing firms and employment are computed within each location group. Employment is total labor at the end of the year. Mean employment is reported for all firms and separately for foreign-invested and domestic firms. *Source:* Data from VES 2016.

Classifying manufacturing firms as light or pollution-intensive shows that zones do not screen out the polluting industries: pollution-intensive firms account for 19% of manufacturing firms inside zones and 15% outside.⁶ Their employment share is lower inside zones than outside, but this reflects firm size rather than composition: light manufacturing in zones is dominated by large foreign-invested export assembly operations, while the pollution-intensive producers are smaller,

⁶The classification assigns each firm by its primary sector. Pollution-intensive sectors are the top quartile of the energy-based intensity ranking constructed from the VES energy module: non-metallic minerals, paper, coke and refined petroleum, basic metals, and textiles. Light manufacturing comprises all other manufacturing sectors (Appendix B).

less often foreign-invested, and concentrated in sectors such as cement and steel. The size gap is concentrated among the foreign-invested firms, whose light-manufacturing operations are more than twice the size of their pollution-intensive counterparts (Table 1, Panel B).

This sorting is consistent with at least three explanations, none mutually exclusive. The design of FDI promotion points foreign investment toward light export manufacturing (Akinci & Crittle, 2008; Athukorala & Tran, 2010): investment incentives during the study period favored export-oriented production (e.g., Decree 24/2000/ND-CP, Investment Law 59/2005/QH11, Decree 108/2006/ND-CP). Comparative advantage likely works in the same direction: foreign investors are drawn to Vietnam largely for labor-intensive export assembly (Hollweg et al., 2017; McCaig & Pavcnik, 2018; Athukorala & Nguyen, 2022), while the pollution-intensive industries sell mostly to the domestic market, where construction demand grew rapidly over the period. The domestic ownership of the pollution-intensive producers, in turn, reflects the state-owned legacy of heavy industry (Nguyen et al., 2016): these industries developed under ministry master plans and state general corporations, with foreign participation limited to joint ventures.⁷

3 DATA AND SUMMARY STATISTICS

3.1 Data

To examine the environmental impacts of industrial zones, I combine information on zone locations and opening dates with satellite-derived measures of industrial combustion and air pollution, together with reanalysis wind fields that support the identification strategy. This subsection describes the analysis grid and the construction of the treatment variable and the two outcome variables, along with the underlying data sources. The main study period covers 2000 to 2020.

Spatial unit and administrative geography. The analysis is conducted on a regular grid of 0.05 by 0.05 degree cells covering Vietnam, each approximately 5 by 5 kilometers. Each cell in the 0.05 degree grid contains arrays of finer 0.01 degree sub-cells, which I use for aggregating higher-resolution inputs and for assigning cells to administrative units. Provincial boundary shapefiles come from the Humanitarian Data Exchange.⁸ Because provincial boundaries changed during the study period through splits and mergers, I harmonize administrative units to their 2000s equivalents using a consistent-identifier mapping. I assign each sub-cell to a province by point-in-polygon and assign the parent cell to the province containing the largest number of its sub-cells, resolving the assignment for cells overlapping multiple provinces. The province is the relevant administrative unit in this setting because environmental regulations during the study period are administered by provincial authorities (Section 2).

⁷Prime Minister Decisions 90/TTg and 91/TTg (1994) established the framework for the state general corporations, under which VICEM in cement and VNSTEEL in steel were organized. Sector master plans include Decision 108/2005/QD-TTg for cement and Decision 145/2007/QD-TTg for steel.

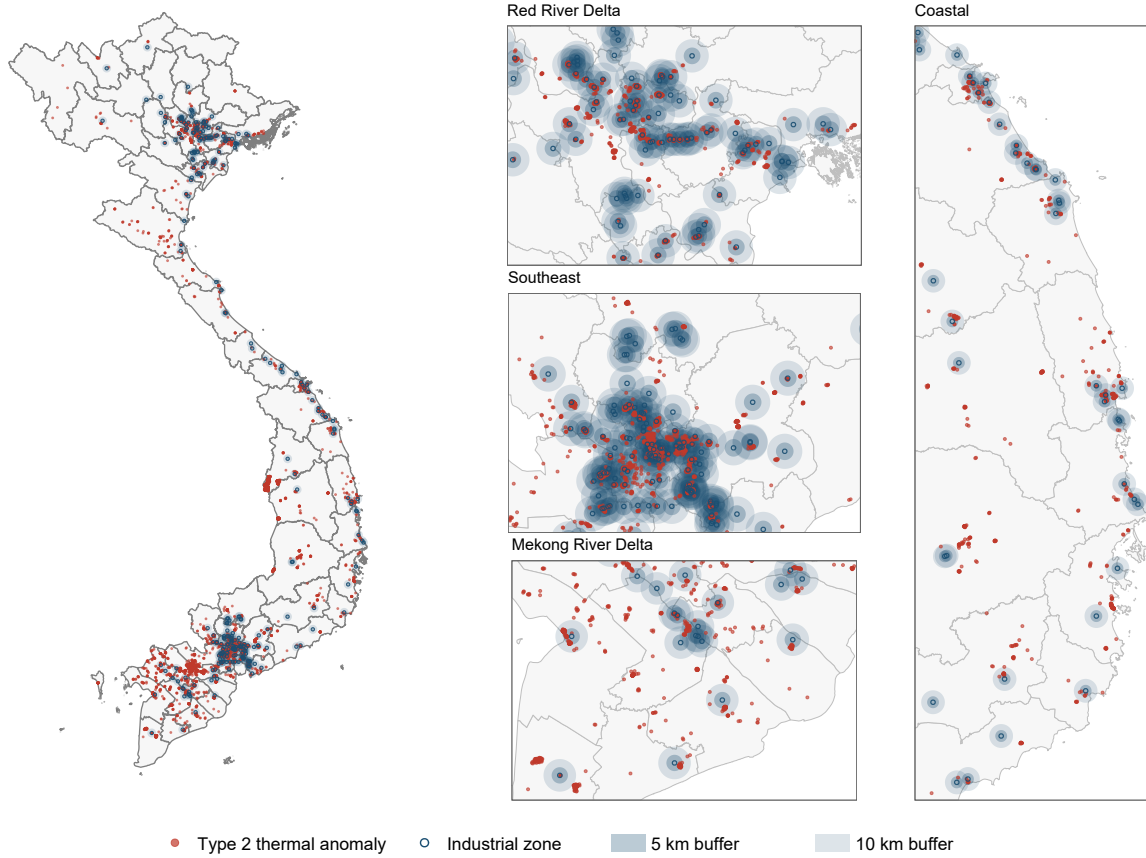
⁸<https://data.humdata.org/dataset/cod-ab-vnm>

Treatment: proximity to an operating industrial zone. A cell is treated when any industrial zone within a given distance of it is in operation, so that treatment varies across cells with their distance to zones and over time as new zones open. Distance to the nearest zone is used only to assign cells to the distance bins of the distance-based analysis. Zone locations come from the Ministry of Planning and Investment database, which records each zone’s name, administrative location, establishment date, and operational status. The database does not report geographic coordinates, and I georeference each zone using Google Maps. The analysis sample comprises zones in operation as of June 2023.

Industrial combustion at the source. The first outcome is the intensity of industrial combustion within a cell, measured from satellite-detected thermal anomalies. Fire detection data come from the NASA FIRMS MODIS Collection 6.1 Active Fire Product (Giglio et al., 2016; NASA FIRMS, 2021). Each detection is recorded with geographic coordinates, date, fire radiative power, and a type classification distinguishing presumed vegetation fires (Type 0), active volcanoes (Type 1), other static land sources (Type 2), and offshore detections (Type 3). Type 2 identifies hotspots detected on sixteen or more days in a calendar year or located in urban areas (Giglio et al., 2021). Because industrial combustion sources such as gas flares, steel mills, and cement kilns tend to operate continuously, this persistence criterion captures most of them. Type 2 has been used in the recent literature as a satellite proxy for local industrial activity (Yang et al., 2024). As seen in Figure 2, Type-2 detections concentrate around the areas with the densest industrial zone footprints in Vietnam, including the Red River Delta in the north, the Southeast and Mekong River Delta in the south, and the central and southern coastal provinces, supporting the interpretation of Type-2 as an industrial proxy for this setting.

The classification does not distinguish industrial from non-industrial persistent thermal sources. In the Vietnamese context, the main non-IZ sources likely captured by Type 2 are informal industrial activities such as brick kilns and rice-husk furnaces, as well as forest clearing in upland areas. The same figure illustrates this: Type-2 detections in the Mekong River Delta and along the interior stretches of the Coastal region extend well beyond the vicinity of industrial zones, reflecting these non-IZ sources rather than IZ-related combustion. Volcanic and offshore detections are negligible in Vietnam and excluded. I aggregate point detections to cell-year counts of Type 2 anomalies and use Type 2 as the measure of industrial activity at the source. As will be shown in Section 4, the estimated increase in combustion is highly concentrated within 5 km of an operational zone. Consequently, my primary on-source specifications define treatment using this 5 km radius, focusing attention on the precise areas where Type-2 detections are most plausibly attributable to zone-related activity. Because non-industrial sources of Type-2 detections, such as rice-husk furnaces, are often heavily concentrated in specific regions like the Mekong River Delta, I verify that the estimates are not driven by these localized measurement artifacts by conducting leave-one-out robustness checks that drop one province at a time. Furthermore, the use of cell and year fixed effects helps absorb diffuse, persistent non-industrial combustion across the rest of the sample.

Figure 2: Type-2 Thermal Anomalies and Industrial Zones in Vietnam



Notes: The figure overlays MODIS Collection 6.1 Type-2 thermal anomaly detections (NASA FIRMS, 2021) (pooled over 2004 to 2019) with the locations of industrial zones in operation as of June 2023. Type-2 identifies hotspots detected on sixteen or more days in a calendar year or located in urban areas (Giglio et al., 2021). The left panel shows the national distribution. The four right panels zoom into the Red River Delta, Southeast, Mekong River Delta, and Coastal provinces. Each zone is plotted at its centroid together with 5 and 10 km buffers. Provincial boundaries are harmonized to their 2002 equivalents. *Source:* Industrial zone locations are from the Ministry of Planning and Investment database, geo-referenced using Google Maps.

Ambient air pollution. The second outcome is monthly mean surface-level PM_{2.5}, a standard measure of ambient fine particulate matter and the pollutant most commonly used to assess the health and welfare burden of industrial air pollution. Industrial combustion contributes to ambient PM_{2.5} both directly, through primary particulate emissions, and indirectly, through gaseous precursors that form secondary particulates in the atmosphere (Hallquist et al., 2009; Huang et al., 2014; G. Wang et al., 2016). PM_{2.5} data are drawn from the V6.GL.02.04 satellite-derived product of Shen et al. (2024), which combines aerosol optical depth retrievals from multiple satellite platforms with GEOS-Chem chemical transport modeling and calibrates the resulting geophysical estimate to ground-based monitors. The product reports monthly surface-level concentrations at 0.01 degree native resolution, approximately one kilometer over Vietnam. I aggregate the data to the 0.05 degree analysis grid by taking cell-level means. Because airborne pollutants are transported by wind, identifying the effect of zone establishment on PM_{2.5} requires accounting for

atmospheric circulation. Section 5 describes the wind-based exposure measure used in the empirical analysis.

Other data. The PM_{2.5} specifications require controls for meteorological conditions that independently affect ambient concentrations. Precipitation removes suspended particles through wet deposition, and temperature governs boundary-layer mixing and the rate at which gaseous precursors convert into secondary aerosol. I control for both using monthly mean 2-meter temperature (entered as a quadratic) and monthly total precipitation from ERA5-Land (Muñoz-Sabater et al., 2021), which shares the reanalysis platform used for the wind fields. Translating the cell-level PM_{2.5} estimates into population-level health burdens requires spatially resolved population counts, and I use the Gridded Population of the World, Version 4 (Center for International Earth Science Information Network (CIESIN), Columbia University, 2017), which reports year-2000 population at approximately 1 km resolution. Cell-level elevation from the Shuttle Radar Topography Mission (Farr et al., 2007) enters the covariate balance tests. District-level baseline characteristics used in these balance tests and as controls interacted with time fixed effects are constructed from the 2002 Vietnam Household Living Standards Survey (VHLSS). Finally, the analysis of industrial composition and mechanisms draws on the Vietnam Enterprise Survey (VES). I use the full VES panel to characterize the sectoral mix of zone activity and to estimate changes in the number of manufacturing firms operating near zones. The test for within-industry fuel switching relies on physical quantities of nine fuel types, which are reported only in the 2009 and 2011–2015 rounds of the VES; I convert these to tons of coal equivalent using default net calorific values from A. Garg et al. (2006) and volume-to-mass conversion factors from United Nations Department of Economic and Social Affairs (2022). The same energy records underpin the pollution-intensity classification used throughout the paper, constructed as fossil-fuel energy per unit of output at the industry level; Appendix B details the construction and compares the resulting ranking with the U.S.-based classification of Shapiro and Walker (2018).

3.2 Summary Statistics

Table 2 reports district-level characteristics in 2002 to illustrate that areas with industrial zones differ systematically from areas without. Although the analysis itself is conducted at the grid-cell level, I aggregate to districts here because the household-survey variables that capture local economic conditions are constructed at the district level. Districts are classified into three groups by their exposure history to industrial zones during 2000–2020: never exposed within 5 km of an operational zone (column 1), first exposed via a zone established during 2003–2020 (column 2), and exposed via a zone established in or before 2003 (column 3). For columns (1) and (2), 2002 values are pre-treatment. For column (3), they reflect both pre-existing characteristics and changes since the zone became operational, and these districts are excluded from the cell-level event-study estimation.

The three groups differ along nearly every economic dimension. Moving from never-exposed to newly-exposed and early-exposed districts, the urban population share rises from 16% to 22%

Table 2: District-Level Characteristics in 2002, by Zone Establishment Cohort

	No Zones	Zones	Zones	p-values		
		2003–2020	before 2003	(1) = (2)	(1) = (3)	(2) = (3)
	(1)	(2)	(3)	(4)	(5)	(6)
Urban population share	0.164 [0.259]	0.219 [0.286]	0.344 [0.397]	0.045	0.000	0.002
Minority population share	0.364 [0.405]	0.090 [0.190]	0.035 [0.112]	0.000	0.000	0.003
Share working-age with secondary degree	0.146 [0.138]	0.190 [0.135]	0.268 [0.162]	0.001	0.000	0.000
Share working-age in farming	0.720 [0.248]	0.578 [0.229]	0.421 [0.270]	0.000	0.000	0.000
Rice specialization	0.432 [0.247]	0.443 [0.246]	0.391 [0.302]	0.641	0.149	0.102
Type-2 fires (count)	0.078 [0.695]	0.150 [0.604]	0.255 [0.814]	0.259	0.028	0.217
PM2.5 ($\mu\text{g}/\text{m}^3$)	21.164 [4.250]	22.186 [4.183]	22.000 [4.228]	0.015	0.057	0.708
Number of Districts	361	154	146			

Notes: The table reports district-level summary statistics. The sample includes 661 districts, classified by exposure history to industrial zones (IZs) during 2000–2020, where a district is classified as exposed in a given year if its centroid lies within 5 km of an operational IZ. Treatment timing is set to one year before recorded operational status to absorb site-preparation activity. Column (1) reports districts never exposed during 2000–2020. Column (2) reports districts whose first exposure corresponds to an IZ established during 2003–2020. Column (3) reports always-exposed districts, defined as those exposed via IZs established in or before 2003. Each cell reports the group mean, with the standard deviation in brackets. Columns (4) through (6) report p-values from t-tests of equal means between the indicated group pairs, with heteroskedasticity-robust standard errors. *Source:* Urban population share, the share of working-age members with a secondary degree, the share of working-age members in farming, and rice specialization are constructed from the 2002 Vietnam Household Living Standards Survey. Type-2 fire counts are the sum of MODIS Type 2 (static land source) detections in 2002 (NASA FIRMS, 2021), the detection type used for the combustion outcome in the analysis. PM2.5 is the annual mean surface concentration in 2002 from Shen et al. (2024). Type-2 fire counts and PM2.5 are aggregated from cell-level data to the district level.

and 34%, while the share of working-age members with a secondary degree rises from 15% to 19% and 27%. The share in farming falls from 72% to 58% and 42%, and the minority population share from 36% to 9% and 4%. These patterns indicate that zones were placed in districts further along in structural transformation. Type-2 fire counts, the industrial combustion measure used in the analysis, follow the same ordering, 0.08, 0.15, and 0.26 in columns (1) to (3), although only the difference between columns (1) and (3) is statistically significant. For column (3), the count also reflects the combustion of the zones already operating in 2002. Rice specialization is the only dimension on which none of the three pairwise comparisons is statistically significant, suggesting that selection operated through urbanization and broader sectoral composition rather than agricultural specialization. The remaining pairwise differences are statistically significant at conventional levels, except for PM2.5 between columns (2) and (3). I address this non-random placement in the cell-level analysis through the identification strategies described in the next section.

4 EMPIRICAL ANALYSIS: INDUSTRIAL COMBUSTION AT THE SOURCE

Type-2 thermal anomalies, by construction, register only when high-temperature combustion occurs at the surface within a sensor pixel, and the satellite signal is spatially confined to the cell containing the heat source. Cell-level distance from operational industrial zones therefore provides a direct test of whether zone establishment generates detectable combustion at the source: if so, the effect should be concentrated in cells closest to the zone boundary and attenuate with distance. Because zones became operational over a 25-year period (Figure 1), a standard two-way fixed-effects regression pooling all cohorts may produce biased estimates when treatment effects vary across cohorts and over time (de Chaisemartin & d’Haultfoeuille, 2020; Goodman-Bacon, 2021; Borusyak et al., 2024). I address this in two complementary specifications. The first estimates distance-bin coefficients separately by cohort and aggregates them to characterize the spatial reach of any effect. The second uses the distance bin at which the effect is concentrated to define a binary treatment indicator, restricts the comparison group to never-treated cells, and examines the effect’s evolution over time.

4.1 Distance-based Analysis

I group zone cohorts into two-year bins to ensure adequate cell counts within each cohort regression. Cohort c comprises cells whose nearest industrial zone first becomes operational in year c or $c + 1$, where c is set to one year before recorded operational status to capture site-preparation activity. For each cohort c , I estimate:

$$y_{ipt} = \sum_{b=1}^9 \delta_b^c \cdot \mathbb{I}(t \geq c) \cdot \mathbb{I}(\text{bin}_i = b) + \alpha_i + \mu_t + \varepsilon_{ipt} \quad (1)$$

where y_{ipt} is a log-transformed measure of Type-2 thermal anomaly activity in cell i in province p in year t ; α_i is a cell fixed effect; μ_t is a year fixed effect; and $\mathbb{I}(\text{bin}_i = b)$ indicates cell i ’s static distance bin from its nearest zone in 5 km increments, from $b = 1$ (0–5 km) through $b = 9$ (40–45 km). The omitted category is $b = 10$ (45–50 km). The sample for each cohort regression includes treated cells (those first entering the 0–5 km ring in year c or $c + 1$) together with all cells that are never within 5 km of any operational zone during the study period. These never-treated cells are distributed across distance bins 2 through 10 based on proximity to the nearest zone, so the post-treatment indicator $\mathbb{I}(t \geq c)$ takes value one for all cells after the cohort year. Each coefficient δ_b^c therefore measures the change in fire activity in bin b after zone establishment, relative to the change in the 45–50 km reference bin. Estimating equation (1) separately by cohort, with treatment timing aligned within each cohort, addresses the cohort-pooling problems that arise in standard two-way fixed-effects estimators under staggered adoption with heterogeneous treatment effects (de Chaisemartin & d’Haultfoeuille, 2020; Goodman-Bacon, 2021; Borusyak et al., 2024).

To obtain an overall estimate, I aggregate the cohort-specific coefficients using a weighted av-

erage:

$$\bar{\delta}_b = \frac{\sum_c n_c \cdot \hat{\delta}_b^c}{\sum_c n_c} \quad (2)$$

where n_c is the number of treated cells in cohort c . I report two sets of standard errors. The first clusters at the province level within each cohort regression and treats the cohort estimates as independent. Because the cohort-specific regressions share a common pool of never-treated cells, this approach understates uncertainty. I therefore also report standard errors from a stratified cluster bootstrap that resamples provinces with replacement across all cohort regressions jointly, preserving both within-province correlation and the cross-cohort dependence induced by the shared comparison group.⁹

4.2 Event-Study Specification

Let $\bar{d}$ denote the distance threshold within which the distance-based effect is concentrated, identified from the analysis above. I define treatment as cell i being within $\bar{d}$ of an operational zone, with T_i denoting the year prior to the first recorded year of operational status. The pre-operation timing definition captures site-preparation activity (land clearing, construction, infrastructure installation) that typically precedes the official operational date by approximately one year. I restrict the sample to never-treated cells and cells that first become treated during the analysis window, excluding always-treated cells (those within $\bar{d}$ before the panel begins), and estimate the event-study:

$$y_{ipt} = \sum_{\ell=-5, \ell \neq -1}^{L=9} \beta_\ell \cdot \mathbb{I}(t - T_i = \ell) + \alpha_i + \mu_t + \varepsilon_{ipt} \quad (3)$$

where $\mathbb{I}(t - T_i = \ell)$ is an indicator for event time ℓ relative to first treatment and all other terms are defined as in equation (1). The omitted period is $\ell = -1$, so coefficients β_ℓ measure the change in outcome relative to the year before treatment. I estimate equation (3) by restricting the comparison group to cells never within $\bar{d}$ of an operational zone during the study period, and report province-clustered standard errors.

Identification assumptions. For the coefficients in equations (1) and (3) to be interpreted as causal effects of industrial zone establishment, the following assumptions must hold.

First, the *parallel trends* assumption requires that, conditional on the fixed effects, Type-2 thermal anomaly activity in cells near operational zones would have evolved similarly to that in cells farther from any zone in the absence of treatment. For the distance-based specification, this requires parallel evolution within each cohort between cells in any treated bin and cells in the 45–50 km reference bin. For the dynamic specification, it requires parallel evolution between treated and never-treated cells. Although not directly testable, the assumption can be partially assessed

⁹In each of 200 replications, I resample provinces with replacement within strata defined by cohort, re-estimate equation (1) separately by cohort, and recompute $\bar{\delta}_b$. Cohorts 2011, 2013, and 2015 are pooled with cohort 2009 into a single resampling stratum because of small cell counts. The bootstrap standard error is the standard deviation of $\bar{\delta}_b$ across replications.

through pre-treatment coefficients. The pre-treatment estimates in the dynamic specification (Figure 3, Panel B) are small in magnitude and statistically indistinguishable from zero. To assess sensitivity to bounded violations, I follow Rambachan and Roth (2023) and construct confidence sets under a smoothness restriction by bounding how much the differential trend’s slope can change between consecutive periods.

Second, the *no-anticipation* assumption requires that outcomes in cells approaching treatment are not affected by the future arrival of a zone. The relevant concern is combustion associated with site preparation: land clearing, construction, and access road development could generate detectable thermal anomalies before the zone becomes operational. Treatment is defined to activate in the year prior to recorded operational status, which absorbs site-preparation activity occurring in the immediate pre-operation year. The identifying no-anticipation assumption therefore requires that site preparation does not materially begin two years prior to official operation, preserving the omitted baseline ($\ell = -1$) as a clean representation of the untreated state. I evaluate this by testing for earlier anticipation across event times $\ell \leq -2$. Pre-treatment coefficients in this range are small in magnitude and statistically indistinguishable from zero, consistent with a clean baseline and no detectable anticipation beyond the year already absorbed into the treatment definition (Figure 3, Panel B).

Third, the *stable unit treatment value assumption* (SUTVA) requires that treatment of one cell does not affect outcomes in other cells. With treatment defined as being within 5 km of an operational zone, the concern is that industrial activity in the 0–5 km ring could induce or displace combustion activity in nearby comparison cells, contaminating the comparison group. The distance-based estimates in Figure 3, Panel A provide a first check. The effect is concentrated in the 0–5 km bin, and the coefficients for the bins beyond 10 km are small and indistinguishable from zero. This comparison, however, only detects displacement that differs across bins. If the treated ring drew activity evenly from all comparison bins, every bin (including the reference bin) would fall by the same amount, and the pattern across bins would look no different. This scenario is unlikely to matter for two reasons. First, even if the entire increase in the treated ring had come from the comparison cells, the estimate would overstate the effect by about a tenth, because the treated ring holds one cell for every 22 comparison cells.¹⁰ Second, the manufacturing firms that appear in newly treated districts are almost all new to the survey rather than relocated from other districts (Appendix Table A5).¹¹ However, relocation from beyond 50 km of a zone cannot be detected

¹⁰Under uniform displacement the estimate equals the true effect multiplied by $1 + (N_T/N_C)(\bar{y}_T/\bar{y}_C)$, where N denotes the number of treated and comparison cells and $\bar{y}$ their mean pre-treatment fire counts. In the estimation sample $N_T/N_C = 312/6,819$ and $\bar{y}_T/\bar{y}_C = 0.059/0.023$. The proportional estimate would therefore be overstated by about 12%.

¹¹The enterprise tax code in the VES is constant across rounds, which allows me to track firms across districts from 2001 to 2018. Of the manufacturing firms arriving in newly treated districts after zone establishment, 9% of pollution-intensive and 5% of light manufacturing firms had been observed in another district before, and fewer than 2% came from comparison districts within 50 km. The VES enumerates all state and foreign-invested enterprises and all private enterprises with 10 or more workers in every round, while smaller private enterprises are covered on a sample basis in some rounds. Among firms with 10 or more workers, for which absence from earlier rounds means that the enterprise did not yet exist, the shares are 14% and 7%, with fewer than 3% from comparison districts within 50 km, no higher than among arrivals in districts never near a zone. These shares are lower bounds, because a firm that closes in one district and re-registers under a new tax code in another is counted as a new firm rather than a mover.

with this design, and I return to it in Section 7. Finally, the 5–10 km bin shows a small positive coefficient, suggesting that some combustion activity extends just beyond the treatment ring. As a robustness check, I estimate a donut specification that excludes this bin from the sample, and the results are similar (Appendix Table A1, Column 2).

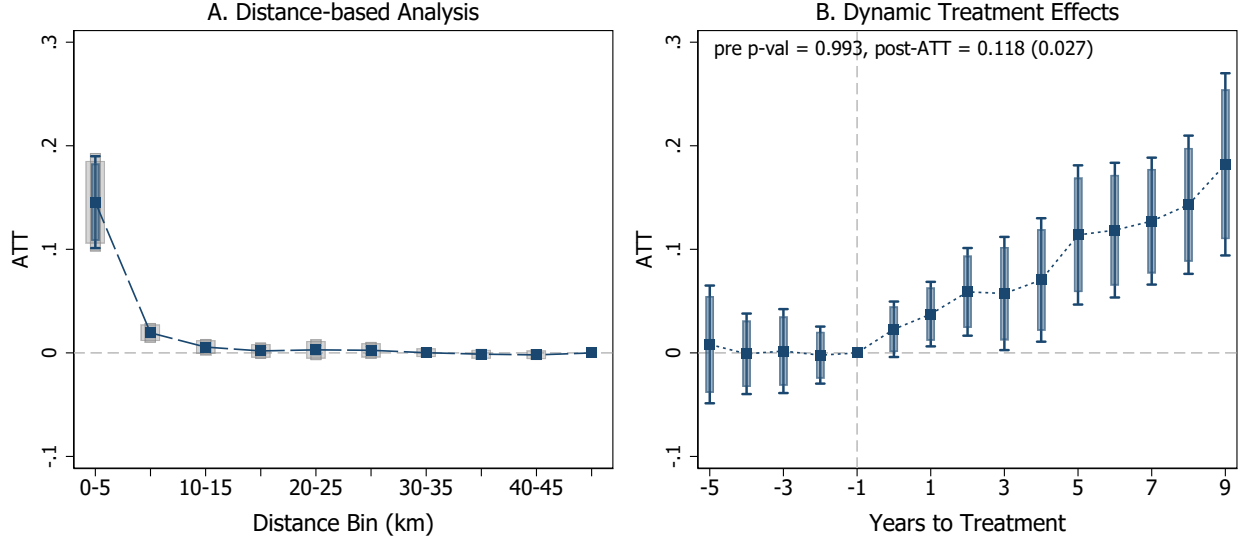
Fourth, under staggered treatment with heterogeneous effects across cohorts, the TWFE estimator can place negative weights on some cohort-specific treatment effects, potentially biasing the average (de Chaisemartin & d’Haultfoeuille, 2020; Goodman-Bacon, 2021; Roth et al., 2023; Borusyak et al., 2024). While the cohort-by-cohort distance specification addresses this issue by construction, the event-study in equation (3) pools all cohorts in a single TWFE regression and is vulnerable to the standard concern. Appendix Figure A1 compares the TWFE event-study path with the heterogeneity-robust estimators of de Chaisemartin and d’Haultfoeuille (2024) and Callaway and Sant’Anna (2021). All three produce nearly identical event-time paths: a flat pre-period, an immediate effect at the year of treatment, and gradually increasing effects over the post-treatment horizon. The post-treatment averages are 0.118 (s.e. 0.027) for TWFE, 0.094 (s.e. 0.022) for de Chaisemartin and d’Haultfoeuille (2024), and 0.096 (s.e. 0.023) for Callaway and Sant’Anna (2021). The pooled estimate is therefore not an artifact of the TWFE weighting problem.

4.3 Results

Figure 3 presents the two sets of estimates, where the outcome is $\ln(1 + \text{Type-2 anomaly})$. The left panel reports the weighted-average distance-bin coefficients $\bar{\delta}_b$ from equation (1). Type-2 thermal anomaly counts in cells within 5 km of an operational zone are approximately 0.14 log points higher than in the 45–50 km reference cells following zone establishment. The effect attenuates sharply with distance. The coefficient for the 5–10 km bin is approximately 0.02 log points ($p < 0.05$ with both analytical province-clustered and bootstrap standard errors), an order of magnitude smaller. All remaining bins between 10 and 45 km are small, precisely estimated, and indistinguishable from zero, with an average coefficient of 0.001 (bootstrap s.e. 0.003) across the seven bins. This pattern is consistent with industrial combustion as a spatially concentrated, point-source phenomenon and differs markedly from the gradual decay observed for dispersed pollutants such as PM2.5 (Section 5).

The right panel reports the dynamic effects $\hat{\beta}_\ell$ from estimating equation (3), using the 0–5 km binary treatment indicator. Pre-treatment coefficients are small and statistically indistinguishable from zero ($p = 0.99$ for the joint test), supporting the parallel trends and no-anticipation assumptions. Following zone establishment, Type-2 thermal anomaly counts increase progressively rather than shifting discretely: the effect is close to zero in the year of operation, reaches approximately 0.1 log points by the fifth year, and 0.17 log points by year nine. The average post-treatment effect, computed as a weighted average of the event-time coefficients, with weights proportional to the number of treated cells contributing to each event time, is 0.12 log points ($p < 0.01$). This trajectory is consistent with industrial zones filling in with tenants over time, so that combustion activity

Figure 3: Industrial Zones and Type-2 Thermal Anomalies: Linear Estimates



Notes: The left panel plots the weighted-average distance-bin coefficients $\bar{\delta}_b$ from equation (1), with 0–5 km through 40–45 km bins and 45–50 km as the reference. The right panel plots the dynamic treatment effects $\hat{\beta}_\ell$ using a binary indicator for whether a cell is within 5 km of an operational industrial zone, with event time ℓ relative to first treatment. Outcome is $\ln(1 + \text{Type-2 anomaly})$ at the cell-year level. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from analytical province-clustered standard errors. Light shaded bands in the left panel are 90 and 95% confidence intervals from a stratified cluster bootstrap with 200 replications. *Source:* Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

within the 0–5 km ring accumulates as the zone matures.¹²

Interpretation and a count-data alternative. The estimates above use $\ln(1 + \text{Type-2 anomaly})$ as the outcome. J. Chen and Roth (2024) show that this transformation does not deliver a percentage-change interpretation when the outcome contains many zeros: the estimated coefficient depends on the units of the outcome and on the constant added inside the log, which can both be problematic when zeros are common. Indeed, the Type-2 detection data have substantial mass at zero, with the share of zero cell-years rising from 84% in the 0–5 km ring to 97% in the 5–10 km ring and 99% or more in all bins beyond 10 km. The coefficients in Figure 3 should therefore be read as evidence that detections are higher near operational zones, not as estimates of a percentage increase.

To obtain estimates with a proper percentage-change interpretation, I re-estimate the dynamic specification (equation 3) using Poisson pseudo-maximum likelihood, which is unit-invariant, accommodates zeros without an ad hoc transformation, and remains consistent under arbitrary distributional misspecification provided the conditional mean is correctly specified (Silva & Tenreiro, 2006; J. Chen & Roth, 2024). The Poisson specification mirrors equation (3) with the same set of fixed effects, but models the conditional mean of fire counts directly as $E[y_{ipt} | \cdot] = \exp(\cdot)$ rather

¹²The effects remain stable when time fixed effects are interacted with district-level baseline characteristics, allowing cells in districts with different pre-treatment compositions to follow flexible trends (Appendix Figure A2).

than log-transforming the outcome.¹³

Table 3 reports estimates of the average effect (post-ATT) of industrial zones on Type-2 thermal anomalies along three dimensions of combustion activity. The post-ATT is computed as the weighted average of post-treatment event-time coefficients (marginal effects), with weights proportional to the number of treated cell-years contributing to each event-time estimate. Column (1) reports a logit model on $\mathbb{I}(\text{TAP})$, an indicator for any Type-2 detection within the cell-year. Columns (2)–(3) use the count of detections, and columns (4)–(5) use the number of distinct days with any detection. All specifications include cell and year fixed effects; columns (3) and (5) add linear interactions of year fixed effects with district baseline characteristics measured in 2002 from the VHLSS.

Table 3: Industrial Zones and Type-2 Thermal Anomalies: Poisson Estimates

	$\mathbb{I}(\text{TAP})$	Type-2 fires (count)		Type-2 fires (days)	
	(1)	(2)	(3)	(4)	(5)
Post-ATT	1.689 (0.332)	1.040 (0.428)	0.837 (0.265)	0.870 (0.347)	0.766 (0.259)
Post-ATT (marginal effect)	0.395 (0.073)	0.566 (0.233)	0.456 (0.144)	0.422 (0.169)	0.372 (0.126)
Pre-treatment mean	0.106	0.233	0.233	0.229	0.229
pre-trend p-value	0.514	0.701	0.820	0.812	0.991
Observations	11,286	11,362	11,362	11,362	11,362
Cell FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Year FE $\times$ Baseline Controls			Yes		Yes
Model	Logit	Poisson	Poisson	Poisson	Poisson

Notes: The table reports the effects of industrial zone establishment on Type-2 thermal anomalies, estimated on a sample of cells within 5 km of an operational industrial zone (treated) and cells never within 5 km of any operational zone during the panel (controls); always-treated cells are excluded. Column (1) is a logit regression where the outcome is an indicator equal to one if any Type-2 detection occurs in the cell-year. Columns (2)–(5) are Poisson PML regressions; columns (2)–(3) use the count of Type-2 detections, and columns (4)–(5) use the number of distinct days with any Type-2 detection. The treatment in all columns is a binary indicator for whether the cell is within 5 km of an operational industrial zone. The post-ATT is the weighted average of post-treatment event-time coefficients (event times 0 through 9), with weights proportional to the number of treated cell-years contributing to each event-time estimate. Marginal effects in column (1) are on the probability scale; marginal effects in columns (2)–(5) are on the count or days scale. The pre-trend p-value is from a joint test that all pre-period coefficients (event times -5 through -2) equal zero. Standard errors are clustered at the province level for columns (2)–(5); column (1) reports unclustered standard errors from `xtlogit, fe`. *Source:* Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

Column (1) shows that the probability of any Type-2 detection rises by 39.5 percentage points ($p < 0.01$) following zone establishment, off a pre-treatment baseline of 10.6%. The preferred Poisson specifications yield post-ATT coefficients of 1.04 for detection counts (column 2) and 0.87 for detection-days (column 4), both significant at the 1% level. Exponentiating these coefficients, zone establishment multiplies the conditional mean of detections by approximately 2.8 and the condi-

¹³Estimated using the `ppmlhdfc` Stata package (Correia et al., 2020), which handles high-dimensional fixed effects and drops cells whose outcome is always zero throughout the panel, as their fixed effect is not identified.

tional mean of detection-days by approximately 2.4. In level terms, the corresponding marginal effects imply increases of 0.57 detections and 0.42 detection-days per cell-year, against pre-treatment means of 0.23 for both measures. The similarity of the count and days marginal effects indicates that the response is concentrated on the extensive margin: zone establishment primarily raises the probability that a cell registers any combustion on a given day, rather than generating multiple detections on already-active days.¹⁴ Pre-trend tests yield p-values between 0.51 and 0.99 across specifications, providing no evidence against parallel trends (Appendix Figure A3).

4.4 Robustness

Columns (3) and (5) of Table 3 add interactions between year fixed effects and district baseline characteristics (urban population share, minority population share, secondary education share, and farming share), allowing cells in districts with different pre-treatment compositions to follow flexible time paths. The estimated effects are modestly smaller but the conclusions remain unchanged. I examine further robustness in Appendix Table A1 and Appendix Figures A4–A5. Excluding the 5–10 km ring from the comparison group to address potential spillover into immediately adjacent cells yields a slightly larger estimate, consistent with removing mildly contaminated controls. Replacing the pre-operation timing convention with treatment activating in the year of recorded operational status produces a somewhat smaller estimate, in line with detectable combustion beginning before the official operational date. The effect remains positive and significant when the treatment ring expands to 10 km. Pre-trend tests do not reject parallel trends in any of these specifications.

The event study dates treatment by the first zone within 5 km, but zones cluster in space: 14% of treated cell-years lie within 5 km of two or more operating zones. Appendix Table A6 shows that the estimates do not depend on the other zones around a cell, whether additional zones within the 5 km ring or zones farther away. The effect is similar when treated cells are restricted to those with a single zone within 5 km or comparison cells to those with no zone within 10 or 25 km, and a second zone within 5 km adds nothing detectable to the first. Weighting all operating zones within 25 km by inverse distance (the aggregation used for the PM2.5 exposure measure in Section 5) gives a small and imprecise coefficient, as expected when the outcome is combustion detected at the source and the effect is confined to the first 5 to 10 km.

The result could also reflect a single province. Dropping one province at a time leaves the estimates tightly clustered around the baseline, with every 95% confidence interval positive (Appendix Figure A4). Parallel trends could fail after treatment in ways the pre-period does not reveal. Under the smoothness restriction $\Delta^{SD}(M)$ of Rambachan and Roth (2023), the post-treatment effect remains significantly positive at the 10% level for deviations from linear pre-trends up to $M \approx 0.005$ (Appendix Figure A5).

The analysis so far establishes that industrial zone establishment generates a localized increase

¹⁴The ratio of the count marginal effect to the days marginal effect, $0.57/0.42 \approx 1.3$, gives the average number of detections per newly active day. A ratio close to one indicates that additional combustion takes the form of new days with a single detection rather than multiple detections on the same day.

in industrial combustion within 5 km of the zone centroids. This result serves two purposes in the analysis that follows. First, it identifies zones as combustion sources and dates their activation. The 5 km radius encompasses zone-registered firms and the supporting activity that clusters near the zone boundary, and the PM2.5 analysis in Section 5 treats this ring as the source area from which emissions are transported downwind. Second, it measures how combustion at the source evolves in aggregate: it accumulates as zones fill. Section 6.1 examines whether it also differs across establishment cohorts and sets these dynamics against their counterparts in PM2.5.

The thermal data do not, however, measure how much fuel an individual zone burns, because the MODIS algorithm registers a fire only when its radiant signal stands out from the surrounding pixels on a cloud-free overpass. At the 1 km resolution of the sensor, this requires a source of the size and temperature of a gas flare, a furnace, or a kiln (Giglio et al., 2016). Most fuel burned in a zone does not meet this condition. The hot-process industries that such sources belong to, non-metallic minerals, basic metals, and coke and refined petroleum, account for 9% of the manufacturing firms inside zones and 3% of their employment in 2016 (Appendix Table D1, Panel A); the remaining firms, in textiles, food, paper, electronics, and assembly, burn coal and oil for steam and process heat at temperatures well below the detection threshold. As a result, a zone with no detections is not necessarily a zone without combustion, and the number of detections at a zone is better read as indicating the presence of high-temperature sources than as measuring the quantity of fuel burned. The exposure measure in Section 5 accordingly counts every operating zone equally, and Appendix E shows that the per-exposure effect is unrelated to the thermal activity observed at the source. The welfare-relevant question is whether the combustion that zones generate translates into ambient pollution exposure for populations living beyond the immediate source area.

5 EMPIRICAL ANALYSIS: AMBIENT AIR POLLUTION

Particulate matter, unlike thermal anomalies confined to the source pixel, disperses with prevailing winds and accumulates at some distance from where it is emitted. Industrial activity contributes to ambient PM2.5 through both direct emissions of primary particles and emissions of gas-phase precursors that form secondary aerosol over hours to days as the plume travels downwind (Hallquist et al., 2009; Huang et al., 2014). These mechanisms generate documented PM2.5 spillovers across district, regional, and national boundaries (e.g., Jia & Ku, 2019). The PM2.5 signal recorded in a given cell therefore reflects emissions from sources that may lie kilometers away in the upwind direction, and distance from an industrial zone is, by itself, a weak proxy for exposure to zone-related pollution. Wind direction breaks the spatial symmetry: if a zone emits meaningful quantities of fine particulates or precursor gases that transform into fine particulates downwind, cells lying downwind would record higher PM2.5 than cells lying upwind of the same zone. A growing literature exploits this asymmetry to identify the effects of spatially fixed pollution sources, including aggregate regional emissions (Deryugina et al., 2019) and highways (An-

derson, 2020), on a range of outcomes. I adopt this approach in the industrial-zone setting by estimating the average effect on cell-level PM2.5 of downwind exposure to nearby zones, thereby recovering the magnitude of pollution attributable to industrial-zone activity.

To do so, I construct a cell-zone-month downwind measure from ERA5-Land reanalysis wind fields (Copernicus Climate Change Service (C3S), 2019; Muñoz-Sabater et al., 2021), using the 10-meter eastward and northward wind components at 0.1 degree resolution. For each calendar day I retain a single snapshot at 06:00 UTC, corresponding to 13:00 local time in Vietnam, to capture wind conditions during the daytime boundary layer mixing regime when industrial emissions and plume dispersion are most active. For each cell-zone pair within 25 km, I compare the direction from the zone to the cell with the direction in which the wind is blowing at the zone on each day. A day is classified as downwind if the two directions are within 45 degrees of each other, and the monthly downwind share for a cell-zone pair is the share of days in the month so classified. Figure 4 illustrates the classification for a single zone.¹⁵ A cell’s exposure measure aggregates this classification across all operational zones within 25 km, weighting each zone’s monthly downwind share by inverse distance. Cells within 5 km of the nearest zone are excluded from the estimation sample because direct emissions dominate exposure regardless of wind direction. This source area, as documented by the Type-2 thermal anomaly results in Section 4, captures emissions from both within-zone manufacturing and proximate economic activity induced by zone establishment.

5.1 Empirical Strategy

To examine the effect of industrial zones on PM2.5, I estimate the following equation:

$$\ln \text{PM2.5}_{ipmt} = \beta E_{imt} + \alpha_i + \mu_{mt} + \varepsilon_{ipmt} \quad (4)$$

where $\ln \text{PM2.5}_{imt}$ is the log of monthly mean PM2.5 concentration in cell i during month m of year t , and

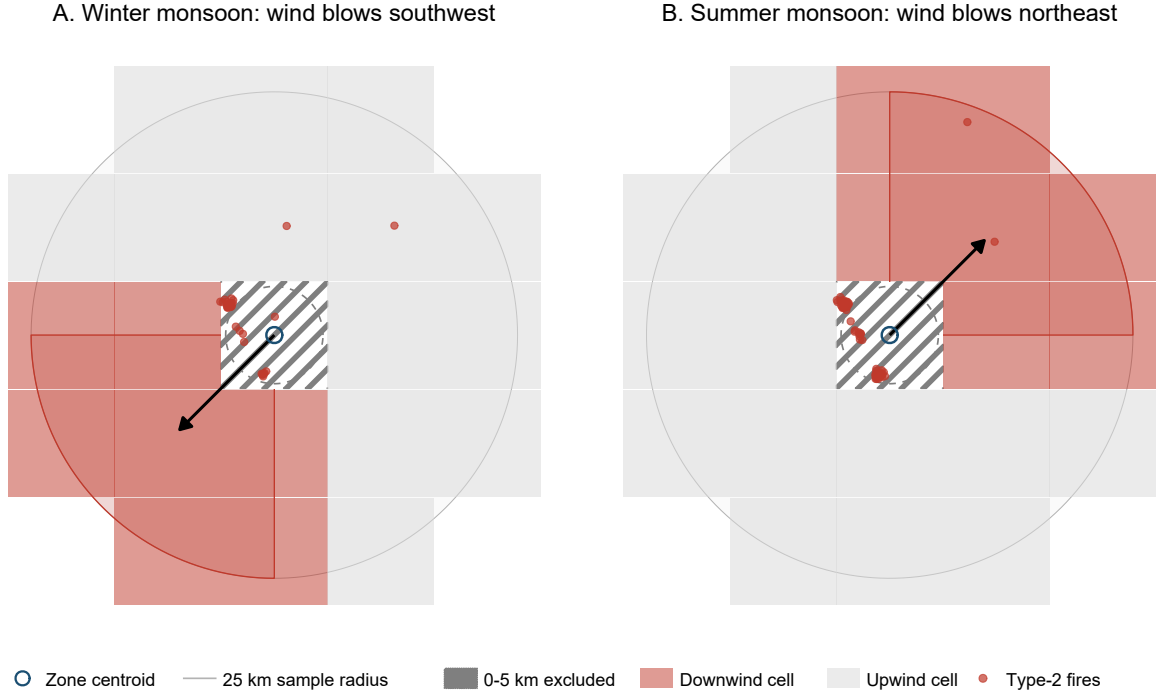
$$E_{imt} = \sum_{j \in \mathcal{J}_i} \mathbb{I}(t \geq T_j) \cdot \text{distance}_{ij}^{-1} \cdot D_{ijmt} \quad (5)$$

is the cell’s downwind exposure measure. The set $\mathcal{J}_i$ comprises industrial zones within 25 km of cell i . For each zone j , T_j is the year prior to its recorded operational status, defined as in equation (3) to absorb site-preparation activity that precedes the official operational date. The variable distance_{ij} is the distance in kilometers from cell i to zone j , and D_{ijmt} is the monthly downwind share for the cell-zone pair defined in Figure 4. The terms α_i and μ_{mt} denote cell and year-month fixed effects, respectively.

The key difference from the fires specification is that the PM2.5 analysis uses monthly data. Year-month fixed effects therefore absorb seasonal pollution patterns and aggregate shocks com-

¹⁵ Although wind direction is given at 0.1° resolution (one value per ERA5 cell), the analysis is conducted on a 0.05° cell grid, so the four 0.05° cells inside a single 0.1° wind cell can receive different classifications when the wedge edge from the zone passes through them. Appendix Figure A6 illustrates this construction at the boundary between the two resolutions.

Figure 4: Wind-based Classification of Cell Exposure to an Industrial Zone



Notes: The figure illustrates the wind-based classification of grid cells around an industrial zone, using a representative zone in the Mekong River Delta as a concrete example. The zone centroid is shown at the center of each panel. The outer circle marks the 25 km sample radius around the zone. The dashed inner circle at 5 km marks the source area, which is excluded from the analysis sample because direct emissions dominate exposure regardless of wind direction. Cells in the 5-to-25 km transport ring are shaded red if the bearing from the zone to the cell falls within $\pm 45^\circ$ of the wind direction at the zone (downwind) and grey otherwise. Cells in the 0-to-5 km source area are shown with diagonal hatching to indicate exclusion. The shaded red wedge depicts this 90° sector at the analysis boundary. Red dots show MODIS Collection 6.1 Type-2 thermal anomaly detections within 25 km of the zone during the illustrated season. Panel A shows the winter monsoon (December to February), when the northeast monsoon carries emissions toward the southwest. Panel B shows the summer monsoon (June to August), when the monsoon reverses. The same cells are classified as downwind in one season and upwind in the other, which is the identifying variation exploited by the wind-based exposure measure. The empirical measure uses daily ERA5-Land wind fields aggregated to the monthly cell-zone downwind share. *Source:* Fire detection data from NASA FIRMS (2021), wind data from Muñoz-Sabater et al. (2021), and industrial zone locations from the Ministry of Planning and Investment database.

mon to all cells. Cell fixed effects absorb, among other time-invariant characteristics, each cell's long-run prevailing wind climatology. Identification comes from within-cell deviations in monthly wind direction rather than from persistent differences in wind exposure across cells. Standard errors are clustered at the province level p .

The coefficient β captures the percentage change in PM2.5 associated with a one-unit increase in downwind exposure, identified from within-cell variation in E_{imt} that remains after absorbing the fixed effects. This variation has two sources: the opening of new zones within 25 km of a cell, which raises the cell's exposure once a zone becomes operational; and the seasonal reversal of monsoon winds, which alternates the position of a given cell between the downwind and upwind sectors of nearby zones across months (Figure 4). The coefficient therefore captures the

reduced-form effect of industrial zone establishment on downwind air quality, operating through all emission channels that the policy activates within the source area defined in Section 4. Emissions from supporting economic activity induced by zone establishment are part of the causal effect of the policy. The event-study evidence below supports this interpretation: the downwind PM2.5 effect is absent before any nearby zone becomes operational and emerges only afterward (Figure 5, Panel A), indicating that the source area’s emissions were triggered by zone establishment rather than pre-existing.

Identification assumptions. For β to be interpreted as the causal effect of downwind exposure on PM2.5, the wind-direction component of E_{imt} must be exogenous to unobserved determinants of PM2.5, conditional on cell and year-month fixed effects.¹⁶ After absorbing cell and year-month fixed effects, the residual variation in D_{ijmt} reflects short-run weather rather than climatological patterns or regional seasonal cycles. This residual variation is plausibly orthogonal to the determinants of zone siting and operational timing, which follow a multi-year central-government approval process initiated years before zones become operational. Consistent with this, Appendix Table A2 reports that across districts within the same province and distance band in the analysis sample, those with higher average downwind exposure do not systematically differ on pre-treatment characteristics (urban share, ethnic minority share, secondary education share, and farm labor share from the 2002 VHLSS), cell elevation, the cell’s share of its province’s 2000 population, or pre-treatment mean log PM2.5. As a further check, I estimate equation (4) replacing log PM2.5 with the cell’s share of provincial population as the outcome. The coefficient on downwind exposure is small and statistically insignificant ($\hat{\beta} = 0.0001, p = 0.34$), indicating that the wind-based measure does not predict demographic changes in the transport ring. I present additional validation tests alongside the main results in the next subsection.

5.2 Results

Table 4 reports the pooled estimates. Column (2) is the preferred specification, estimated on cells in the 5–25 km transport ring around the nearest zone with cell fixed effects, year-month fixed effects, and time-varying weather controls. A one-standard-deviation increase in inverse-distance-weighted downwind exposure raises monthly PM2.5 by 2.7%, equivalent to $0.66 \mu\text{g}/\text{m}^3$ at the unweighted sample mean of $24.4 \mu\text{g}/\text{m}^3$. The standardizing unit, 0.078 km^{-1} , corresponds to a cell fully downwind of a single operational zone at approximately 13 km for the entire month.

The timing of the effect matches zone establishment. Panel A of Figure 5 reports the effect by event time: a positive effect appears once a zone becomes operational and persists across all post-treatment windows, estimated more precisely at longer horizons (blue markers). Two placebo tests show that the effect appears only under the design’s own variation: the actual timing of operations

¹⁶The exposure measure has the structure of a shift-share regressor, where zone operational status $\mathbb{I}(t \geq T_j)$ provides the shifts and the inverse-distance-weighted monthly downwind share $\text{distance}_{ij}^{-1} \cdot D_{ijmt}$ provides the shares. Recent work on shift-share identification establishes that consistent estimation requires exogeneity of only one component (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Here the share component carries the identification.

Table 4: Wind-based Estimates of Industrial Zone Exposure on PM2.5

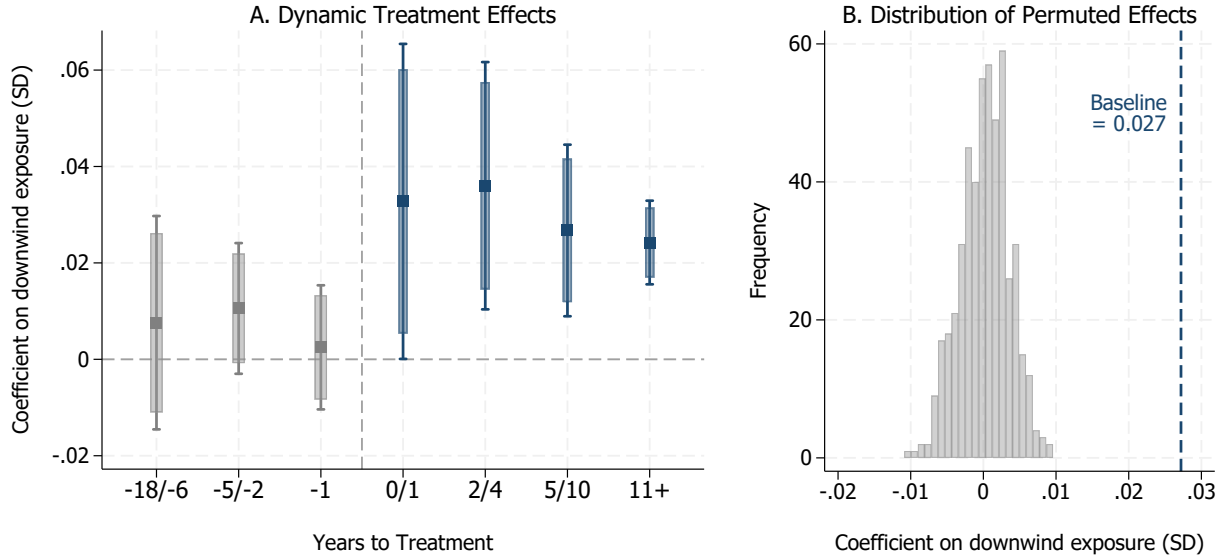
	Dependent Variable is $\ln(\text{PM2.5})$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Downwind Exposure (SD)	0.025 (0.005)	0.027 (0.005)	0.025 (0.005)	0.029 (0.006)	0.021 (0.006)	0.032 (0.007)	0.024 (0.004)	0.030 (0.006)
Implied effect ($\mu\text{g}/\text{m}^3$)	0.61	0.66	0.61	0.71	0.52	0.77	0.59	0.73
Sample SD of exposure	0.078	0.078	0.078	1.152	0.151	0.097	0.054	0.100
Unstandardized coefficient	0.322	0.348	0.322		0.139	0.331	0.445	0.300
Observations	793,212	793,212	793,212	793,212	909,036	1,317,156	793,212	793,212
Adjusted R-squared	0.722	0.739	0.770	0.740	0.742	0.726	0.739	0.739
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month FE $\times$ Baseline Controls			Yes					
Sample	5-25km	5-25km	5-25km	5-25km	0-25km	5-40km	5-25km	5-25km
Wind Aggregate	IDW	IDW	IDW	EW	IDW	IDW	IDW	IDW
Wind Angle $\pm^\circ$	45	45	45	45	45	45	30	60

Notes: The table reports estimates from equation (4). The dependent variable is the log of monthly mean PM2.5 concentration ($\mu\text{g}/\text{m}^3$) in a 0.05 degree cell. The regressor is the cell's monthly downwind exposure to operational industrial zones, scaled by its standard deviation within each column's estimation sample so that coefficients are interpretable as the effect of a one-standard-deviation increase in exposure. Columns (1)–(3) and (5)–(8) construct exposure as the inverse-distance-weighted sum of the monthly downwind shares across operational zones within the relevant buffer (IDW) $E_{imt} = \sum_{j \in \mathcal{J}_i} \mathbb{I}(t \geq T_j) \cdot \text{distance}_{ij}^{-1} \cdot D_{ijmt}$; column (4) replaces the inverse-distance weights with equal weights (EW) $E_{imt} = \sum_{j \in \mathcal{J}_i} \mathbb{I}(t \geq T_j) \cdot D_{ijmt}$. The downwind share for a cell-zone pair is the share of days in the month on which the wind at the zone blew within a specified angular tolerance of the bearing from the zone to the cell, computed from ERA5-Land 06:00 UTC 10-meter wind fields (Copernicus Climate Change Service (C3S), 2019; Muñoz-Sabater et al., 2021). The baseline tolerance is $\pm 45^\circ$ (columns 1–6); columns (7) and (8) use $\pm 30^\circ$ and $\pm 60^\circ$ to assess sensitivity to this definition. Zones are treated as operational from one year before their recorded operational date, consistent with the treatment timing in equation (3). The row “Implied effect ($\mu\text{g}/\text{m}^3$)” reports the level change in PM2.5 from a one-standard-deviation increase in exposure, computed as the coefficient times the sample-mean PM2.5; the row “Sample SD of exposure” reports the within-sample standard deviation of the unstandardized exposure. The row “Unstandardized coefficient” reports the coefficient on the raw (unscaled) exposure measure in each column. For columns (1)–(3) and (5)–(8), the exposure measure is in the same units (inverse-distance-weighted downwind share, km^{-1}), so unstandardized coefficients are directly comparable across these columns. Column (4) uses an equal-weighted (EW) exposure measure with different units; the unstandardized coefficient for column (4) is therefore omitted. All columns except Column (1) include cell fixed effects, year-by-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and a linear term of monthly total precipitation, both from ERA5-Land). Column (3) additionally interacts year-by-month fixed effects with four baseline district characteristics measured in 2002 from the VHLSS: urban population share, ethnic minority share, secondary education share, and farm-labor share. Column (1) reproduces column (2)'s specification without weather controls. Columns (1)–(4) and (7)–(8) restrict the sample to cells in the 5–25 km transport ring around the nearest zone; column (5) adds the 0–5 km source ring, where direct emissions dominate exposure regardless of wind direction; column (6) extends the buffer to 40 km. Standard errors clustered at the province level are reported in parentheses.

and the actual location of sites.

First, when the exposure measure's wind and proximity geometry is applied before any nearby zone operates, the placebo coefficient is 0.008 (s.e. 0.006): no pre-treatment window shows a significant effect, and the year immediately before operation is estimated most precisely and sits near zero (gray markers). Second, when every zone is relocated to a randomly drawn site in its own

Figure 5: Industrial Zones and PM2.5: Dynamic Effects and Permutation Test



Notes: Panel A reports pre-treatment placebo coefficients (gray markers, left of the dashed line) and post-treatment dynamic effects (blue markers, right). Pre-treatment coefficients are estimated by omitting the operational-status indicator from equation (5) so that the exposure measure reflects only the directional and proximity geometry of zones, independent of when they became operational. This placebo exposure is interacted with three pre-treatment window indicators (≤ -6 , -5 to -2 , and -1 years before the first nearby zone becomes operational) and estimated jointly on the pre-treatment sample. Post-treatment coefficients are estimated from separate regressions, each combining all pre-treatment cell-months of treated cells with post-treatment cell-months in a given event-time window; never-treated cells are included in all regressions. Event time is years since the first zone within 25 km became operational. Panel B plots the distribution of placebo coefficients from a permutation test. In each of 500 permutations, every zone is relocated to a randomly drawn point within a district in the same province, subject to a 25 km exclusion buffer from every actual zone in that province; if a within-province placement is infeasible after 40 attempts, the algorithm falls back to a randomly drawn district in the same broader region. Operational years are reshuffled across placebo zones within each province. For each placebo configuration, I recompute E_{imt} following equation (5) and re-estimate equation (4), scaling each placebo E_{imt} by its own within-sample standard deviation. The vertical dashed line marks the baseline estimate. All specifications include cell fixed effects, year-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and monthly total precipitation from ERA5-Land). Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Source: PM2.5 data from Shen et al. (2024); wind data from ERA5-Land (Muñoz-Sabater et al., 2021); industrial zone locations from the Ministry of Planning and Investment.

province (at least 25 km from the province's actual zones) and the operational years are reshuffled among them, the effect is equally absent: across 500 such permutations, the placebo coefficients center tightly on zero (mean 0.0002, standard deviation 0.0034).^{17,18} The baseline estimate of 0.027

¹⁷The exclusion buffer is enforced only against same-province real zones. Fake zones in dense industrial corridors near province boundaries may fall within 25 km of real zones in neighboring provinces, in which case placebo exposure absorbs some signal from genuine downwind exposure. This residual overlap biases the placebo distribution slightly toward the baseline estimate, making the test more conservative than an idealized null.

¹⁸Several small, densely industrialized provinces cannot accommodate all of their zones under this within-province constraint: Bac Ninh, with 10 real zones in approximately 820 km², is the clearest case, but Ha Nam, Thai Binh, Hai Duong, and Ho Chi Minh City also exhibit insufficient candidate area outside the 25 km exclusion buffer of their own zones. For these provinces, the algorithm falls back to drawing from candidate districts in the same broader region (e.g., the Red River Delta for the northern provinces; the Southeast region for Ho Chi Minh City). Across the 500

lies approximately eight standard deviations above their mean, outside the support of the distribution entirely (Panel B). The notes to Figure 5 detail the construction of both tests.

The baseline estimate summarizes the average response of PM2.5 to a one-standard-deviation shift in downwind exposure, not the burden this implies for the people living in the transport ring: cells differ in how much exposure they receive, and population is unevenly distributed across them. I therefore compute the implied level effect of zone exposure for each cell-month, $PM2.5_{imt} \cdot [1 - \exp(-\hat{\beta} \cdot E_{imt} / \sigma_E)]$, and average it over the 5–25 km ring weighted by year-2000 cell population. The result is an increment of 0.70 $\mu\text{g}/\text{m}^3$ per month, 14% of the World Health Organization’s 2021 annual PM2.5 guideline of 5 $\mu\text{g}/\text{m}^3$. It measures the average monthly PM2.5 increment from zone exposure borne by a typical resident of the ring.

The increment is large relative to the deterioration in air quality over the study period and uneven across cells. Population-weighted PM2.5 in the transport ring rose from 22.0 $\mu\text{g}/\text{m}^3$ in 2000–2002 to 27.9 $\mu\text{g}/\text{m}^3$ in 2017–2019, an increase of 5.9 $\mu\text{g}/\text{m}^3$ that took the average resident from below to above Vietnam’s national ambient standard of 25 $\mu\text{g}/\text{m}^3$ (QCVN 05:2023/BTNMT). The zone-attributable increment accounts for approximately 12% of this rise. Across cells, the effect is considerably larger for the heavily exposed: at the 95th percentile of cell-month exposure (2.1 standard deviations), the implied increment is 1.39 $\mu\text{g}/\text{m}^3$, 28% of the WHO annual guideline, and at the 99th percentile (5.1 standard deviations) it reaches 3.34 $\mu\text{g}/\text{m}^3$, or 67% of the guideline. The most exposed cells are those located close to and directly downwind of multiple operational zones simultaneously.

5.3 Robustness

The remaining columns of Table 4 examine sensitivity to alternative specifications and samples. Allowing flexible differential trends by interacting year-month fixed effects with 2002 district-level baseline characteristics leaves the estimate unchanged (column 3). Replacing the inverse-distance-weighted exposure with an equal-weighted sum of monthly downwind shares yields a standardized coefficient of similar magnitude (column 4). Because the two measures have different units, only the standardized comparison is informative. Including cells in the 0–5 km source ring attenuates the per-unit effect of downwind exposure, with the unstandardized coefficient declining from 0.348 to 0.139, consistent with direct emissions dominating exposure within the source area regardless of wind direction. The standardized coefficient is more stable because the within-sample standard deviation of exposure approximately doubles when source cells are included (column 5). Extending the buffer to 5–40 km leaves the per-unit effect essentially unchanged (unstandardized coefficient 0.331 versus 0.348 in the baseline), with the modest increase in the standardized coefficient reflecting the wider dispersion of exposure rather than a strengthening of the underlying relationship (column 6). Narrowing the angular tolerance to $\pm 30^\circ$ or widening it to $\pm 60^\circ$ produces standardized coefficients close to the baseline, although the per-unit effect varies modestly with

permutations, on average 81 of 287 zones (28%) are placed through this regional fallback, with the remaining 206 zones (72%) placed strictly within province.

the tolerance (columns 7–8).

Appendix Figure A7 reports leave-one-province-out estimates, which cluster tightly around the baseline with all 95% confidence intervals excluding zero. In the baseline specification, the year-month fixed effects absorb national shocks, and regional pollution trends could remain. Appendix C presents additional results using alternative sets of fixed effects: region-by-year-month and province-by-year-month. These specifications compare a cell with neighbors that share its surroundings, and the comparison excludes the pollution transported from surrounding zones, part of the local externality the baseline measures. The composite coefficient is accordingly smaller but remains positive and statistically significant under both. A decomposition by distance separates the cell’s own exposure from the transported component. The effects of exposure within 50 km are similar across the three sets of fixed effects, and the smaller composite coefficients reflect the excluded transport rather than a weaker effect (Appendix Table C1).

5.4 Heterogeneity in the Pollution Effect

The pooled estimate in Table 4 combines zones observed at different ages and established under different regimes into a single coefficient. This subsection asks how much the effect varies over a zone’s life and how much across establishment cohorts. I first estimate equation (4) separately by event-time window, where event time is years since the first zone within 25 km became operational: each window’s sample pairs the pre-treatment observations of treated cells with their post-treatment observations in that window, and the coefficient is identified from the within-cell transition from zero to positive exposure. The point estimates decline across event time: the standardized coefficient is 0.040 in years 0–4, 0.027 in years 5–10, and 0.024 beyond year 10, and the implied increment per standard deviation of exposure falls from 0.96 to 0.65 and 0.61 $\mu\text{g}/\text{m}^3$ (Table 5, Panel A). Because the longer event-time windows are identified from progressively earlier cohorts, the decline could reflect changes within zones as they age or differences across the cohorts themselves.

Panel B of Table 5 separates changes over a zone’s life from differences across cohorts by estimating the same event-time windows separately by establishment cohort. A cell’s cohort is the year the first zone within 25 km became operational, the same year that defines its event time, grouped into the three regulatory periods of Section 2: before 2003, 2003 to 2009, and after 2009. Every cohort is observed in the 0–4 and 5–10 windows, and the post-2009 cohort is not observed beyond year 10.

The variation lies across cohorts rather than over a zone’s life. Within cohorts, the effect does not decline: the pre-2003 cohort’s coefficient stays between 0.013 and 0.021 across the three windows, and the 2003–2009 cohort’s between 0.042 and 0.049, ranges that are small relative to the standard errors. The pooled decline in Panel A therefore reflects the changing cohort composition of the windows: the 11+ window draws most of its treated cell-months from the pre-2003 cohort, the cohort with the smaller per-exposure effect of the two populating that window, and none from the post-2009 cohort. Across cohorts, the differences are stable: the 2003–2009 cohort’s coefficient

Table 5: Industrial Zones and PM2.5: Dynamic Effects, Overall and by Establishment Cohort

	Dependent Variable is ln(PM2.5)			
	Pooled	Event-time Window		
	(1)	0–4	5–10	11+
	(1)	(2)	(3)	(4)
Panel A: All Zones				
Downwind Exposure (SD)	0.027 (0.005)	0.040 (0.012)	0.027 (0.009)	0.024 (0.004)
		[147,228]	[225,324]	[299,808]
Implied effect ($\mu\text{g}/\text{m}^3$)	0.65	0.96	0.65	0.61
Mean PM2.5 ($\mu\text{g}/\text{m}^3$)		24.0	24.5	25.2
Observations	793,212	268,080	346,176	420,660
Adjusted R-squared	0.739	0.749	0.747	0.739
Panel B: By Establishment Cohort				
Pre-2003 cohort	0.023 (0.004)	0.013 (0.019)	0.017 (0.007)	0.021 (0.004)
	[322,164]	[24,300]	[86,028]	[211,836]
2003–2009 cohort	0.047 (0.013)	0.045 (0.012)	0.042 (0.016)	0.049 (0.013)
	[321,480]	[106,140]	[127,368]	[87,972]
Post-2009 cohort	0.017 (0.021)	-0.007 (0.018)	0.052 (0.031)	
	[28,716]	[16,788]	[11,928]	
Cell FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes
Sample	5-25km	5-25km	5-25km	5-25km

Notes: The table reports estimates from equation (4) for different event-time windows, where event time is years since the first zone within 25 km became operational. Panel A pools all zones. Column (1) reproduces the pooled specification from Table 4, column (2). Columns (2)–(4) restrict the sample to treated cells, combining all their pre-treatment observations with the post-treatment observations falling within the indicated event-time window; post-treatment observations outside the window are excluded. Observation counts across columns (2)–(4) therefore overlap and do not sum to column (1). The row “Implied effect” reports the level change in PM2.5 from a one-standard-deviation increase in exposure, computed as the coefficient times the mean PM2.5 concentration among treated cells in the relevant event-time window. The row “Mean PM2.5” reports this mean in $\mu\text{g}/\text{m}^3$. Treated cell-month observations contributing to the post-treatment variation in each window are reported in brackets, in both panels. Panel B estimates the column specifications separately by establishment cohort, grouping treated cells by the year the first zone within 25 km became operational (before 2003, 2003 to 2009, and after 2009, matching the regulatory periods of Section 2). In Panel B, column (1) combines all post-treatment windows of the cohort. Exposure is standardized on the common estimation sample throughout, and coefficients are comparable across all cells of the table. The estimation sample contains no pre-treatment observations for the pre-2003 cohort, whose zones largely predate the panel; the estimates for this cohort are identified from monthly variation in downwind exposure after treatment. The post-2009 cohort is not observed beyond event year 10. All columns include cell fixed effects, year-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and monthly total precipitation from ERA5-Land). Standard errors clustered at the province level are reported in parentheses.

is at least twice the pre-2003 cohort’s in every window. The post-2009 cohort shows no effect in its

first five years (-0.007 , s.e. 0.018) and an imprecise 0.052 (s.e. 0.031) in years 5 to 10. Its estimates rest on the fewest treated cell-months, and as of 2023 its median occupancy rate is 34% and 63% of its zones remain under construction (Appendix Table A3). Both the small pooled effect and the rise from the first window to the second are at least partly consistent with zones that are still filling in.

6 DISCUSSION

In this section, I examine what drives the estimated pollution effects and what they imply for health. Section 6.1 asks which margin of zone activity generates the variation across cohorts and places: its scale, its industrial composition, or its emissions technique. Section 6.2 translates the pollution into a health burden and compares it with the income gains from zones.

6.1 Industrial Composition and the Pollution Intensity of Zones

Combustion at the source rises with zone age (Figures 3 and A3), while the pollution generated per unit of downwind exposure stays flat within cohorts and differs sharply across them (Section 5.4). The contrast suggests that the emission intensity of zone activity is largely set at entry. To interpret this variation, I adapt the scale-composition-technique decomposition from the trade-and-environment literature to a place-based policy (Grossman & Krueger, 1995; Antweiler et al., 2001; Shapiro & Walker, 2018). That literature decomposes a change in pollution into the volume of activity (scale), the industrial mix of that activity (composition), and emissions per unit of output within industries (technique), and uses it to study how trade and growth reshape a country's aggregate emissions. The evidence below suggests that the variation is predominantly driven by composition, with no support for technique on the margins the energy data measure and no gradient in zone scale.

Technique. A technique effect would arise if firms within the same industry adopt cleaner fuels as zones mature or as standards tighten, thereby lowering the pollution intensity of zone activity independently of any change in industrial mix. I test for this using firm-level energy consumption from the VES (2009 and 2011–2015), defining the fossil fuel share as the ratio of combustion-generating energy to total energy in tons of coal equivalent (Section 3). Columns (1)–(4) of Table 6 regress this share on zone-cohort indicators. Firms near pre-2003 zones show a lower fossil fuel share than firms in never-treated districts in the absence of industry controls (column 1), but the difference disappears once industry fixed effects are added (column 3): the lower share reflects the industries located near these zones rather than cleaner fuel choices within industry. In the specification with industry-by-year fixed effects (column 4), the fuel composition of energy use does not differ across zone cohorts: all cohort coefficients are small and statistically indistinguishable from zero.

Columns (5)–(7) of Table 6 extend the test to the other margins the energy records measure, under the industry-by-year specification: whether a firm uses any fossil fuel, fossil fuel consumption

Table 6: Technique Margins and Combustion across Zone Cohorts

	Fossil fuel share of energy use				Any fossil fuel use	Fossil TCE per output	ln(Total TCE per output)	Fire- days
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pre-2003 Zone Cohort	-0.111 (0.030)	-0.020 (0.013)	0.008 (0.009)	0.006 (0.008)	0.014 (0.015)	-0.0012 (0.0005)	-0.013 (0.080)	
2003–2009 Zone Cohort	-0.005 (0.038)	0.014 (0.016)	0.019 (0.015)	0.018 (0.015)	0.055 (0.019)	0.0005 (0.0005)	0.069 (0.071)	0.451 (0.152)
Post-2009 Zone Cohort	0.063 (0.056)	0.016 (0.033)	0.008 (0.017)	0.008 (0.017)	0.048 (0.017)	0.0019 (0.0021)	0.240 (0.118)	0.056 (0.063)
Year-(Month) FE	Yes	Yes	Yes					Yes
Province FE		Yes	Yes	Yes	Yes	Yes	Yes	
Industry FE			Yes					
Cell FE								Yes
Industry × Year FE				Yes	Yes	Yes	Yes	
Firm size controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Weather controls								Yes
Unit	Firm-Y	Firm-Y	Firm-Y	Firm-Y	Firm-Y	Firm-Y	Firm-Y	Cell-Y
Observations	625889	625888	625888	625885	625885	611999	611999	20672
Model	OLS	OLS	OLS	OLS	OLS	OLS	OLS	Poisson

Notes: Columns (1)–(7) report coefficients on zone-cohort indicators from firm-level regressions, with firms in never-treated districts as the omitted category; all are weighted by firm employment and include firm-size controls. In columns (1)–(4) the dependent variable is the fossil fuel share of total energy consumption, the ratio of energy from coal, briquettes, gasoline, kerosene, mazut, diesel, and LPG to total energy in tons of coal equivalent (TCE), converted using net calorific values from A. Garg et al. (2006) and volume-to-mass factors from United Nations Department of Economic and Social Affairs (2022); columns (2)–(4) progressively add province, industry, and industry-by-year fixed effects. Columns (5)–(7) report the other margins the energy records measure, under the industry-by-year specification of column (4): an indicator for positive fossil fuel consumption, defined for firms with positive energy consumption (employment-weighted never-treated mean 0.59); fossil fuel consumption in TCE per million dong of gross turnover in 2010 prices, winsorized at the top 2.5% within two-digit industry (never-treated mean 0.0047); and the log of total energy consumption in TCE per million dong of gross turnover in 2010 prices. Columns (6)–(7) restrict the sample to firms reporting positive gross turnover; revenue is deflated with the World Bank CPI (2010 = 100). Industry fixed effects are at the two-digit level of the Vietnamese industrial classification, and differences within these industries may reflect product mix as well as technology. Column (8) reports post-treatment average marginal effects from the Poisson fire-days specification (equation 3) estimated separately by cohort; pre-2003 zones are excluded as always-treated in the fire panel. The corresponding cohort effects on PM_{2.5} are in Table 5, Panel B. Standard errors clustered at the province level in parentheses. Firm-level data are from the Vietnam Enterprise Survey for 2009 and 2011–2015.

per unit of output, and total energy consumption per unit of output, the latter two in 2010 prices. For technique on these margins to generate the cohort pattern, firms near pre-2003 and post-2009 zones would have to run cleaner or more efficient operations than firms near 2003–2009 zones in the same industry and year. Firms near post-2009 zones do not: they are about as likely to use fossil fuel as firms near 2003–2009 zones, about 5 percentage points above the never-treated baseline for both, and they use about 27% more total energy per unit of output than firms in never-treated districts, consistent with zones still filling in rather than operating cleaner technology. The one estimate in the direction of cleaner operation is the fossil fuel intensity of output near pre-2003 zones, about one quarter below the never-treated mean. However, the difference is too small to

account for the twofold gap in per-exposure effects between the pre-2003 and 2003–2009 cohorts. Part of it may also reflect product mix rather than technology, since the industry fixed effects are at the two-digit level.

The energy records do not measure emissions per unit of fuel: combustion technology and end-of-pipe abatement remain unobserved. Three pieces of indirect evidence speak to this margin. First, abatement that improves as zones age would lower the per-exposure effect over a zone’s life, but the within-cohort profiles in Panel B of Table 5 show no decline. Second, if the emission standards had changed how firms near zones operate, the change would likely appear at their introduction. I interact the cohort indicators of Table 6 with indicators for the years from 2011 and from 2014 onward, when the 2009 emission standards and the 2013 PM2.5 standard took effect. Firms near pre-2009 zones show no change on any of the four measured margins relative to firms in the same industry and year in never-treated districts (Appendix Table A4). Third, if firms near zones abated, larger firms would be the most likely to do so. Compliance carries fixed costs, which create economies of scale in abatement and would make larger firms cleaner within their industry. The firm-size gradients in the regressions of Table 6 run the other way. In the specifications with industry-by-year fixed effects, firms with 200 or more workers have a fossil fuel share about 19 percentage points above micro firms and are 41 percentage points more likely to burn fossil fuel, while using about 64% less total energy per unit of output. Within the pollution-intensive industries, the fuel-share gradient is steeper, about 24 percentage points. The pattern points to scale economies in combustion rather than in abatement: solid fuels require the boilers and kilns that larger producers operate, while smaller firms rely on purchased electricity.

Yet none of the three tests is perfect. The within-cohort profiles cannot separate abatement from a zone’s growth: an improvement that coincided with expanding activity would leave no visible decline. The standards comparison nets out abatement common to all firms, because the standards were applied nationally. End-of-pipe abatement is unobserved throughout, and a difference across cohorts that is constant over time would remain undetected. The evidence therefore supports a narrower conclusion: on the margins the data measure, nothing points to abatement. This is consistent with the weak enforcement on operating firms documented in Section 2.

Scale. A scale effect would arise if larger or more active zones generated more pollution per unit of downwind exposure. I measure the scale of a zone with four proxies: (i) planned area, (ii) occupied area, measured as leased industrial area and as the occupancy rate interacted with planned area, (iii) tenant activity, measured as the total employment, employment per leased hectare, and revenue of the secondary projects operating in the zone, and (iv) combustion observed at the source, measured as mean annual Type-2 detections. For the area and activity measures, I interact downwind exposure with indicators for the quartile of the nearest zone’s proxy, standardizing exposure on the full estimation sample so that the coefficients are comparable across quartiles. None of the area or activity measures produces a gradient consistent with a scale effect, with the coefficients ranging from 0.018 to 0.040 (Appendix Figures A8 and A9). Because the exposure measure aggregates all zones within 25 km while the quartile test uses only the nearest zone, I classify cells

instead by the inverse-distance weighted average of each proxy across all zones within 25 km, and this classification also yields no gradient (Appendix Figure A10).

Combustion at the source shows the same absence of a gradient. Across terciles of the nearest zone’s mean annual Type-2 detections, the per-exposure effect is 0.021 in the tercile with no detections, 0.038 in the middle, and 0.027 in the top (Appendix Table E1, Panel B). Zones that never register a detection raise downwind PM2.5 as much as zones that register several a year, as expected if detections mark the presence of hot-process industries rather than the quantity of fuel burned (Section 4). However, these measures of zone activity are not exogenous, and the comparisons are descriptive. Across all four proxies, larger or more active zones do not generate more pollution per unit of downwind exposure.

Composition. A composition effect would arise if the industries zones admit, rather than the emission intensity of firms within those industries or the scale of zones, drove the pollution effect. The industries entering each cohort’s zones at establishment are not observed directly: the VES identifies a firm’s location inside a zone only from 2010, and the earliest zones predate the firm records. The evidence is therefore indirect and proceeds in three steps: the activity zones add to nearby districts (i), whether the effect is larger where the admitted mix is dirtier (ii), and whether the mix inside zones changes over a zone’s life (iii). The cohort pattern is then read against these results.

First, to characterize the composition of the additional firms observed near zones, I estimate an event study for the number of manufacturing firms operating in districts near industrial zones:

$$y_{dt} = \sum_{k=-5}^9 \beta_k \mathbb{I}\{e_{dt} = k\} + \alpha_d + \mu_t + \varepsilon_{dt}, \quad (6)$$

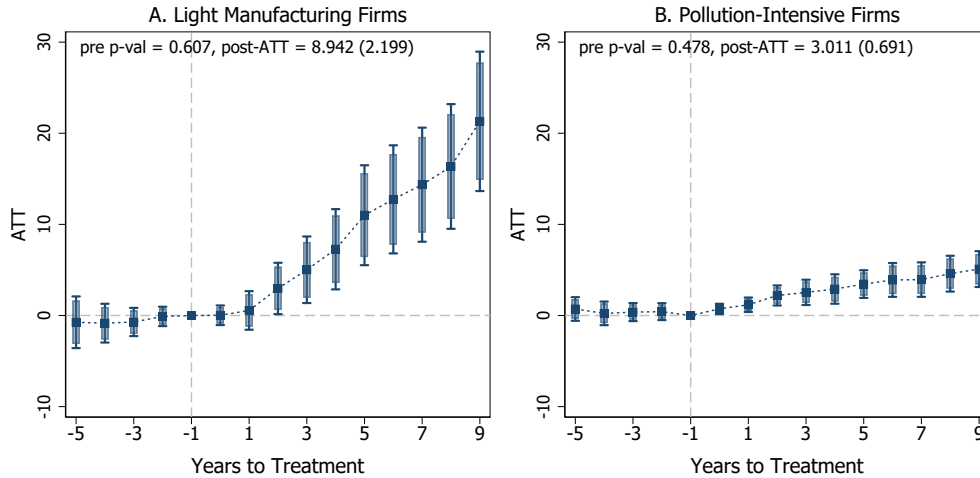
where y_{dt} is the number of, or employment in, light or pollution-intensive manufacturing firms in district d of province p in year t ; α_d and μ_t are district and year fixed effects; and $e_{dt} = t - T_d$ is event time relative to the year T_d in which the first industrial zone within 5 km of district d became operational, with $k = -1$ omitted as the reference period. Estimation uses the heterogeneity-robust estimator of de Chaisemartin and d’Haultfoeuille (2024), with standard errors clustered at the province level. Because the estimator compares districts around the opening of their first nearby zone with never-treated districts, identification comes from zones becoming operational during the sample period, most of which belong to the 2003–2009 cohort. Light manufacturing accounts for an average post-establishment increase of approximately 9 firms per district. Pollution-intensive firms, the top quartile of the VES energy-based intensity ranking (Section 2 and Appendix B), also increase, by about 3 firms per district; the effect is statistically significant but far smaller than the light-manufacturing response (Figure 6).¹⁹ The conclusion is qualitatively sim-

¹⁹Appendix Figure A11 reports the corresponding employment estimates, where the contrast is sharper: roughly 1,700 additional light-manufacturing workers per district against fewer than 200 in pollution-intensive industries. District-level employment is noisier than firm counts, and the light-manufacturing employment series displays a pre-trend.

ilar under the Shapiro and Walker (2018) classification and under alternative cutoffs of the VES ranking (Appendix Figure B1).

Because the VES firm records are repeated cross sections, these counts measure the net change in the number of firms operating, and an increase near zones could in principle reflect relocation rather than new activity. However, tracking firms by their tax code shows that the manufacturing firms arriving in districts newly within 5 km of a zone are almost all new to the survey rather than observed in another district before (Appendix Table A5). The distance-band estimates give the same answer: the pollution-intensive response, estimated as in equation (1), is confined to districts within 15 km of a new zone, about 3 to 5 firms per district, and is small beyond that distance, with no decline in the surrounding districts comparable to the increase near zones (Appendix Figure A12). The incremental activity near zones is concentrated in less pollution-intensive sectors, even as total combustion expands.

Figure 6: Industrial Zone Establishment and Manufacturing Firm Counts



Notes: This figure reports event-study estimates of industrial zone establishment on the number of light manufacturing firms (Panel A) and pollution-intensive manufacturing firms (Panel B) in nearby districts. The corresponding employment estimates are in Appendix Figure A11. The outcome y_{dt} is the firm count in district d in year t . Event time $e_{dt} = t - T_d$ is defined relative to the year in which the first industrial zone within 5 km of district d became operational, with $k = -1$ as the reference period. Pollution-intensive industries are the top quartile of the energy-based intensity ranking constructed from the VES energy module: non-metallic minerals, paper, coke and refined petroleum, basic metals, and textiles (Appendix B). All other manufacturing sectors are classified as light manufacturing. Estimation uses the heterogeneity-robust estimator of de Chaisemartin and d'Haultfoeuille (2024). The pre-trend p -value reports a joint test of the null that all pre-treatment coefficients equal zero. Post-ATT is the average treatment effect across all post-treatment periods, with the standard error in parentheses. All specifications include district and year fixed effects. Standard errors are clustered at the province level. Outer bars show 95% confidence intervals and inner bars show 90% confidence intervals. Data are from the Vietnam Enterprise Survey for firm counts and the Ministry of Planning and Investment for industrial zone locations.

Second, if zones admit the industries that surround them, the mix would be dirtier where the base is pollution-intensive, and the per-exposure effect would be larger there. I classify cells by whether their district's baseline (2001–2003) pollution-intensive manufacturing share is above or below the national median. The comparison is interpretable only where the baseline window pre-dates zone entry, and Table 7 therefore reports it within the 2003–2009 cohort. The per-exposure

effect in high-share districts is roughly twice that in low-share districts (0.080 versus 0.038 per standard deviation), and the gap is positive at every horizon, persisting beyond a decade. However, the baseline mix is not exogenous, and the comparison is descriptive. Where the surrounding industrial base was pollution-intensive, zones established between 2003 and 2009 show a higher PM2.5 response per unit of exposure, consistent with the composition channel.

Table 7: Industrial Zones and PM2.5: Dynamic Effects by Baseline Pollution Intensity
2003–2009 Cohort

	Dependent Variable is $\ln(\text{PM}_{2.5})$			
	Event-time Window			
	Pooled (1)	0–4 (2)	5–10 (3)	11+ (4)
Downwind Exposure (SD), Low	0.038 (0.013)	0.031 (0.018)	0.025 (0.017)	0.043 (0.012)
Downwind Exposure (SD), High	0.080 (0.015)	0.077 (0.022)	0.092 (0.018)	0.077 (0.016)
Difference, High – Low	0.042 (0.015) [303,480]	0.046 (0.030) [99,900]	0.067 (0.018) [119,880]	0.033 (0.016) [83,700]
Cell FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes
Sample	5-25km	5-25km	5-25km	5-25km

Notes: Estimates from equation (4) for different event-time windows, where event time is years since the first zone within 25 km became operational, on the cells of the 2003–2009 establishment cohort, defined as in Table 5, Panel B, with sample construction following that table. Cells are classified by whether their district’s baseline (2001–2003) pollution-intensive manufacturing share is above or below the national median. The pollution-intensive classification is the VES energy-based measure (Appendix B). The comparison is restricted to the 2003–2009 cohort because its zones enter after the baseline window. The pre-2003 cohort is omitted because its zones predate the window, and the split does not measure the base that preceded them. The post-2009 cohort is omitted because its short exposure history and small cell counts make the split by baseline share too imprecise to be informative. Downwind exposure is standardized on the common estimation sample and interacted with the high/low indicator; the two coefficients are therefore directly comparable. The rows “Difference, High – Low” report the gap between the two coefficients with its standard error, and the bracketed figures are the treated cell-months in each window. All columns include cell fixed effects, year-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and monthly total precipitation from ERA5-Land). Standard errors clustered at the province level in parentheses.

Third, the VES asks firms whether they are located in a zone. I use the 2011 and 2013–2016 rounds, in which the question was asked of nearly all firms (Appendix D). Over these years the mix of firms inside zones changed little: pollution-intensive industries account for 19 to 20% of manufacturing firms inside zones and for about 10% of their employment in every year, among large and small firms alike (Appendix Table D1, Panel A). Where the in-zone mix of a district did shift toward lighter industries, the PM2.5 effect of exposure did not decline. I classify the cells of the pre-2009 cohorts into terciles by the 2011–2016 change in the fuel intensity of the in-zone

mix in the district hosting their nearest zone. The change in the per-exposure effect from the years through 2011 to the years from 2017 onward is the same in the tercile whose mix became lightest as in the tercile whose mix became heaviest, and the result holds under each alternative construction of the shift (Appendix Table D1, Panel B, and Appendix Table D2). The shifts that occurred consisted of lighter industries growing beside pollution-intensive plants that remained in place. The result shows that shifts of this kind do not lower the effect. It does not measure what the exit of those plants would do. The composition channel therefore points to the industries each cohort of zones admitted rather than to changes in the mix of firms within zones over their life. This is the firm-side counterpart of the within-cohort stability in Panel B of Table 5.

Variation across cohorts. The cohort differences in both ambient PM2.5 (Section 5.4) and combustion at the source are consistent with the composition channel these results support. On PM2.5, the 2003–2009 cohort’s per-exposure effect is 0.047 per standard deviation, against 0.023 for the pre-2003 cohort and 0.017 for post-2009 zones (Table 5, Panel B). Column (8) of Table 6 estimates the fire-days effect separately by cohort, and shows the same pattern: 0.451 for the 2003–2009 cohort and 0.056 for post-2009 zones, the latter statistically indistinguishable from zero (the pre-2003 cohort is excluded as always-treated in the fire panel). The pattern is not a monotone decline across regulatory regimes but a peak in the 2003–2009 cohort and a null for the most recent one.

The institutional timeline supports this reading: the 2003–2009 cohort was established during Vietnam’s period of rapid industrialization, under the framework Law on Environmental Protection but before the QCVN technical standards of 2009, when the entry of pollution-intensive manufacturing was least constrained. The standards did not lower the burden from zones already operating: for the pre-2009 cohorts, the per-exposure effect is 0.025 (s.e. 0.005) in the years through 2009, 0.021 (0.008) in 2010–2013, and 0.028 (0.005) from 2014 onward, net of the growth of these zones over the same years. Post-2009 zones were approved subject to the 2009 standards, which apply at the design stage. Their approval also coincided with the reorientation of Vietnamese foreign investment toward electronics and light assembly, and their near-zero effect on both combustion and ambient PM2.5 is consistent with a cleaner composition of activity from the outset.²⁰ The post-2009 cohort is also still filling in: as of 2023, its median occupancy rate is 34%, against 93% for the 2003–2009 cohort and near-full occupancy for the pre-2003 cohort, and 63% of its zones remain under construction (Appendix Table A3). Part of its small effect therefore reflects activity that has not yet materialized.

Taken together, the variation in the pollution effect across cohorts and across the places where zones are sited reflects the industrial composition of zone activity rather than the within-industry

²⁰Two forces plausibly contribute to the cleaner composition of the most recent cohort. The first is the tightening of entry standards, which could screen pollution-intensive industries out of newer zones at the approval stage. The second is the reorientation of Vietnamese manufacturing and foreign investment toward electronics and light assembly over the same period (Hollweg et al., 2017; Athukorala & Nguyen, 2022), which would shift the mix of entrants toward cleaner sectors independently of regulation. Because the two coincide in calendar time, the available data cannot separate them. They need not be separated here, however: both operate through the composition of entering firms rather than the emission intensity of firms already operating, so the cleaner cohort reflects a shift in industrial mix under either account.

emission intensity of firms or the physical scale of zones. This differs from the finding of Shapiro and Walker (2018) that the decline in U.S. manufacturing emissions was driven by the technique effect, attributed to an effectively enforced increase in the implicit pollution tax. Where firm-level emissions enforcement is limited, the environmental footprint of place-based industrial policy depends more on which industries the policy attracts than on the emission intensity of the firms operating within them. The near-zero effect of post-2009 zones indicates that place-based development and environmental quality are not inherently in conflict. The lever in this setting is what a zone admits when its projects are approved, not standards on the firms already inside it.

6.2 Health Burden

To provide a sense of the welfare magnitude, I translate the PM_{2.5} estimates into illustrative health costs using standard dose-response parameters from the epidemiological literature. The translation rests on three assumptions that go beyond the empirical design. First, the wind-based identification isolates only the directional component of zone-attributable pollution, the increment that varies with a cell's position in the downwind plume. Pollution from zones that disperses uniformly or accumulates as elevated background concentration is absorbed by cell and year-month fixed effects, and the population-weighted increment of 0.70 $\mu\text{g}/\text{m}^3$ is therefore a partial measure of the total burden. The health costs derived from it can be interpreted as a lower bound.

Second, the dose-response parameters used below are estimated from cross-sectional variation in chronic, long-run average exposure (Ebenstein et al., 2017), whereas the wind-based estimate reflects episodic, directional variation in plume exposure. Under a linear dose-response, the timing of exposure within the month does not matter. Under the concave exposure-response curves used in the Global Burden of Disease, an increment that arrives in spikes on a few downwind days causes less damage than the same increment spread evenly. The translation treats it as evenly spread and would then overstate the burden. The overstatement is unlikely to be large: under a logarithmic curvature, the shape of the Global Exposure Mortality Model (Burnett et al., 2018), even if the entire monthly increment arrived on a single day, the implied burden would fall by about a quarter.²¹ Third, the monetary figures rest on a value of a statistical life transferred from the United States to Vietnam, scaled by relative income.

With these caveats, I apply the dose-response relationship from Ebenstein et al. (2017), in which each additional 10 $\mu\text{g}/\text{m}^3$ of PM₁₀ reduces life expectancy by 0.64 years, and convert it to PM_{2.5} terms with the Air Quality Life Index methodology, which yields 0.098 life-years per $\mu\text{g}/\text{m}^3$ of

²¹The Global Exposure Mortality Model specifies the log hazard ratio as proportional to $\ln(1 + z/\alpha)$ in the concentration z above a counterfactual (Burnett et al., 2018). The estimated population-weighted increment of 0.70 $\mu\text{g}/\text{m}^3$ is a monthly average. Suppose it came entirely from a single day. The concentration on that day would then be $0.70 \times 30 = 21 \mu\text{g}/\text{m}^3$ above the sample-mean baseline of 24 $\mu\text{g}/\text{m}^3$, and the other days of the month would be unchanged. Under a linear dose-response, the damage from that day is proportional to the change in concentration relative to the baseline, $21/24 = 0.88$. Under a logarithmic curve, it is proportional to the change in the log of concentration, $\ln(45/24) = 0.63$. The two coincide for small changes and diverge as the change grows. The concentrated day would therefore cause $0.63/0.88 \approx 0.72$ of the damage that the linear translation assigns to the month. If the increment came instead from a quarter of the month's days, the ratio would be 0.95. A curve more concave than the logarithmic one would lower these ratios, and a milder one would raise them.

PM2.5.²² At the point estimate, this implies a reduction of approximately 25 days of life expectancy per exposed individual ($0.70 \times 0.098 \times 365 \approx 25$). The 95% confidence interval on the PM2.5 coefficient gives a range of approximately 15 to 33 days. The burden is unevenly distributed: at the 95th percentile of cell-month exposure, the implied reduction is approximately 50 days, twice the population-weighted average.

Approximately 46 million people reside in the 5–25 km transport ring as of 2000, implying an aggregate burden of roughly 3.2 million person-years of life expectancy lost across the exposed population. To value this burden, I use the benefit-transfer value of a statistical life for Vietnam: \$342,000 in 2015 dollars under an income elasticity of 1.0, a common default for transfers to lower-income countries (Viscusi & Masterman, 2017), or about \$14,800 per life-year.²³ At this value, the aggregate burden amounts to an illustrative present value on the order of \$47 billion, and the 25 days per exposed person to about \$44 per year over a 40-year horizon. Under an income elasticity of 1.5, the reference case for low- and middle-income countries (Robinson et al., 2019), both figures fall by about four fifths, to roughly \$9 billion and \$8 per person per year.

For comparison, Pham (2026) estimates that household labor income per capita rises by about 7% in districts near zones, roughly \$30 to \$50 per person per year. With the three assumptions above, which understate the cost on one side and overstate it on the other, the health cost per exposed person ranges from roughly one fifth of the income gain to roughly its size.²⁴ A complete welfare evaluation would also require the fiscal costs of the program (foregone corporate income tax, duty-free imports of capital equipment, land-rent exemptions, and publicly built zone infrastructure), and these are not reported at any level of aggregation: Vietnam publishes no information on tax expenditures (Truong & Le, 2016), and provincial budgets do not separate zone-related spending. I therefore interpret the comparison as placing the environmental cost on the same order as the income gains rather than as a cost-benefit result.

7 CONCLUSION

This paper estimates the environmental cost of a place-based industrial policy in a setting where official measurement of industrial emissions is limited. It proceeds in two steps that link the source of the pollution to the population exposed to it: satellite detections of industrial combustion mea-

²²The conversion assumes a PM2.5-to-PM10 ratio of 0.65, so that $10 \mu\text{g}/\text{m}^3$ of PM2.5 corresponds to approximately $15.4 \mu\text{g}/\text{m}^3$ of PM10, implying $0.64 \times (10/6.5) \approx 0.98$ life-years per $10 \mu\text{g}/\text{m}^3$ of PM2.5. The assumed ratio is consistent with Vietnam's ambient monitoring network, which reports PM2.5-to-PM10 ratios of 0.65 to 0.75 across six reference-grade stations spanning the north-south gradient (Ministry of Natural Resources and Environment, 2021).

²³The value adjusts a U.S. base VSL of \$9.6 million by GNI per capita with an income elasticity of 1.0. It converts to a value of a statistical life-year of approximately \$14,800 at a 3% discount rate ($\text{VSLY} = \$342,000 / 23.1$, where 23.1 is the annuity factor over 40 years of remaining life expectancy). The 25 days per person, valued at this rate, amount to about \$1,000 in present value, or about \$44 per year over the same horizon. The figures are also sensitive to the base VSL and the discount rate.

²⁴The income gain converts the estimates of Pham (2026), in 2010 Vietnamese dong, to 2015 dollars. The gain accrues to households in districts within about 15 km of a zone, whereas the exposed population is that of the 5–25 km ring, and the comparison treats the two populations as interchangeable. The concavity bound above leaves this range essentially unchanged.

sure activity at the source, and wind-driven variation in downwind exposure identifies its contribution to the ambient PM_{2.5} borne by the nearby population. In Vietnam, zone establishment more than doubles the frequency of detected combustion within 5 km, and the transported pollution raises ambient PM_{2.5} for the 5–25 km transport ring by a population-weighted 0.70 $\mu\text{g}/\text{m}^3$ per month. Applying standard dose-response estimates, the implied reduction in life expectancy is approximately 25 days per exposed individual, aggregating to 3.2 million person-years across the 46 million residents of the transport ring.

The average effect masks substantial variation across establishment cohorts. Within a cohort, the per-exposure effect persists essentially unchanged beyond a decade of operation, while combustion at the source rises over the same horizon. Across cohorts, the pattern is not a simple decline but a peak for zones established between 2003 and 2009, under framework environmental legislation but before binding technical standards, and a null for those established afterward, which were still filling with tenants at the end of the study period. Decomposing the variation into scale, composition, and technique effects, I find that it operates through composition rather than within-industry abatement or zone scale: firm-level energy data show no within-industry differences across cohorts, and the effect is persistently larger where the surrounding industrial base is more pollution-intensive. Whether the cleaner composition of newer zones is driven by tighter entry standards or by the reorientation of Vietnamese manufacturing toward electronics and light assembly cannot be separated with the available data, since the two coincide in time. Either way, under the weak firm-level enforcement that prevailed throughout the period, it is likely the mix of industries a zone attracts, not abatement within firms, that shapes its environmental footprint.

The estimates measure the local environmental footprint of zones, the pollution imposed on the populations around them, and not the net national effect of the policy. The two differ by what the activity inside zones would have emitted, and to whom, had the zones not been built, and the sign of the difference is not determined in advance: a firm that moves from a dense inner district to a zone on the urban edge lowers exposure at the origin and raises it at the destination, while a firm that expands on entering a zone emits more than it did before. Q. Chen et al. (2025) document the analogous problem for environmental regulation in China, where production shifted from regulated plants to unregulated plants of the same business group and the reduction measured at the regulated plants overstated the national reduction. The data show little sign that zones draw activity from their surroundings: combustion in the comparison bins is unchanged after a zone opens, and pollution-intensive firm counts decline only slightly beyond 15 km, at most about a tenth of the near-zone increase (Section 4, Appendix Figure A12). They cannot detect activity that relocates from beyond that distance. Relocation of polluting establishments into zones was official policy in the two largest cities, though the programs were directed at wastewater and residential nuisance rather than combustion and moved few large sources.²⁵ The enterprise-

²⁵Ho Chi Minh City listed about 1,400 polluting establishments in residential areas for relocation to industrial zones and outlying districts in 2002 (Decision 80/2002/QD-UB), the national plan for seriously polluting establishments set out 439 cases for 2003–2007 (Decision 64/2003/QD-TTg), and Hanoi listed 117 inner-city facilities for relocation by 2020 (Decision 130/QD-TTg of 2015). Of the 284 production establishments under the national plan, 28 were relocated

level tracking finds few firms moving into treated districts (Appendix Table A5), though the VES records enterprises rather than plants, and a plant opened in a zone while another closes elsewhere is not observed as a move. For the populations near zones, the cost is the same whether the activity is new or relocated. For a national accounting, the part of the footprint that would otherwise have been emitted elsewhere in Vietnam remains unidentified.

These findings carry three implications. The first concerns the evaluation of place-based industrial policy: environmental costs are a quantitatively meaningful component of its welfare impact, large enough to belong in cost-benefit assessments alongside the productivity, employment, and income effects that have been the focus of the existing literature. The second is that these costs are not fixed features of industrial zones. The null effect for the most recent cohort suggests that place-based industrial development and environmental quality need not be in conflict, though their record as they reach full occupancy remains to be seen. The third concerns environmental regulation where enforcement on operating firms is weak, a condition documented in other developing countries as well (Duflo et al., 2013; Greenstone & Hanna, 2014). In Vietnam, the variation lay in what zones admitted, not in how operating firms responded to the standards. Whether the same holds elsewhere is an empirical question, but the result suggests that in such settings the environmental record of a zone program is largely decided when its projects are approved.

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and most of the remainder were required to build wastewater treatment. The Ho Chi Minh City list drew on the industries barred from residential areas, among them dyeing, tanning, electroplating, chemicals, and slaughtering. Implementation was slow in both cities, and the cement, steel, and tobacco plants on the lists had for the most part not moved a decade later.

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Supplementary Materials for Online Publication

The Environmental Cost of Place-Based Industrial Policy: Evidence from Industrial Zones in Vietnam

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A APPENDIX TABLES AND FIGURES

A.1 Appendix Tables

Table A1: Robustness Checks: Industrial Zones and Type-2 Thermal Anomalies (Days)

	Baseline	Donut - (5-10) km	Treatment 5 km, operation	Treatment 10 km
	(1)	(2)	(3)	(4)
Post-ATT (marginal effect)	0.422 (0.169)	0.498 (0.206)	0.279 (0.134)	0.314 (0.136)
pre-trend p-value	0.812	0.851	0.542	0.531
Observations	11,362	8,588	11,362	9,690
Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Model	Poisson	Poisson	Poisson	Poisson

Notes: The table reports robustness checks for the Poisson estimates of industrial zones on Type-2 thermal anomalies (days), corresponding to the preferred specification in column (4) of Table 3. Column (1) reproduces the baseline specification: treatment is defined as a cell being within 5 km of an operational industrial zone in the year prior to the recorded operational date, with cells more than 5 km from any operational zone throughout the study period as the comparison group. Column (2) reports a donut specification that excludes cells in the 5–10 km ring from the sample, addressing potential SUTVA violations from spillovers into immediately adjacent cells. Column (3) defines treatment as activating in the year of recorded operational status rather than the year prior, removing the pre-operation timing adjustment that absorbs site-preparation activity. Column (4) defines treatment as a cell being within 10 km of an operational zone (in the year prior to operation), expanding the treatment ring beyond the 5 km threshold used in the baseline. The post-ATT is computed as the weighted average of post-treatment marginal effects (event times 0 through 9), with weights proportional to the number of treated cell-years contributing to each event-time estimate. Marginal effects are computed via Stata's `margins, dydx(*)` command following Poisson PML estimation. The pre-trend p-value is from a joint test that all pre-period coefficients (event times –5 through –2) equal zero. Standard errors clustered at the province level in parentheses. Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

Table A2: Directional Wind Exposure and Pre-Treatment Characteristics

	Year 2002				Year 2000		Pre-treatment
	Urban Share	Minority Share	Secondary Education Share	Farm Labor Share	Population Share	Elevation	ln(PM2.5)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Downwind Exposure (SD)	0.004 (0.034)	0.001 (0.008)	0.009 (0.007)	-0.018 (0.013)	0.000 (0.000)	10.521 (9.097)	-0.004 (0.014)
Observations	1,071	1,071	1,071	1,071	1,071	1,071	636
Adjusted R-squared	0.143	0.556	0.280	0.328	0.404	0.763	0.952
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance Band FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports OLS regressions of pre-treatment characteristics on the standardized downwind exposure variable E_{imt} defined in equation (5). The unit of observation is a district $\times$ distance band cell; outcomes measured at the cell level are averaged within district $\times$ distance band. The sample comprises district $\times$ distance band observations within 25 km of any industrial zone, excluding the innermost 0–5 km ring. In Columns (1)–(4), outcomes are district-level urban share, ethnic minority share, secondary education share, and farm labor share from the 2002 VHLSS. In Column (5), the outcome is the cell-level share of its province’s total population in 2000. In Column (6), the outcome is cell-level elevation in meters. In Column (7), the outcome is the cell-level mean of log PM2.5 across all panel years strictly before the cell’s first industrial zone within 25 km, with PM2.5 from Shen et al. (2024). All regressions include province and distance band fixed effects. Standard errors clustered at the province level in parentheses.

Table A3: Industrial Zone Characteristics by Establishment Cohort

	Pre-2003	2003–2009	Post-2009
Median planned area (ha)	197	236	262
Median leased area (ha)	124	100	40
Median occupancy rate	0.996	0.929	0.341
Share with occupancy below 0.25	0.013	0.117	0.426
Share under construction in 2023	0.013	0.089	0.631

Notes: The table reports characteristics of industrial zones by establishment cohort, for the 395 zones in the analysis sample with a recorded operation year: 75 established before 2003, 179 in 2003 to 2009, and 141 after 2009. Areas are in hectares; leased area is the area occupied by tenant firms. Occupancy rates, leased areas, and construction status come from the 2023 vintage of the Ministry of Planning and Investment zone database and therefore reflect conditions three years after the end of the study period; occupancy among post-2009 zones during the study period is lower than reported here. Under construction denotes zones whose 2023 status is construction rather than operation. *Source:* Data from the Ministry of Planning and Investment zone database (2023).

Table A4: Technique Margins of Firms near Pre-2009 Zones after the Emission Standards

	Pre-2003 cohort ×		2003–2009 cohort ×		Joint <i>p</i>
	$\mathbf{1}(t \geq 2011)$	$\mathbf{1}(t \geq 2014)$	$\mathbf{1}(t \geq 2011)$	$\mathbf{1}(t \geq 2014)$	$(t \geq 2014)$
Fossil fuel share of energy use	-0.006 (0.012)	-0.010 (0.008)	-0.004 (0.021)	-0.005 (0.015)	0.42
Any fossil fuel use	0.013 (0.028)	-0.024 (0.018)	-0.005 (0.019)	0.010 (0.021)	0.15
Fossil fuel TCE per unit of output	0.0006 (0.0006)	0.0002 (0.0004)	-0.0008 (0.0016)	-0.0000 (0.0005)	0.73
ln(total TCE per unit of output)	0.034 (0.065)	-0.022 (0.049)	-0.104 (0.094)	-0.022 (0.081)	0.87

Notes: Each row is one regression of the outcome on the zone-cohort indicators of Table 6, their interactions with indicators for the years from 2011 and from 2014 onward, and firm-size controls, with province and industry-by-year fixed effects (column (4) of Table 6), employment weights, and standard errors clustered at the province level, on the VES energy rounds of 2009, 2011, and 2013–2015 (625,885 firm-years for the first two outcomes and 611,999 for the last two). The omitted category is firms in never-treated districts; the interactions for the post-2009 cohort are included but not reported. The years from 2011 are those in which the 2009 technical emission standards were in force, and the years from 2014 those after the 2013 ambient PM2.5 standard and the 2014 revision of the Law on Environmental Protection. The last column reports the *p*-value of the joint test that the two 2014-onward interactions for the pre-2009 cohorts are zero. Outcomes and sample as in columns (4)–(7) of Table 6.

Table A5: Origins of Manufacturing Firms Arriving in Districts near Industrial Zones

	Arrivals (count)	Movers (%)	From comparison districts within 50 km (%)	From another zone ring (%)	From another province (%)	Movers' share of employment (%)
<i>Panel A: All firms</i>						
Districts newly within 5 km of an operating zone, after establishment						
Pollution-intensive industries	2,927	9.4	1.7	7.0	1.6	18.3
Light industries	15,099	4.6	1.0	3.3	1.3	10.1
Districts never within 5 km of a zone						
Pollution-intensive industries	3,777	6.6	1.4	4.5	1.1	17.3
Light industries	23,793	6.9	2.0	4.2	1.4	16.4
<i>Panel B: Firms with 10 or more workers at arrival</i>						
Districts newly within 5 km of an operating zone, after establishment						
Pollution-intensive industries	1,668	14.1	2.3	10.9	2.4	19.2
Light industries	6,391	7.3	1.6	5.2	2.2	10.7
Districts never within 5 km of a zone						
Pollution-intensive industries	1,905	9.4	2.3	6.0	1.8	18.4
Light industries	8,425	10.8	3.4	6.3	2.7	17.6

Notes: The table follows manufacturing firms across districts in the Vietnam Enterprise Survey 2001–2018 using the enterprise tax code, whose first nine digits are constant across survey rounds. An arrival is the first year a firm is observed in a district. A mover is an arrival that was observed in a different district in an earlier year, and its origin is the district of that last earlier observation. Other arrivals are first observations in the survey. Districts newly within 5 km of an operating zone are those first within 5 km between 2003 and 2016, the cohorts of equation (1), and their arrivals are counted from the year of establishment onward. Movers “from comparison districts within 50 km” have an origin district that is never within 5 km of a zone and lies within 50 km of one. Movers “from another zone ring” have an origin district that was already within 5 km of an operating zone. Panel A includes all firms. Panel B restricts arrivals to firms with 10 or more workers, which the survey enumerates in every round together with all state and foreign-invested enterprises, whereas smaller private enterprises are covered on a sample basis for the full questionnaire in some rounds. Pollution-intensive industries follow the classification of Appendix B. The mover shares are lower bounds on relocation: a firm that closes in one district and re-registers under a new tax code in another is counted as a new arrival rather than a mover. Excluding enterprises ever listed with units in more than one district (0.3% of firms) lowers the shares in Panel A for districts newly within 5 km to 8.7 and 4.3%.

Table A6: Industrial Zones and Type-2 Thermal Anomalies (Days): Multiple Zones around a Cell

Panel A: Restricting the sample (event study)

	Baseline (1)	Treated cells: one zone within 5 km (2)	Comparison cells: no zone within 10 km (3)	Comparison cells: no zone within 25 km (4)
Post-ATT (marginal effect)	0.422 (0.169)	0.352 (0.161)	0.498 (0.206)	0.400 (0.227)
pre-trend p-value	0.812	0.197	0.851	0.904
Observations	11,362	10,811	8,588	4,617
Treated cells	108	79	108	108
Comparison cells	490	490	344	135
Cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Model	Poisson	Poisson	Poisson	Poisson

Panel B: Exposure to all zones (pooled Poisson)

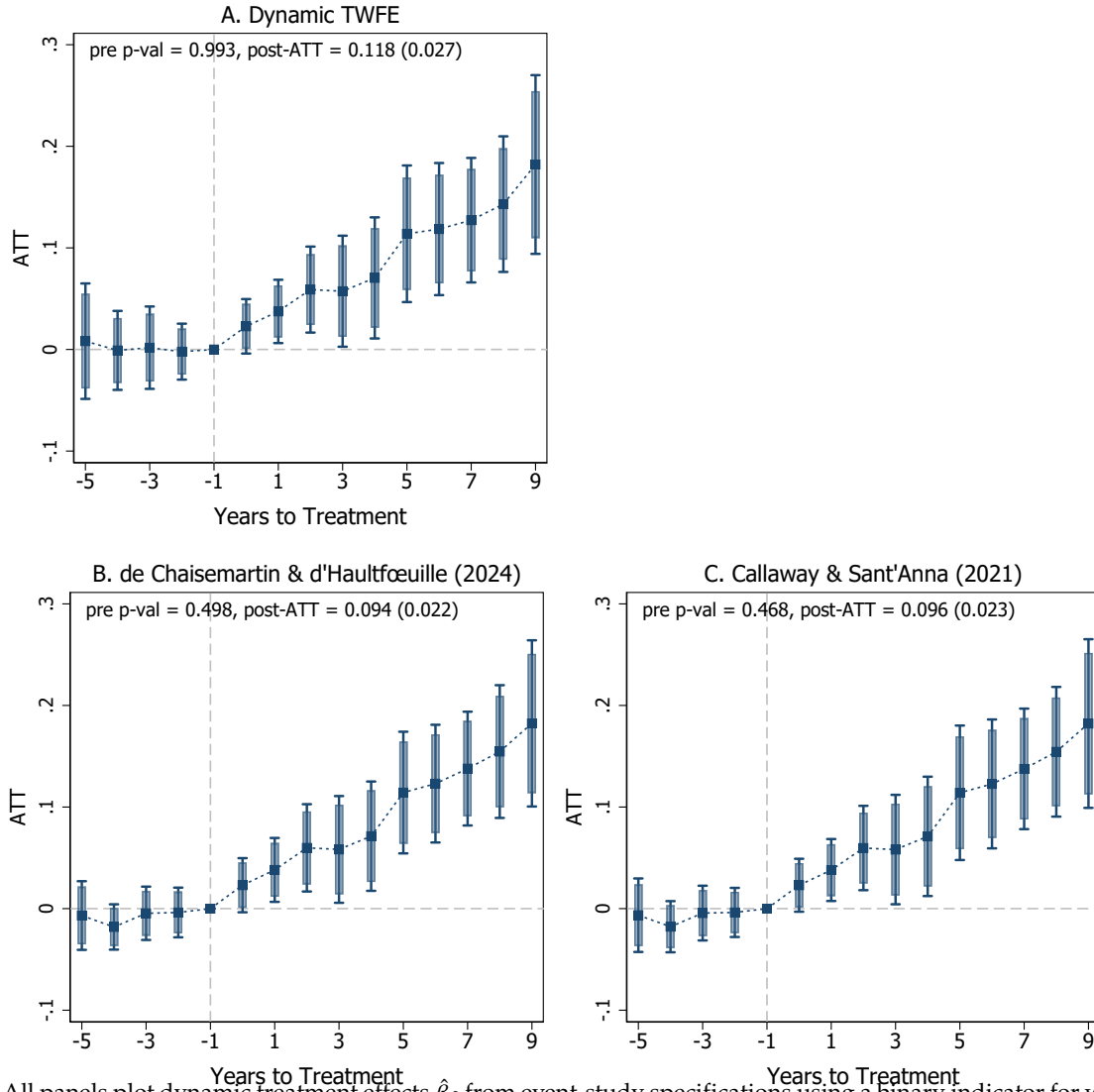
	(1)	(2)	(3)
Post × one operating zone within 5 km	0.765 (0.326)		
Post × two or more operating zones within 5 km	0.584 (0.483)		
Post (any operating zone within 5 km)		0.712 (0.325)	
Distance-weighted number of operating zones 5–25 km away (per s.d.)		0.067 (0.157)	
Distance-weighted number of operating zones within 25 km (per s.d.)			0.116 (0.122)
Observations	11,362	11,362	11,362
Cell FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Model	Poisson	Poisson	Poisson

Notes: The table examines whether the event-study estimates of Table 3, column (4), depend on the number and proximity of the zones around a cell. Treatment is being within 5 km of any operating zone in the year before its recorded operational date. Panel A re-estimates the event study on restricted samples: column (2) keeps treated cells that never have more than one zone within 5 km, and columns (3) and (4) keep comparison cells that never have an operating zone within 10 or 25 km. Post-ATT marginal effects, pre-trend tests, and standard errors are computed as in Table 3. Panel B replaces the event-time indicators with a post-treatment indicator on the baseline sample and reports Poisson coefficients. Column (1) splits the indicator by the number of operating zones within 5 km of the cell in that year. Columns (2) and (3) use the number of operating zones in each 5 km band around the cell, weighted by the inverse of the band midpoint (2.5, 7.5, 12.5, 17.5, and 22.5 km) and standardized to unit variance, over 5–25 km in column (2), alongside the 5 km indicator, and over 0–25 km in column (3), in place of it. Standard errors clustered at the province level in parentheses.

A.2 Appendix Figures

Figure A1: Industrial Zones and Type-2 Thermal Anomalies: Linear Estimates

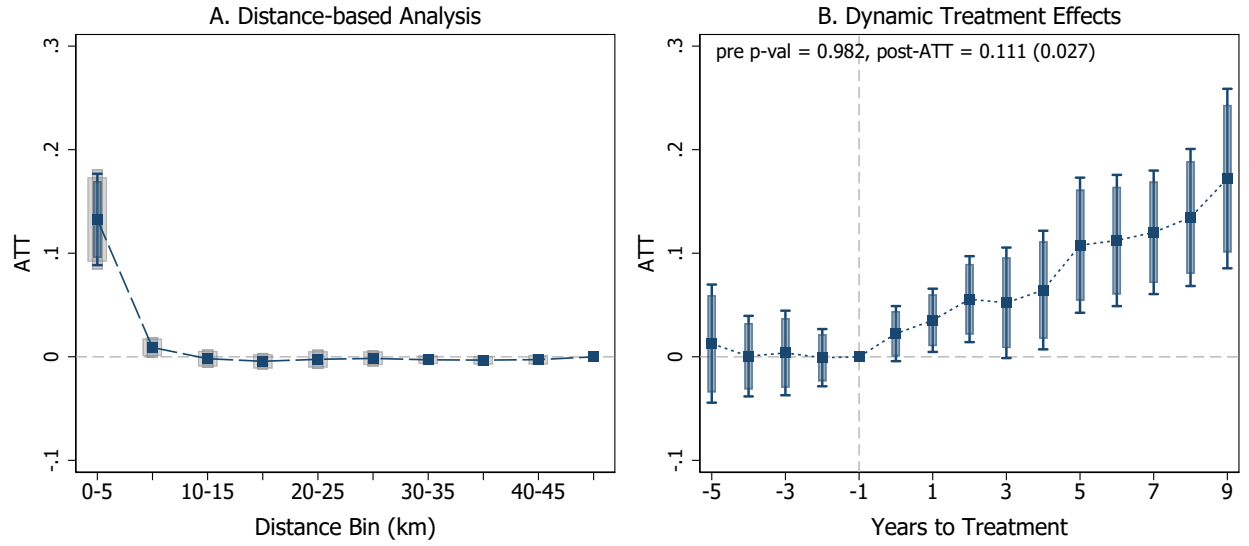
Robustness to heterogeneity-robust estimators



Notes: All panels plot dynamic treatment effects $\hat{\beta}_\ell$ from event-study specifications using a binary indicator for whether a cell is within 5 km of an operational industrial zone, with event time ℓ relative to first treatment. Outcome is $\ln(1 + \text{Type-2 detections})$ at the cell-year level. Panel A uses a two-way fixed-effects specification with event-time indicators pooled across cohorts and never-treated cells as the comparison group, with the post-ATT computed as a weighted average of post-treatment event-time coefficients, with weights proportional to the number of treated cells at each event-time. Panels B and C use the heterogeneity-robust estimator of de Chaisemartin and d'Haultfoeuille (2024) and Callaway and Sant'Anna (2021), which aggregate cohort-specific dynamic effects with non-negative weights. All specifications include cell fixed effects and year fixed effects, and endpoints $\ell = -5$ and $\ell = 9$ bin all event times outside the plotted range. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Source: Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

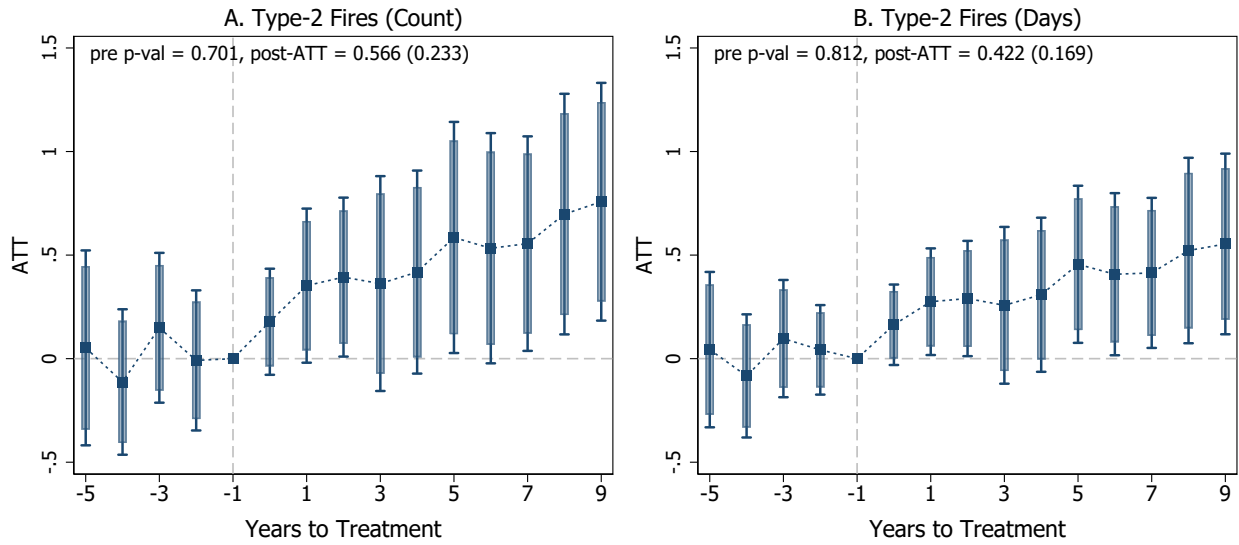
Figure A2: Industrial Zones and Type-2 Thermal Anomalies: Linear Estimates

Robustness to including time FE interacted with baseline characteristics



Notes: The left panel plots the weighted-average distance-bin coefficients $\bar{\delta}_b$ from equation (1), with 0–5 km through 40–45 km bins and 45–50 km as the reference. The right panel plots the dynamic treatment effects $\hat{\beta}_\ell$ using a binary indicator for whether a cell is within 5 km of an operational industrial zone, with event time ℓ relative to first treatment. Outcome is $\ln(1 + \text{Type-2 detections})$ at the cell-year level. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from analytical province-clustered standard errors. Light shaded bands in the left panel are 90 and 95% confidence intervals from a stratified cluster bootstrap with 200 replications. *Source:* Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

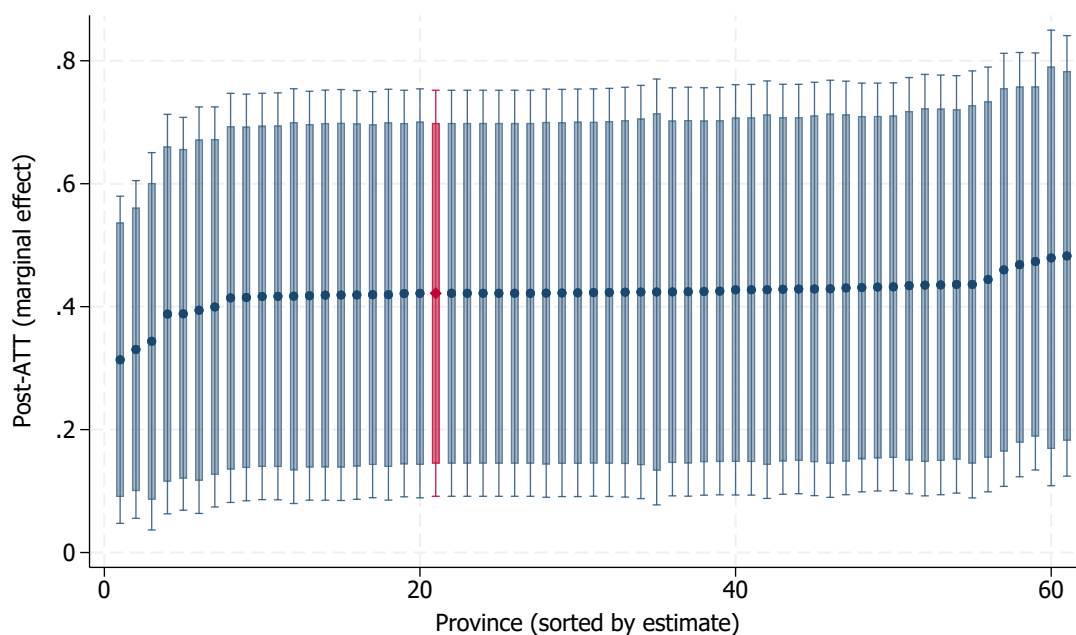
Figure A3: Industrial Zones and Type-2 Thermal Anomalies: Poisson Event-Study



Notes: Both panels plot dynamic treatment effects (marginal effects) $\hat{\beta}_\ell$ from event-study specifications with a Poisson model, using a binary indicator for whether a cell is within 5 km of an operational industrial zone, with event time ℓ relative to first treatment. Outcome is number of Type-2 detections (Panel A) and number of days with Type-2 detections (Panel B) at the cell-year level. The post-ATT is computed as a weighted average of post-treatment event-time marginal effects, with weights proportional to the number of treated cells at each event-time. Both specifications include cell fixed effects and year fixed effects, and endpoints $\ell = -5$ and $\ell = 9$ bin all event times outside the plotted range. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Source: Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

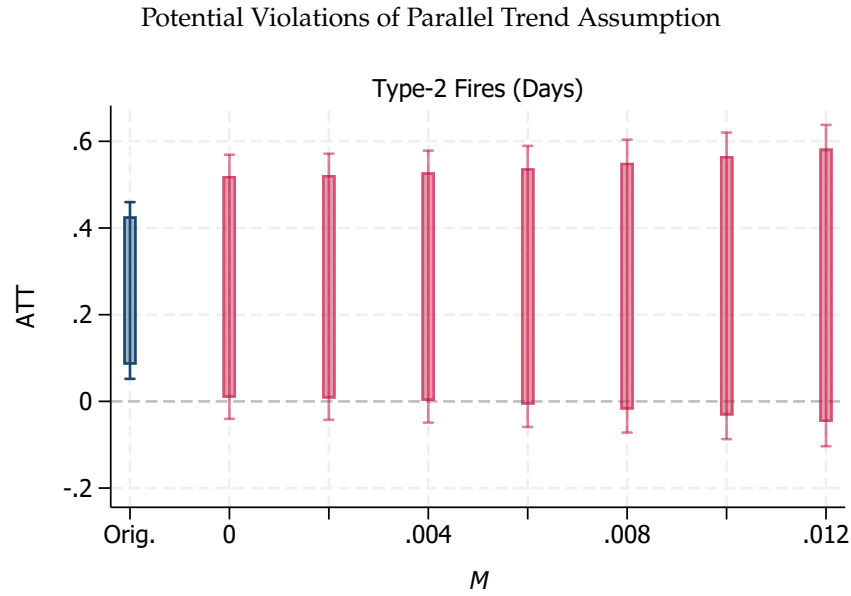
Figure A4: Industrial Zones and Type-2 Thermal Anomalies (Days)

Leave-One-Province-Out Distribution of Post-ATT Estimates



Notes: The figure reports leave-one-province-out estimates of the post-treatment marginal effect of industrial zones on Type-2 thermal anomaly days. Each point on the x-axis corresponds to a separate Poisson event-study regression in which one province is dropped from the sample, with provinces sorted along the x-axis by the size of the resulting estimate. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals. The cranberry marker shows the baseline estimate using the full sample (column 4 of Table 3). All specifications include cell fixed effects, year fixed effects, and district baseline characteristics interacted with time fixed effects, with standard errors clustered at the province level. The post-ATT is computed as the weighted sum of post-treatment event-time marginal effects (event times 0 through 9), with weights proportional to the number of treated cell-years contributing to each event-time estimate. *Source:* Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

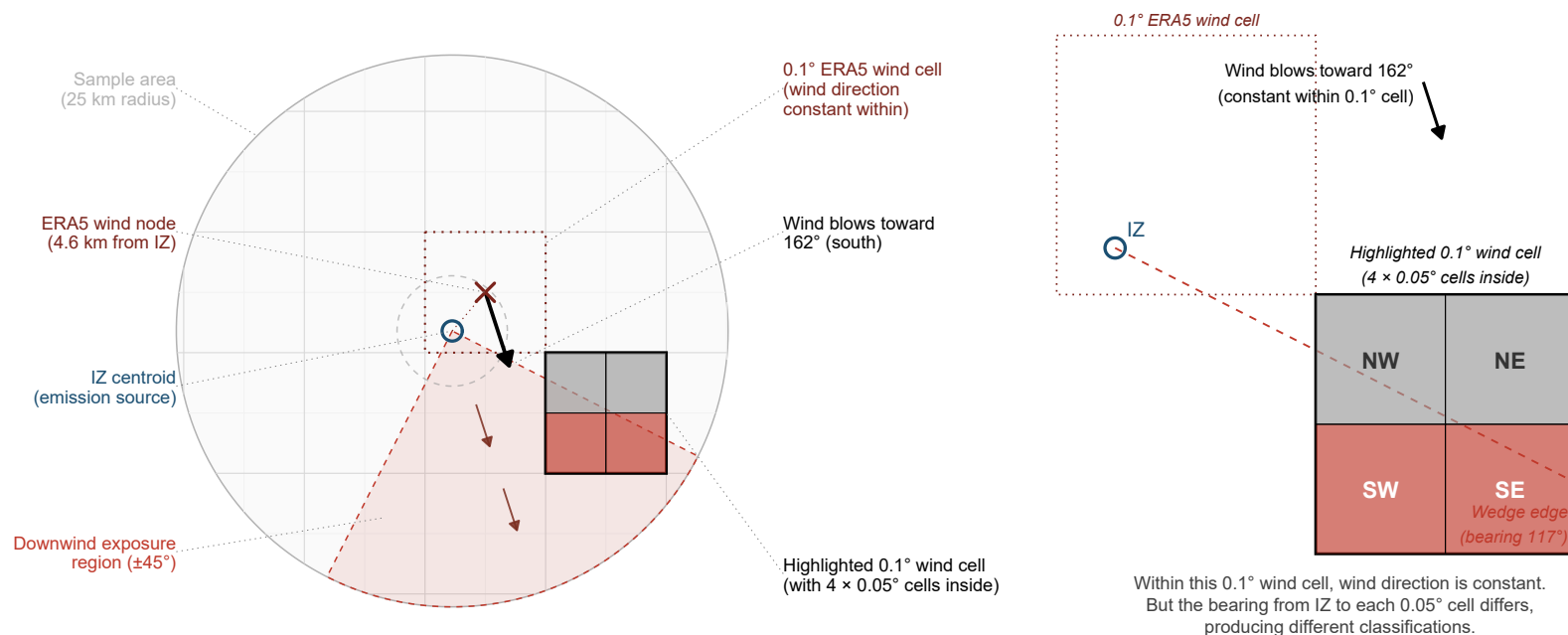
Figure A5: Industrial Zones and Type-2 Thermal Anomalies (Days)



Notes: The figure reports the sensitivity analysis of Rambachan and Roth (2023) for the post-treatment effect at event time $\ell = 2$ in the days-with-fire specification. The leftmost bar ("Orig.") shows the original 90/95% confidence interval from the baseline Poisson event-study estimate. The remaining bars show 90/95% robust confidence intervals under the smoothness restriction $\Delta^{SD}(M)$, which bounds deviations from linearity to no more than M between consecutive periods. $M = 0$ allows for linear violations of parallel trends, and larger values of M allow for greater deviations from linearity. The post-treatment effect remains positive and statistically significant at the 10% level for M up to approximately 0.005, indicating that the result is robust to modest deviations from a linear pre-trend. The baseline specification is the preferred Poisson estimate from column (5) of Table 3, with cell fixed effects and year fixed effects. Standard errors are clustered at the province level. Source: Data are from the NASA FIRMS MODIS Collection 6.1 Active Fire Product, 2000–2020.

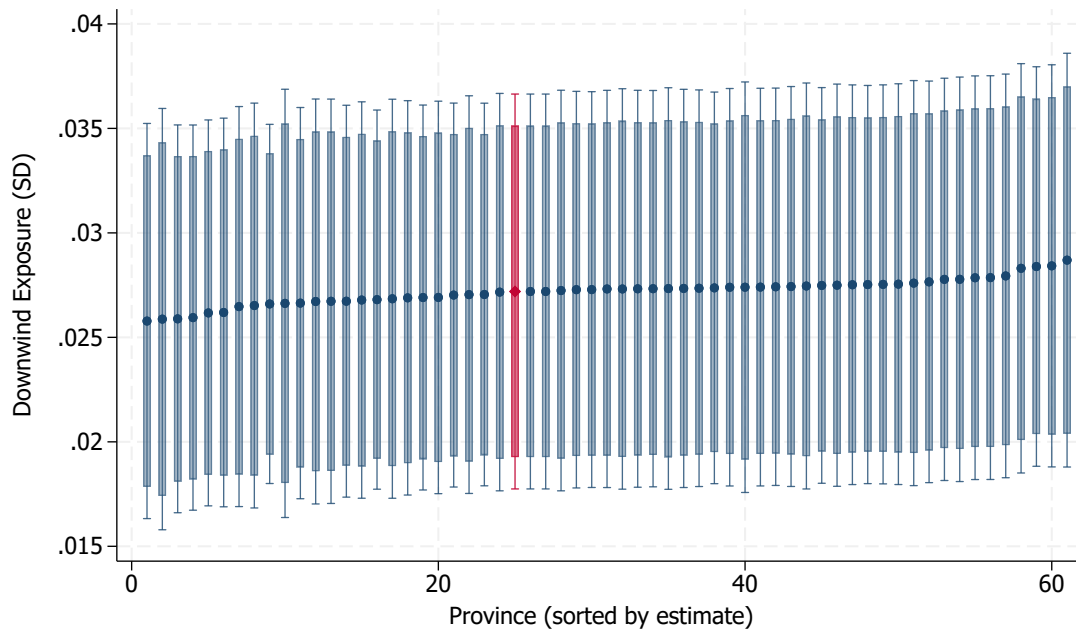
Figure A6: Wind-Based Downwind Classification: Schematic at 0.1° and 0.05° Resolutions

A. Overview at 25 km scale: 0.1° wind cells (red) and 0.05° analysis cells (inside highlighted cell) B. Wind direction is constant at 0.1°, classification varies at 0.05°



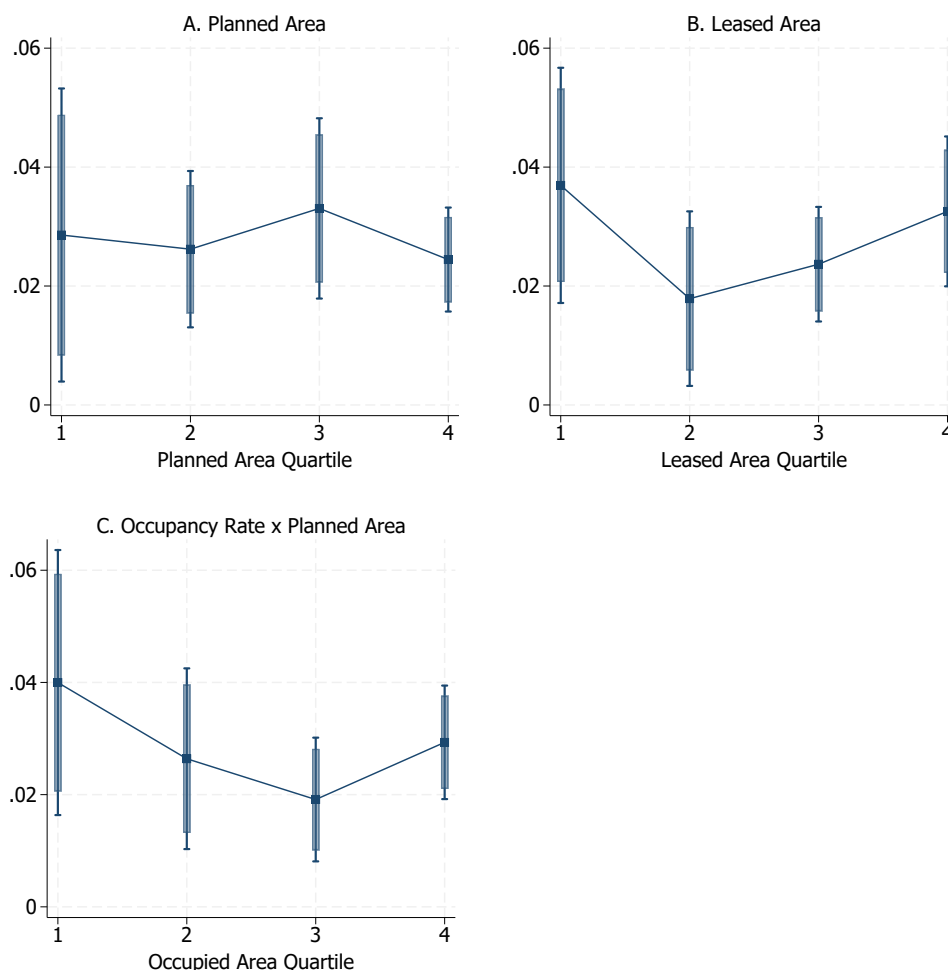
Notes: The figure illustrates the construction of wind-based downwind exposure to industrial zones, highlighting the resolution mismatch between ERA5-Land wind data (0.1° native resolution) and the analysis grid (0.05°). Panel A shows the design at the 25 km sample scale around a representative industrial zone (IZ). The IZ centroid, at the center, is the emission source. The ERA5 wind node is the nearest 0.1° grid node to the IZ centroid; in this schematic the offset is 4.6 km. The 0.1° ERA5 wind cell containing both the IZ and its assigned ERA5 node is outlined with a red dotted line, and the full 0.1° and 0.05° grids are drawn faintly across the sample area. The downwind exposure region is the $\pm 45^\circ$ wedge anchored at the IZ apex, oriented in the direction the wind blows toward (162° in this schematic, i.e., approximately due south). The highlighted 0.1° wind cell in the southeast contains four 0.05° analysis cells, color-coded by downwind classification. Panel B zooms in on the highlighted 0.1° cell, with the IZ's 0.1° wind cell shown for context in the upper-left. The wind direction is constant within each 0.1° cell, but the bearing from the IZ to each of the four 0.05° cells inside the highlighted 0.1° cell differs, so the wedge edge (bearing $117^\circ = 162 - 45$) splits the four cells into two downwind (SW, SE) and two upwind (NW, NE). All geometric parameters (offset between IZ and ERA5 node, wind direction, position of the highlighted 0.1° cell) are chosen for visual clarity rather than drawn from data.

Figure A7: Industrial Zones and PM2.5: Leave-One-Province-Out Estimates



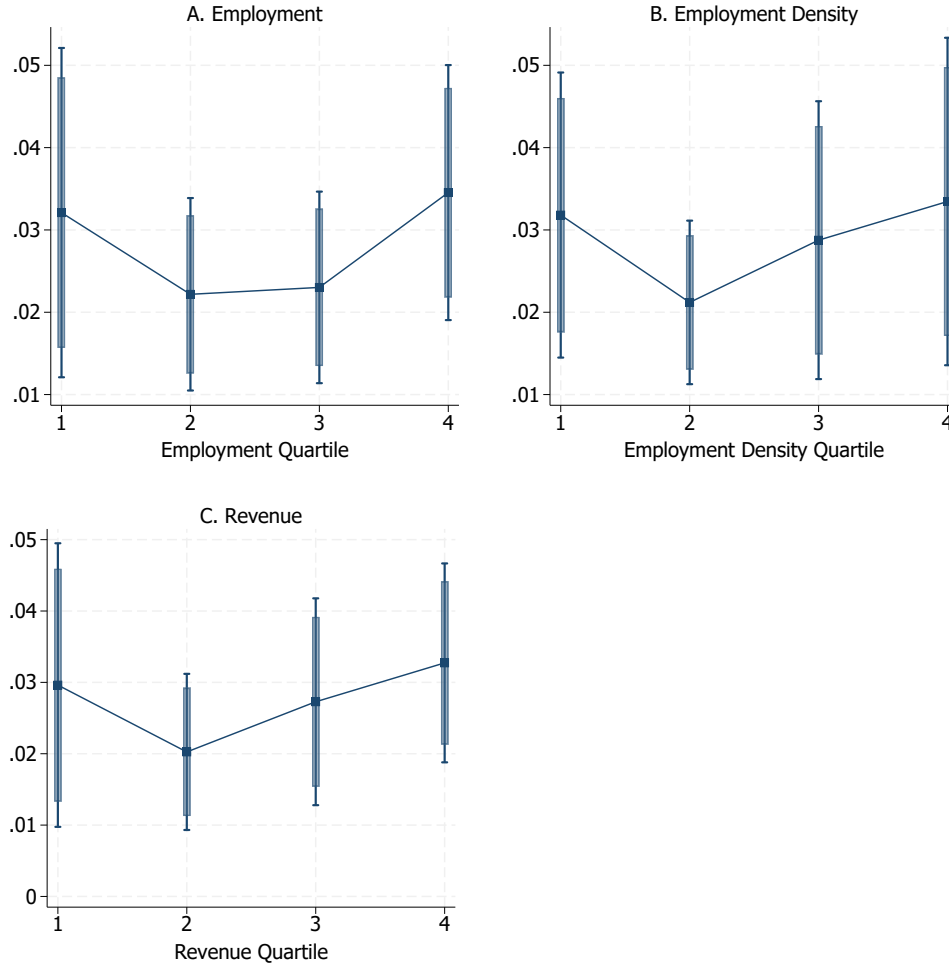
Notes: The figure reports leave-one-province-out estimates of the effect of downwind exposure to industrial zones on $\ln(\text{PM}_{2.5})$. Each point on the x-axis corresponds to a separate regression of equation (4) in which one province is dropped from the sample, with provinces sorted along the x-axis by the size of the resulting estimate. The exposure measure is re-standardized within each leave-one-out sample. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals. The cranberry marker shows the baseline estimate using the full sample (column 2 of Table 4). All specifications are estimated on cells in the 5–25 km transport ring and include cell fixed effects, year-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and a linear term in monthly precipitation from ERA5-Land). Standard errors are clustered at the province level. *Source:* Data are from Shen et al. (2024) for PM_{2.5}, ERA5-Land for wind (Muñoz-Sabater et al., 2021), and the Ministry of Planning and Investment for industrial zone locations.

Figure A8: Industrial Zones and PM2.5: Heterogeneity by Zone Area



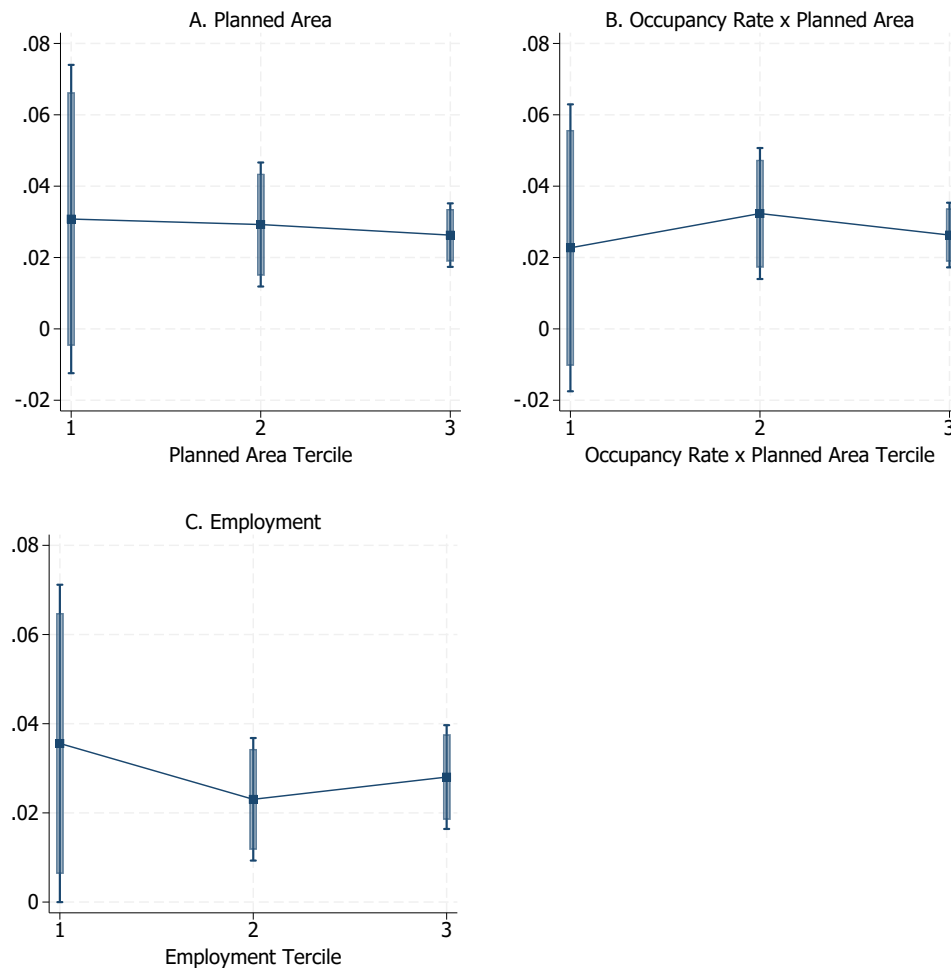
Notes: The figure reports the effect of downwind exposure to industrial zones on $\ln(\text{PM}_{2.5})$, estimated separately by quartile of the nearest zone's area. Panel A defines quartiles over planned area. Panels B and C use occupied-area measures from the 2023 database of the Ministry of Planning and Investment: leased industrial area in Panel B and the occupancy rate interacted with planned area in Panel C. Leased area and occupancy are measured in 2023, after the study period ends; every zone in the estimation sample matches the database, so the three panels use the same estimation sample. Each marker is the coefficient on the interaction of standardized downwind exposure with the corresponding area-quartile indicator, from a single regression per panel. Downwind exposure is the inverse-distance-weighted downwind share across operational zones within 25 km, standardized on the full estimation sample so that the coefficients are comparable across quartiles. Quartiles are defined over zones, with quartile 1 the smallest and quartile 4 the largest. The specification includes cell fixed effects, year-month fixed effects, and weather controls (a quadratic in monthly mean 2-meter temperature and monthly total precipitation from ERA5-Land), estimated on cells in the 5–25 km transport ring. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Data are from Shen et al. (2024) for $\text{PM}_{2.5}$, ERA5-Land for wind (Muñoz-Sabater et al., 2021), and the Ministry of Planning and Investment for industrial zone locations, planned and leased areas, and occupancy rates.

Figure A9: Industrial Zones and PM2.5: Heterogeneity by Zone Activity



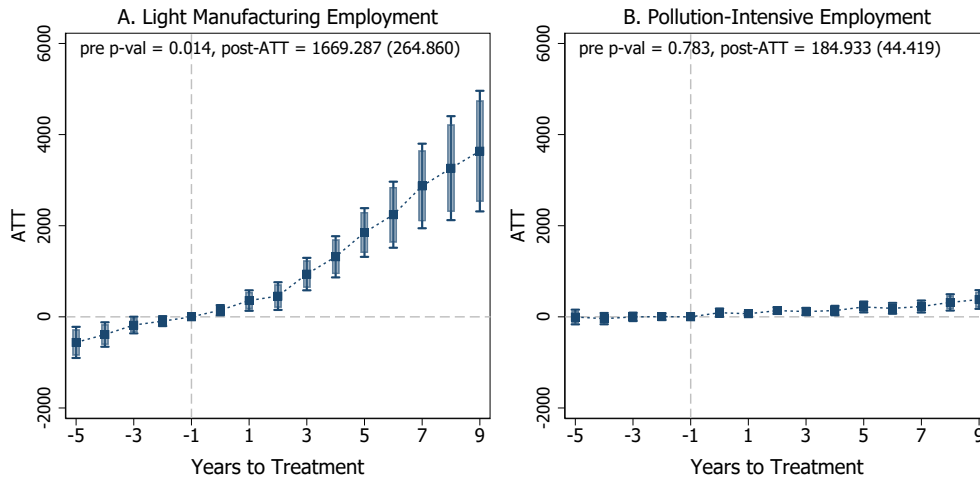
Notes: The figure repeats the scale test of Appendix Figure A8 with activity measures of the secondary projects operating in each zone, from the 2023 registry of the Ministry of Planning and Investment. Secondary projects are the tenant investment projects in a zone, as distinct from the zone's own infrastructure-development project. Panel A defines quartiles over the total number of workers in secondary projects (foreign and domestic combined), Panel B over workers per leased hectare, and Panel C over the total revenue of secondary projects. All measures are administrative totals reported by zone management boards as of June 2023, after the study period ends; they capture the end-of-period stock of activity rather than its evolution during the sample. Zones reporting no operating tenants are assigned to the bottom quartile; the density quartiles exclude zones with no leased area. Each marker is the coefficient on the interaction of standardized downwind exposure with the corresponding quartile indicator, from a single regression per panel; quartiles are defined over zones, with quartile 1 the lowest and quartile 4 the highest. Specifications, sample, and inference follow Appendix Figure A8. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Data are from Shen et al. (2024) for PM2.5, ERA5-Land for wind (Muñoz-Sabater et al., 2021), and the Ministry of Planning and Investment zone registry for activity measures.

Figure A10: Industrial Zones and PM2.5: Heterogeneity by Weighted Characteristics of All Zones within 25 km



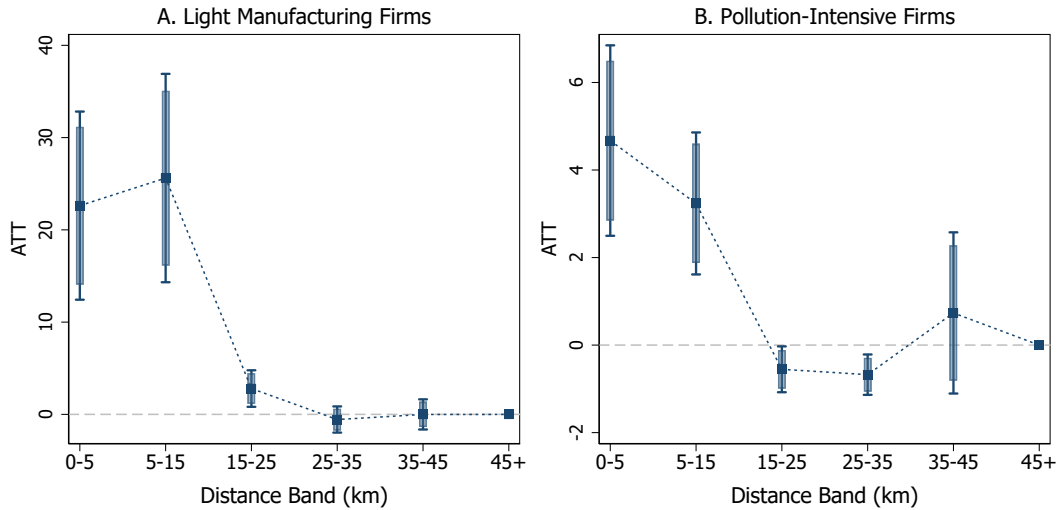
Notes: The figure repeats the scale tests of Appendix Figures A8 and A9 with cells classified by the characteristics of all zones within 25 km rather than the nearest zone, matching the aggregation of the exposure measure. For each cell, the zone characteristic is averaged over all zones within 25 km with inverse-distance weights; Panel A uses planned area, Panel B the occupancy rate interacted with planned area, and Panel C the total number of workers in secondary projects. Groups are terciles of the weighted characteristic, defined over the cells in the estimation sample: cells surrounded by little zone activity contribute limited variation in downwind exposure, and terciles keep every group precisely estimated. Occupancy, leased area, and employment are measured in 2023, as in Appendix Figures A8 and A9. Each marker is the coefficient on the interaction of standardized downwind exposure with the corresponding tercile indicator, from a single regression per panel. Cells whose surrounding zones do not match the registry are excluded; the estimation sample is 868,896 cell-months, 96% of the nearest-zone sample. Specifications and inference follow Appendix Figure A8. Navy markers are point estimates, navy bars are 90% confidence intervals, navy caps are 95% confidence intervals from province-clustered standard errors. Data are from Shen et al. (2024) for PM2.5, ERA5-Land for wind (Muñoz-Sabater et al., 2021), and the Ministry of Planning and Investment zone registry.

Figure A11: Industrial Zone Establishment and Manufacturing Employment



Notes: The figure reports event-study estimates of industrial zone establishment on employment in light manufacturing (left panel) and pollution-intensive manufacturing (right panel) in nearby districts, the employment counterpart of the firm-count estimates in Figure 6. Specifications, classification, estimator, and inference follow Figure 6. District-level employment is noisier than firm counts, and the light-manufacturing employment series displays pre-treatment movement; the firm-count estimates are therefore the primary evidence on the composition of net entry. Outer bars show 95% confidence intervals and inner bars show 90% confidence intervals. Source: Data are from the Vietnam Enterprise Survey for employment and the Ministry of Planning and Investment for industrial zone locations.

Figure A12: Industrial Zone Establishment and Manufacturing Firm Counts by Distance Band



Notes: The figure estimates equation (1) with the district as the unit of analysis and the number of light (Panel A) or pollution-intensive (Panel B) manufacturing firms as the outcome, in levels. For each two-year cohort, treated districts are those first within 5 km of an operating zone in the cohort years, comparison districts are those never within 5 km of any zone, assigned to bands by the distance to their nearest zone, and the reference band is 45 km and beyond. Bands beyond 5 km are 10 km wide, because districts are larger than the cells of Section 4, so that each band contains at least 34 comparison districts. Coefficients are averaged across cohorts with weights proportional to the number of treated districts, as in equation (2), and standard errors are clustered at the province level. Classification, sample years (2001–2018), and data follow Figure 6. Outer bars show 95% confidence intervals and inner bars show 90% confidence intervals. Source: Data are from the Vietnam Enterprise Survey and the Ministry of Planning and Investment for industrial zone locations.

B POLLUTION-INTENSITY CLASSIFICATION FROM THE VIETNAM ENTERPRISE SURVEY

This appendix describes the construction of the pollution-intensity classification used in the paper and compares the resulting industry ranking with the U.S.-based classification of Shapiro and Walker (2018).

B.1 Construction

The classification is built from the energy module of the Vietnam Enterprise Survey, which reports physical quantities of nine energy types (electricity, coal, briquettes, gasoline, kerosene, mazut, diesel, LPG, and natural gas) consumed for production in the 2009 and 2011–2015 survey rounds. I convert quantities to tons of coal equivalent (TCE) using default net calorific values from Garg et al. (2006) and volume-to-mass conversion factors from United Nations Department of Economic and Social Affairs (2022). Fossil energy is total TCE excluding electricity. The measure therefore captures on-site combustion, the process that generates the thermal anomalies and primary particulate emissions studied in the paper.

For each firm, pollution intensity is fossil TCE per unit of output, measured as gross turnover deflated to 2010 prices with the World Bank CPI series. I drop observations with nonpositive revenue and winsorize the firm-level intensity ratios at the top and bottom 2.5% within each industry, pooling across survey rounds. The industry-level intensity is the revenue-weighted mean of the winsorized firm ratios, equivalent to industry fossil TCE per unit of industry output. The estimation sample comprises approximately 141,000 manufacturing firm-year observations.

The classification unit is a set of 20 manufacturing categories that can be measured consistently over the full 2001–2018 firm panel, spanning the change in Vietnam’s industrial classification from VSIC 1993 to VSIC 2007.¹ Pollution-intensive industries are the top quartile of the intensity ranking, five of the 20 categories: non-metallic mineral products, paper and paper products, coke and refined petroleum products, basic metals, and textiles. The ranking is insensitive to construction choices: rank correlations between the baseline intensity and variants using 1% winsorization tails, unweighted means, unwinsorized ratios of sums, or intensity per worker range from 0.84 to 0.96.

B.2 Comparison with the U.S.-based classification

Table B1 reports the full ranking alongside the emissions intensity of Shapiro and Walker (2018), which is estimated from U.S. data and widely used in the literature, and therefore provides an external benchmark. The comparison shows two features. First, the four highest-ranked categories under the Vietnamese energy data, non-metallic minerals, paper, coke and refined petroleum, and basic metals, are the core of the U.S.-based classification. Second, the classifications differ below the top. The U.S.-based intensities are defined for broader sector groups: printing and

¹Recycling (VSIC 1993) and repair and installation of machinery (VSIC 2007) do not correspond to a consistent industry across the two classification vintages and are left unclassified.

recorded media is classified as pollution-intensive only because it belongs to the paper group, yet it ranks last, 20th of 20, in the Vietnamese energy data. Drugs and chemicals belongs to the chemicals group but ranks 8th, reflecting that chemical production pollutes primarily through process emissions and water effluent rather than fuel combustion, the margin relevant for the combustion-based outcomes studied here. Conversely, textiles ranks 5th in Vietnam, consistent with the coal-fired boilers used in dyeing and finishing, and with the inclusion of dyeing among the production types with pollution risk in Appendix IIa of Decree 40/2019/ND-CP.² Overall, the rank correlation between the two intensity measures is 0.38: agreement is concentrated at the polluting tail, the part of the ranking that determines the classification, and weaker in the middle of the distribution.

The two classifications differ more for individual firms than in aggregate shares. The Vietnamese classification marks 16.5% of manufacturing firm-year observations as pollution-intensive against 24.1% under the U.S.-based classification, and the two agree for 84% of observations; the pollution-intensive share of manufacturing employment is 13 to 14% under either. Appendix Figure B1 re-estimates the firm-count event studies of Figure 6 under the U.S.-based classification and under alternative cutoffs of the Vietnamese ranking, the top four categories (dropping textiles) and the top eight (adding food and beverages, other transport equipment, and drugs and chemicals, which fall just below the quartile cutoff).

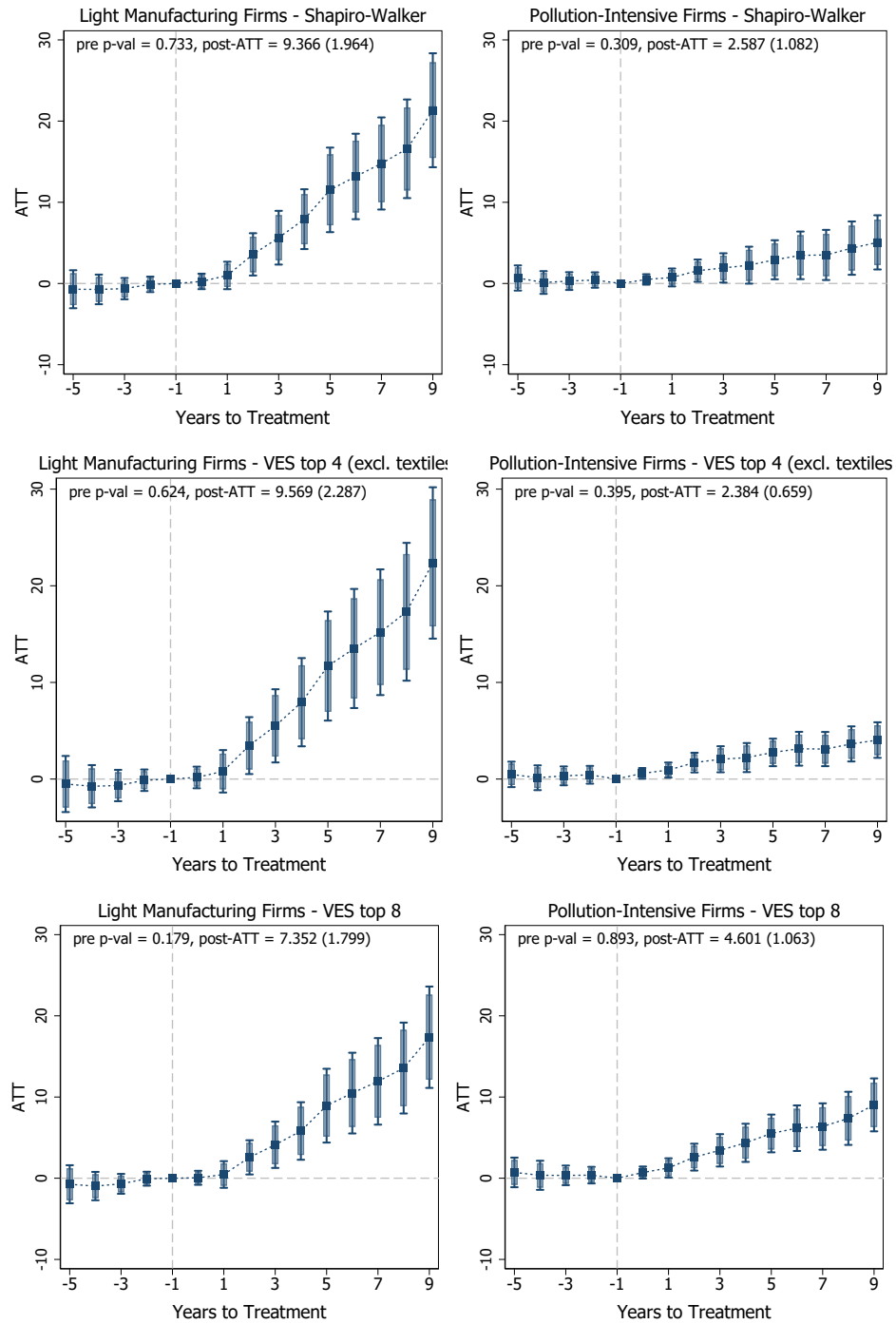
²At the more disaggregated two-digit VSIC 2007 level, available for the 2009–2015 energy rounds, beverages ranks fifth of 23 manufacturing industries, reflecting coal-fired boilers in breweries; beer and liquor production also appears in Appendix IIa of Decree 40/2019/ND-CP. In the panel-consistent categories, beverages is merged with the much larger food sector and the combined category ranks sixth.

Table B1: Pollution-Intensity Rankings: VES Energy-Based versus U.S.-Based

Manufacturing category	VES energy-based		Shapiro and Walker (2018)		Pollution-intensive	
	Intensity	Rank	Intensity	Rank	VES	SW
Non-metallic mineral products	0.0513	1	19.91	2.0	✓	✓
Paper and paper products	0.0071	2	14.61	3.5	✓	✓
Coke and refined petroleum products	0.0051	3	13.88	5.0	✓	✓
Metals	0.0037	4	36.57	1.0	✓	✓
Textiles	0.0027	5	1.44	15.0	✓	
Food and beverage	0.0022	6	2.60	10.5		
Other transport equipment	0.0021	7	1.26	17.0		
Drug and other chemical products	0.0021	8	13.42	6.0		✓
Rubber and plastic products	0.0010	9	3.13	8.0		
Wearing apparel	0.0009	10	1.44	15.0		
Fabricated metal products	0.0008	11	1.24	18.0		
Wood and wood products	0.0007	12	6.75	7.0		
Electronic, computer and optical products	0.0006	13	1.52	12.5		
Machinery and equipment nec	0.0006	14	1.00	20.0		
Leather and related products	0.0005	15	1.44	15.0		
Motor vehicles	0.0004	16	1.03	19.0		
Tobacco	0.0004	17	2.60	10.5		
Furniture and others	0.0003	18	3.06	9.0		
Electrical appliances	0.0003	19	1.52	12.5		
Recorded media	0.0002	20	14.61	3.5		✓

Notes: The table reports pollution intensity by manufacturing category for the 20 categories measured consistently over the 2001–2018 firm panel, sorted by the VES energy-based ranking. The VES intensity is fossil-fuel energy in tons of coal equivalent per million constant-2010 dong of gross output, computed from firm-level energy and revenue records in the 2009 and 2011–2015 VES rounds, winsorized at 2.5% tails within category and aggregated with revenue weights. The Shapiro and Walker (2018) intensity is emissions per unit of output estimated from U.S. data, matched to the Vietnamese manufacturing categories at the level of broader sector groups. The final two columns mark the pollution-intensive classification under each measure: VES = top quartile of the energy-based ranking (used throughout the paper); SW = the U.S.-based classification of Shapiro and Walker (2018). The Spearman rank correlation between the two intensity measures is 0.38. *Source:* Vietnam Enterprise Survey and Shapiro and Walker (2018).

Figure B1: Firm Counts under Alternative Pollution-Intensity Classifications



Notes: The figure re-estimates the firm-count event studies of Figure 6 under three alternative pollution-intensity classifications. The top row uses the U.S.-based classification of Shapiro and Walker (2018). The middle row uses the top four categories of the VES energy-based ranking (excluding textiles). The bottom row uses the top eight categories (adding food and beverages, other transport equipment, and drugs and chemicals). Specifications, estimator, and inference follow Figure 6. Source: Vietnam Enterprise Survey and Ministry of Planning and Investment.

C THE SPATIAL REACH OF ZONE EXPOSURE AND ALTERNATIVE FIXED EFFECTS

This appendix decomposes the PM2.5 estimate in Table 4, column (2), by the distance between a cell and the zones whose exposure affects it, and uses the decomposition to interpret estimates of equation (4) under alternative sets of fixed effects. Let d_{ij} denote the distance between the centroids of cells i and j , and define three distance bands around cell i : $\mathcal{N}_1(i) = \{j \neq i : d_{ij} \leq 25\}$, $\mathcal{N}_2(i) = \{j : 25 < d_{ij} \leq 50\}$, and $\mathcal{N}_3(i) = \{j : 50 < d_{ij} \leq 100\}$, with distances in km. The average downwind exposure of band k in month m of year t is

$$R_{imt}^{(k)} = \frac{1}{|\mathcal{N}_k(i)|} \sum_{j \in \mathcal{N}_k(i)} E_{jmt}, \quad k = 1, 2, 3 \quad (\text{C1})$$

where E_{jmt} is the downwind exposure of cell j from equation (5) and $|\mathcal{N}_k(i)|$ is the number of cells in the band. I divide each band average by the standard deviation that scales the own-cell measure, so that all coefficients below are per standard deviation of exposure. Adding the band averages to equation (4) gives

$$\ln \text{PM2.5}_{ipmt} = \kappa_0 E_{imt} + \sum_{k=1}^3 \kappa_k R_{imt}^{(k)} + \alpha_i + \mu_{mt} + \varepsilon_{ipmt} \quad (\text{C2})$$

where κ_k is the effect of average exposure in band k , conditional on the cell's own exposure and on exposure in the other bands.

Equation (4) omits the band averages. Its coefficient β therefore combines the own-cell effect with the band effects, weighted by the extent to which band exposure varies with own-cell exposure. Let δ_k denote the coefficient from a regression of $R_{imt}^{(k)}$ on E_{imt} with the same controls and fixed effects as equation (4). Because equations (4) and (C2) are estimated on the same sample with the same controls and fixed effects, the omitted variable bias formula (Angrist & Pischke, 2009, Section 3.2.2) holds exactly:

$$\beta = \kappa_0 + \sum_{k=1}^3 \kappa_k \delta_k. \quad (\text{C3})$$

The term $\kappa_k \delta_k$ is the contribution to β of exposure in band k .

Panel A of Table C1, column (1), reports estimates of equation (C2) on the sample and with the fixed effects of Table 4, column (2). Conditional on its own exposure, a cell's PM2.5 increases with average exposure in the 0–25 km band (0.031) and in the 25–50 km band (0.048). The coefficient on the 50–100 km band is imprecisely estimated, and average exposure in a 100–200 km band has no explanatory power when added as a fourth regressor.

The estimates of δ_k in this specification are 0.40, 0.10, and 0.01. Under equation (C3), the estimate of 0.027 in Table 4 consists of 0.010 from the cell's own exposure, 0.012 from exposure within 25 km, 0.005 from exposure at 25–50 km, and less than 0.001 from exposure beyond 50 km. The exposure that affects a cell's PM2.5 is therefore located within about 50 km of the cell,

Table C1: PM2.5 Estimates under Alternative Fixed Effects: Decomposition by Distance Band

	Dependent Variable is $\ln(\text{PM}_{2.5})$		
	(1)	(2)	(3)
Panel A: Distance profile of exposure effects, equation (C2)			
Own cell	0.010 (0.004)	0.007 (0.002)	0.004 (0.001)
0–25 km band	0.031 (0.011)	0.024 (0.006)	0.030 (0.004)
25–50 km band	0.048 (0.015)	0.053 (0.018)	0.043 (0.008)
50–100 km band	0.023 (0.025)	0.015 (0.027)	0.060 (0.015)
Share of 25–50 km band variation absorbed	71%	77%	85%
Panel B: Estimate from equation (4)			
Downwind Exposure (SD)	0.027 (0.005)	0.017 (0.003)	0.009 (0.001)
Implied effect ($\mu\text{g}/\text{m}^3$)	0.66	0.42	0.22
Observations	793,212	793,212	793,212
Adjusted R-squared	0.739	0.869	0.947
Cell FE	Yes	Yes	Yes
Year-Month FE	Yes		
Region $\times$ Year-Month FE		Yes	
Province $\times$ Year-Month FE			Yes
Weather Controls	Yes	Yes	Yes

Notes: The table reports estimates on the estimation sample of Table 4, column (2): cells in the 5–25 km transport ring, monthly, with weather controls (a quadratic in monthly mean 2-meter temperature and monthly total precipitation from ERA5-Land) and downwind exposure standardized within the sample. Column (1) includes cell and year-month fixed effects; columns (2) and (3) replace the year-month fixed effects with region-by-year-month and province-by-year-month fixed effects. Panel A reports equation (C2), which adds the average downwind exposure of the panel cells in three distance bands around each cell (equation (C1)), scaled by the same standard deviation as the own-cell measure. Band coefficients are not comparable across rows, because the bands contain different numbers of cells. The bottom row reports the share of the variation in 25–50 km band exposure absorbed by each set of fixed effects, computed as one minus the ratio of the within-transformed to the raw standard deviation. Panel B reports the coefficient on the cell's own downwind exposure from equation (4); the implied effect converts the coefficient to $\mu\text{g}/\text{m}^3$ at the sample-mean PM_{2.5} of 24.4 $\mu\text{g}/\text{m}^3$. Panels A and B are connected by the identity in equation (C3). Standard errors clustered at the province level in parentheses.

consistent with the two distances used in the research design: the 25 km radius of the exposure measure and the 50 km outer bound of the estimation panel. Alternative partitions of the bands, at 0–20/20–40/40–80 km and at 0–15/15–30/30–60 km, yield a similar pattern.

Columns (2) and (3) replace the year-month fixed effects with region-by-year-month and province-by-year-month fixed effects (7 regions and 58 provinces in the estimation sample). Panel B reports the corresponding estimates of equation (4). The coefficient remains positive and statistically significant, at 0.027 per standard deviation of exposure with year-month fixed effects, 0.017 with region-by-year-month fixed effects, and 0.009 with province-by-year-month fixed effects. Equation (C3) holds in each column and separates two possible sources of the decline: a smaller effect of exposure at each distance (the κ_k) or a weaker relationship between band exposure and own-cell exposure (the δ_k).

The estimates in Panel A indicate the latter. The band coefficients differ little across columns,

at 0.031, 0.024, and 0.030 for the 0–25 km band and 0.048, 0.053, and 0.043 for the 25–50 km band. The effect of exposure at these distances is thus similar whether it is identified from comparisons across the whole country or from comparisons among cells in the same province and month, which argues against a confounder operating at the region or province level as the source of the band effects.

What differs across the columns is the relationship between band exposure and own-cell exposure. Cells in the same province share most of their surroundings: the bottom row of Panel A shows that province-by-year-month fixed effects absorb 85% of the variation in 25–50 km band exposure, compared with 77% for region-by-year-month and 71% for year-month fixed effects, whereas the own-cell measure retains 47% of its variation in the tightest specification. The estimate of δ_2 falls from 0.10 in column (1) to 0.04 in column (2) and -0.03 in column (3). With the κ_k similar across columns, the composite coefficient declines accordingly.

The coefficient in equation (4) measures the effect of zone exposure inclusive of pollution transported from surrounding zones whose exposure varies with the cell's own, which is the object relevant for the local externality the paper studies. Region-by-year-month and province-by-year-month fixed effects remove this transported component from the comparison, and their lower coefficients likely understate the full local effect rather than correct it. The effect of a one-standard-deviation increase in exposure lies between 0.017 and 0.027 log points, or 0.42 to 0.66 $\mu\text{g}/\text{m}^3$ at the sample mean. The province-by-year-month estimate of 0.009 sits below this range because its δ_2 is negative.

D COMPOSITION OF FIRMS INSIDE ZONES OVER TIME

Section 6.1 draws its composition evidence from firms operating in districts near zones. This appendix turns to firms operating inside zones. In its 2010 to 2016 rounds, the VES asks each firm whether it is located in a zone. Because the question covers all zone types and is answered by the firm itself, the reported in-zone firms span a broader set of zones than the empirical analysis. I use the 2011 and 2013–2016 rounds, in which the question was asked of nearly all firms (in 2010 and 2012, when the detailed questionnaire went to a sample, it was asked of about one third), and keep manufacturing firms reporting a zone location. Pollution-intensive industries are defined as in the rest of the paper (Appendix B).

Panel A of Table D1 reports the national composition of in-zone manufacturing by year. The mix of firms inside zones did not shift toward light manufacturing over the period the data cover. The pollution-intensive share is 0.20 of firms in 2011 and 0.19 in 2016, and about 0.10 of employment in both years. The share changes little among firms with 100 or more workers and declines slightly among smaller firms. The industries whose processes run hot enough to register as satellite thermal anomalies (non-metallic minerals, basic metals, and coke and refined petroleum) account for about a tenth of in-zone manufacturing firms and 3 to 4% of their employment in every year (the last two rows of Panel A).

Panel B tests whether the PM_{2.5} effect of exposure declined where the in-zone mix did change. The VES records that a firm is in a zone but not which one, and most districts in the industrial core host several zones. The unit of analysis is therefore the district. The sample comprises districts hosting zones that were operational by 2009 and with at least five in-zone manufacturing firms in both 2011 and 2016: 94 districts. These two years are the first and last rounds in which the zone question was asked of nearly all firms, and the reference years of the 2012 and 2017 Economic Censuses, in which every enterprise receives the full questionnaire.

Since light industries also burn fuel, a shift in the mix should be measured with every industry weighted by its own intensity rather than by a pollution-intensive share. A share gives lighter industries a weight of zero, and it declines when light manufacturing grows around unchanged heavy plants. I therefore measure the shift as the 2011–2016 change in the predicted fuel use per worker of a district’s in-zone manufacturing firms, where predicted fuel use is employment multiplied by the fossil-fuel intensity per worker of the firm’s two-digit industry from the ranking in Appendix B. Expressed per worker, the measure reflects the composition of in-zone employment rather than its size. I sort districts into terciles of the change. The mean change is –0.6 log points in the first tercile, whose mix moved most toward lighter industries, and +0.6 in the third.

Because the terciles differ in the level of the effect, a comparison of their post-shift effects alone cannot show whether the effect fell where the mix became lighter. The comparison is therefore a difference-in-differences: for each tercile, the effect before the shift, the effect after it, and their difference. I assign cells to the district hosting their nearest zone. Within each tercile, using the cells of the pre-2003 and 2003–2009 cohorts, I regress log PM_{2.5} on standardized downwind exposure and on its interaction with an indicator for the later period, with the cell and year-month

Table D1: Composition of Manufacturing Firms inside Zones, 2011–2016

Panel A: In-zone manufacturing firms, national

	2011	2013	2014	2015	2016
Firms	5,580	6,671	7,049	7,712	8,730
Employment (thousands)	1,935	2,413	2,660	2,914	3,209
Pollution-intensive share of firms	0.200	0.202	0.190	0.191	0.190
Pollution-intensive share of employment	0.096	0.099	0.095	0.091	0.095
Share among firms with 100+ workers	0.164	0.176	0.168	0.163	0.168
Share among firms with <100 workers	0.231	0.223	0.210	0.214	0.206
Hot-process share of firms	0.098	0.101	0.092	0.089	0.091
Hot-process share of employment	0.042	0.042	0.039	0.035	0.034

Panel B: PM2.5 effect of exposure by change in the in-zone mix of the host district, 2011–2016

	Districts sorted by the 2011–2016 change in predicted fuel use per worker of in-zone firms		
	(1) Decreased most	(2) Middle tercile	(3) Increased most
Mean change in log predicted fuel use per worker	-0.60	-0.07	0.57
Districts (zones)	32 (57)	31 (73)	31 (73)
Cells	586	325	515
Effect of exposure on ln(PM2.5), pre-2009 cohorts:			
Years 2011 and earlier	0.030 (0.013)	0.008 (0.006)	0.032 (0.007)
Years 2017 and later	0.038 (0.010)	0.020 (0.006)	0.040 (0.004)
Change, 2017 and later minus 2011 and earlier	0.008 (0.007)	0.012 (0.005)	0.008 (0.005)

Notes: Panel A reports manufacturing firms that report a zone location in the Vietnam Enterprise Survey, in the rounds in which the zone question was asked of nearly all firms (5,580 firms in 2011 and 8,730 in 2016). Pollution-intensive industries are the top quartile of the VES energy-based intensity ranking (Appendix B); hot-process industries are non-metallic minerals, basic metals, and coke and refined petroleum. Panel B sorts districts hosting zones operational by 2009, with at least five in-zone manufacturing firms in both 2011 and 2016 (94 districts hosting 203 zones), into terciles of the 2011–2016 change in the predicted fuel use per worker of their in-zone manufacturing firms, where predicted fuel use is employment multiplied by the fossil-fuel intensity per worker of the firm’s two-digit industry (Appendix B); the first row reports the mean change in log points. Each cell in the 5–25 km transport ring is assigned to the district hosting its nearest zone. For each tercile, the three coefficients come from one regression of $\ln(\text{PM}_{2.5})$ on standardized downwind exposure and its interaction with an indicator for the years from 2017 onward, with cell and year-month fixed effects and weather controls as in Table 5, Panel B, estimated on cells of the pre-2003 and 2003–2009 cohorts in the tercile, with pre-treatment cell-months always included and post-treatment cell-months restricted to the years through 2011 and from 2017 onward. The rows report the effect in the earlier period, the effect in the later period, and their difference. Standard errors clustered at the province level in parentheses.

fixed effects and weather controls of equation (4). The coefficient on exposure is the effect in the earlier period, the sum of the two coefficients is the effect in the later period, and the coefficient on the interaction is the change between them. The earlier period is the post-treatment months through 2011, before the interval over which the shift is measured. The later period is the months

from 2017 onward, after it. The months of 2012–2016 are excluded, and pre-treatment months are always included. I define the periods in calendar time rather than zone age because the sorting variable is a change over a fixed calendar interval.

In the first tercile, the per-exposure effect is 0.030 in the years through 2011 and 0.038 from 2017 onward, a change of 0.008 (s.e. 0.007). The change is the same in the third tercile and similar in the second (Panel B). The effect therefore did not decline in the districts whose in-zone mix moved most toward lighter industries, and the change in the effect is unrelated to the direction of the shift. The terciles differ instead in the level of the effect. It is highest in the third tercile, whose in-zone mix is the most fuel-intensive by 2016, and lowest in the second, whose zones host the largest concentrations of light assembly.

The shift in the first tercile did not come from pollution-intensive firms leaving or contracting. Across the districts of this tercile, the number of pollution-intensive in-zone firms rose from 230 to 301 and their employment from 39 to 46 thousand between 2011 and 2016, while employment in other manufacturing rose from 194 to 456 thousand. Predicted fuel use of the pollution-intensive firms changed little. That of the other industries rose by roughly two thirds. The mix moved toward lighter industries because light manufacturing grew around the heavy plants, not because they closed, and the lighter entrants added fuel use of their own. Where the number or employment of pollution-intensive in-zone firms fell elsewhere in the sample, the declines were small, and no district lost its pollution-intensive base.

To check whether the conclusion depends on how the shift is measured, I estimate

$$\ln \text{PM}_{imt} = \beta_1 E_{imt} + \beta_2 E_{imt} P_t + \gamma_1 E_{imt} \Delta S_d + \gamma_2 E_{imt} P_t \Delta S_d + \mathbf{X}'_{imt} \boldsymbol{\theta} + \alpha_i + \lambda_{mt} + \varepsilon_{imt} \quad (\text{D1})$$

where E_{imt} is standardized downwind exposure, $P_t = \mathbf{1}(t \geq 2017)$, ΔS_d is the shift measure of the district hosting cell i 's nearest zone, standardized to unit variance across the 94 districts, and the sample, controls, and fixed effects are otherwise those of Panel B, with pre-treatment months always included and the post-treatment months of 2012–2016 excluded. The effect of exposure is β_1 in the years through 2011 and $\beta_1 + \beta_2$ from 2017 onward. The coefficient γ_1 measures how the early-period effect varies with the shift, and γ_2 how the change in the effect between the two periods varies with it. If the effect declined where the mix became lighter, γ_2 would be positive. Table D2 reports γ_1 and γ_2 for five constructions of ΔS_d : the change in the pollution-intensive share of in-zone employment, the log changes in the number and in the employment of pollution-intensive firms, and the log changes in predicted fuel use in total and per worker. Across the five measures γ_2 lies between -0.008 and 0.001 with standard errors between 0.003 and 0.005 : a one-standard-deviation larger shift toward lighter industries alters the change in the effect between the two periods by less than 0.01 against a level of 0.023 , and not in the direction of a composition-driven decline. Extending the later period to 2014 onward, which adds three years during which the shift is still being measured, gives γ_2 between -0.005 and 0.001 . Neither the construction of the shift measure nor the choice of period affects the interpretation.

Two caveats apply. First, the shifts observed over 2011–2016 were moderate. Pollution-intensive

Table D2: PM2.5 Effect of Exposure and Alternative Measures of the Shift in the In-Zone Mix

Measure of the shift ΔS_d	γ_1 (Exposure $\times \Delta S_d$)	γ_2 (Exposure $\times 1(t \geq 2017) \times \Delta S_d$)
Change in pollution-intensive share of employment	0.000 (0.005)	0.001 (0.005)
Log change in number of pollution-intensive firms	-0.003 (0.007)	-0.001 (0.004)
Log change in employment of pollution-intensive firms	-0.003 (0.007)	0.001 (0.004)
Log change in predicted fuel use	0.003 (0.007)	-0.008 (0.004)
Log change in predicted fuel use per worker	0.003 (0.005)	-0.003 (0.003)

Notes: Each pair of rows reports the coefficients γ_1 and γ_2 from one estimate of equation (D1), in which the shift measure ΔS_d is the 2011–2016 change for the district hosting the cell’s nearest zone, standardized to unit variance across the 94 districts of Table D1, Panel B. The measures are the change in the share of in-zone manufacturing employment in pollution-intensive industries; the log change in the number of pollution-intensive in-zone firms; the log change in their employment; and the log changes in the predicted fuel use of all in-zone manufacturing firms, in total and per worker. The sample is that of Panel B of Table D1, pooling the three terciles and keeping the months through 2011 and from 2017 onward (1,426 cells, 239,568 cell-months). For the per-worker measure, $\beta_1 = 0.023$ (0.006) and $\beta_2 = 0.010$ (0.004). Standard errors clustered at the province level in parentheses.

firms rarely left the older zones, and the shifts toward lighter industries came from the growth of light manufacturing beside them. The results show that shifts of this kind do not lower the effect. They do not show what the exit of pollution-intensive firms would do, because it did not occur in these years. Second, the results are descriptive, since which firms enter and leave a zone is not exogenous.

With these caveats, the firm-level data show two results. Nationally, the pollution-intensive share of in-zone firms and of their employment changed little between 2011 and 2016 (Panel A). Across districts hosting older zones, shifts in the in-zone mix are unrelated to the change in the PM2.5 effect of exposure, although the level of the effect tracks the level of the mix (Panel B and Table D2). Both results are consistent with the within-cohort stability of the effect in Table 5: a zone’s pollution burden is set by the industries admitted when it fills and persists thereafter.

E DOWNWIND EXPOSURE WEIGHTED BY OBSERVED COMBUSTION AT THE SOURCE

The exposure measure in equation (5) counts every operating zone within 25 km of a cell equally. This appendix examines how the PM2.5 effect of exposure relates to the thermal activity observed at the source, measured from the Type-2 thermal anomalies of Section 4.

Equation (5) is a sum of zone contributions,

$$E_{imt} = \sum_{j \in \mathcal{J}_i} c_{ijmt}, \quad c_{ijmt} = \mathbb{I}(t \geq T_j) \cdot \text{distance}_{ij}^{-1} \cdot D_{ijmt} \quad (\text{E1})$$

where zone j contributes once it is operating, with a weight that declines with its distance from cell i and rises with the share of the month the cell spends downwind of it. The weighted measure multiplies each contribution by the zone's observed combustion F_j ,

$$E_{imt}^F = \sum_{j \in \mathcal{J}_i} c_{ijmt} F_j. \quad (\text{E2})$$

F_j is the zone's mean annual Type-2 detections over its operating years in 2002–2020, counted in the 0.05-degree cells within 5 or 10 km of the zone centroid, with each cell attributed to its nearest zone. Zones with no detections receive weight zero, and they are 40% of zones at 5 km and 29% at 10 km. Because F_j is fixed over time, month-to-month variation in E_{imt}^F comes from the downwind shares, as in E_{imt} , and setting $F_j \equiv 1$ returns E_{imt} itself. Alternative weights replace the detection count with fire radiative power or fire-days, or enter $\log(1 + F_j)$ in place of F_j .

The question is how a zone's effect varies with its observed combustion. Equation (4) assigns every zone the same effect per unit of contribution. Allowing each zone its own coefficient gives

$$\ln \text{PM2.5}_{imt} = \sum_{j \in \mathcal{J}_i} \beta_j c_{ijmt} + \mathbf{X}'_{imt} \boldsymbol{\theta} + \alpha_i + \mu_{mt} + \varepsilon_{imt}. \quad (\text{E3})$$

Estimating equation (4) with E_{imt}^F in place of E_{imt} imposes $\beta_j = \phi F_j$: a zone's effect is proportional to its observed combustion, and a zone without detections has none. The linear form $\beta_j = \beta + \phi F_j$ nests both cases, the proportional model at $\beta = 0$ and equation (4) at $\phi = 0$. Under the linear form, equation (E3) becomes

$$\ln \text{PM2.5}_{imt} = \beta E_{imt} + \phi E_{imt}^F + \mathbf{X}'_{imt} \boldsymbol{\theta} + \alpha_i + \mu_{mt} + \varepsilon_{imt} \quad (\text{E4})$$

where β is the effect of exposure to a zone with no detections and ϕ is the additional effect per unit of observed combustion.

Table E1, Panel A, reports two regressions for each weighting. The first uses the weighted measure as the only exposure regressor, which imposes $\beta = 0$. The second is equation (E4), which includes both measures. The sample is that of Table 4, column (2), and each measure is standardized within it. In the first regression, the coefficient on the weighted measure is between 0.013 and

0.023 per standard deviation, below the 0.027 for E_{imt} . In the second, β is 0.026 for every weighting and ϕ is between 0.001 and 0.002, with a standard error of the same size. Exposure to a zone with no detections raises PM2.5 by as much as exposure to any other zone, and additional observed combustion adds nothing. The first regression gives a lower coefficient because it forces zones without detections to have no effect, a restriction the data do not appear to support.

Panel B sorts cells into terciles of their nearest zone's F_j and interacts E_{imt} with the tercile indicators. The effect does not rise across terciles: 0.021, 0.038, and 0.027 for detections within 10 km, with a similar pattern for radiative power. Zones with no detections raise downwind PM2.5 as much as zones in the top tercile, which record about four detections a year.

The per-exposure effect is therefore unrelated to the thermal activity observed at the source. This is the pattern the composition of zone tenants would predict. Thermal detections register the hot-process industries, which account for 9% of in-zone manufacturing firms and 3% of their employment (Appendix Table D1). The remaining firms burn fuel at temperatures below the detection threshold and contribute to ambient PM2.5 without contributing to detections. The thermal data record whether a zone hosts high-temperature sources, not how much it emits. Across zones, the PM2.5 effect follows the industrial composition documented in Section 6.1 rather than the intensity of detections.

Table E1: Downwind Exposure Weighted by Observed Combustion at the Source

Panel A: Weighted exposure measures

Weight F_j on each zone's contribution	Weighted measure only	Both measures, equation (E4)	
	Coefficient (1)	β (2)	ϕ (3)
None (baseline exposure, Table 4, column 2)	0.027 (0.005)		
Mean annual detections within 5 km	0.014 (0.004)	0.026 (0.005)	0.002 (0.004)
Mean annual detections within 10 km	0.014 (0.005)	0.026 (0.005)	0.002 (0.004)
Mean annual radiative power within 5 km	0.013 (0.004)	0.026 (0.005)	0.002 (0.004)
Mean annual radiative power within 10 km	0.014 (0.004)	0.026 (0.005)	0.002 (0.004)
Mean annual fire-days within 5 km	0.015 (0.004)	0.026 (0.005)	0.002 (0.005)
log(1 + mean annual detections within 10 km)	0.022 (0.005)	0.026 (0.007)	0.001 (0.008)
log(1 + mean annual radiative power within 5 km)	0.023 (0.005)	0.026 (0.008)	0.001 (0.010)
Observations	793,212		

Panel B: Effect of baseline exposure by tercile of the nearest zone's observed combustion

	Lowest (1)	Middle (2)	Highest (3)
Nearest zone's mean annual detections within 10 km	0.021 (0.005)	0.038 (0.009)	0.027 (0.007)
Tercile mean of the weight	0.0	0.3	3.9
Cells	1,095	1,112	1,077
Nearest zone's mean annual radiative power within 5 km	0.025 (0.005)	0.045 (0.015)	0.026 (0.006)
Tercile mean of the weight	0.0	1.5	31.1
Cells	1,648	557	1,079

Notes: Panel A estimates equation (4) with the weighted measure of equation (E2) in place of E_{imt} (column (1)) and equation (E4) (columns (2) and (3)), on the sample of Table 4, column (2): cells in the 5–25 km transport ring, monthly, with cell and year-month fixed effects and weather controls. Each measure is standardized by its standard deviation in the sample, so coefficients are per standard deviation of the respective measure. Weights are defined in the text; zones with no detections receive weight zero. Panel B interacts E_{imt} with indicators for terciles of the nearest zone's weight, defined over the cells in the estimation sample, and reports the tercile mean of the weight and the number of cells; cells whose nearest zone is not in the fire file are excluded (748,752 cell-months). Standard errors clustered at the province level in parentheses.

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